WHAT IS VALUE STREAM MAP (VSM)?



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WHAT IS LEAN?

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LEAN IS A PROCESS IMPROVEMENT TECHNIQUE THAT FOCUSES ON WASTE ELIMINATION

LEAN MANAGEMENT CONSISTS OF SEVERAL TOOLS AND TECHNIQUES THAT HELP IDENTIFY AND ELIMINATE WASTE FROM YOUR BUSINESS PROCESS

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WHAT IS VALUE STREAM MAP (VSM)?

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VSM IS A METHOD IN WHICH YOU:

- **CREATE A CURRENT STATE MAP**
- **ANALYZE IT**
- **DESIGN THE FUTURE STATE MAP**

VSM IS A VISUALIZATION TOOL

IT IDENTIFIES & ELIMINATES WASTE

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WHAT IS VALUE STREAM MAP (VSM)?

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THE NAME VALUE STREAM MAP IS DERIVED FROM THREE WORDS:

VALUE	STREAM	MAP
INDICATES THE VALUE-ADDING ACTIVITIES AT EVERY STEP OF THE STREAM	REFERS TO THE COMPLETE FLOW OF YOUR MANUFACTURING OR SERVICE OPERATIONS	INDICATES THE VALUE-ADDING ACTIVITIES AT EVERY STEP OF THE STREAM

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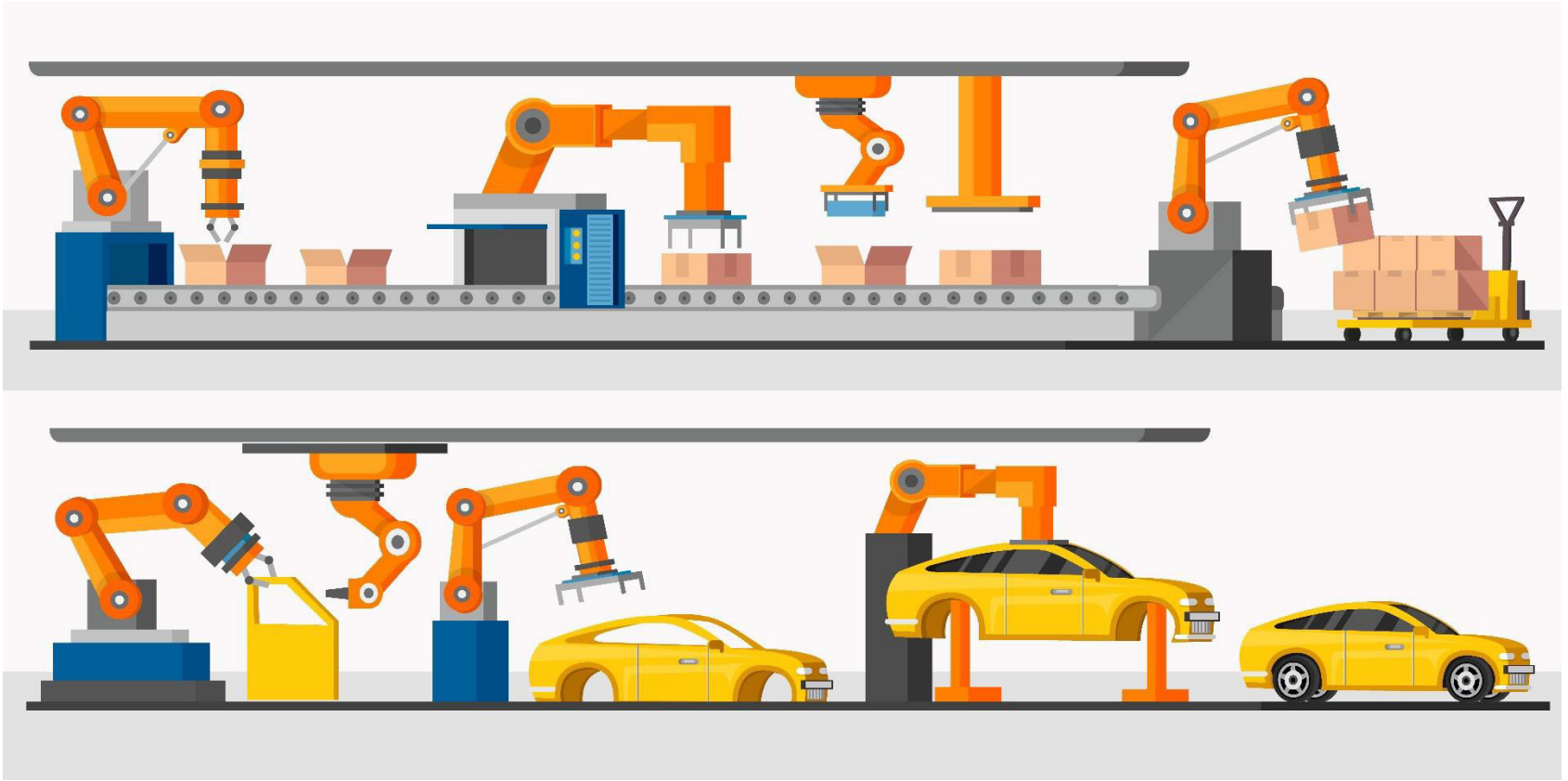
RELATIONSHIP OF CONTINUOUS FLOW & VSM

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CONTINUOUS FLOW IS THE ULTIMATE PRODUCTION FORM

VSM IS A STRONG SUPPORTER OF CONTINUOUS FLOW

VSM HELPS IDENTIFY WHERE CONTINUOUS FLOW CAN BE IMPLEMENTED



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INTERCOMPANY & INTRACOMPANY VSM

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IT IS IMPERATIVE TO MAP THE INTERCOMPANY AND THE INTRACOMPANY VALUE-ADDING PROCESSES

INTERCOMPANY: MAPPING BETWEEN 2+ ORGANIZATIONS

INTRACOMPANY: MAPPING WITHIN YOUR ORGANIZATION



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VALUE STREAM & VALUE CHAIN

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VALUE CHAIN MAKES USE OF ALL ACTIVITIES OF ALL PARTIES INVOLVED

VALUE STREAM ONLY CITES SPECIFIC PARTS OF THE COMPANY THAT ADD VALUE TO THE END-PRODUCT UNDER CONSIDERATION



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VALUE ADDED AND NON-VALUE ADDED ACTIVITIES

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VALUE ADDED ACTIVITIES ARE THOSE THAT THE CUSTOMER IS WILLING TO PAY FOR

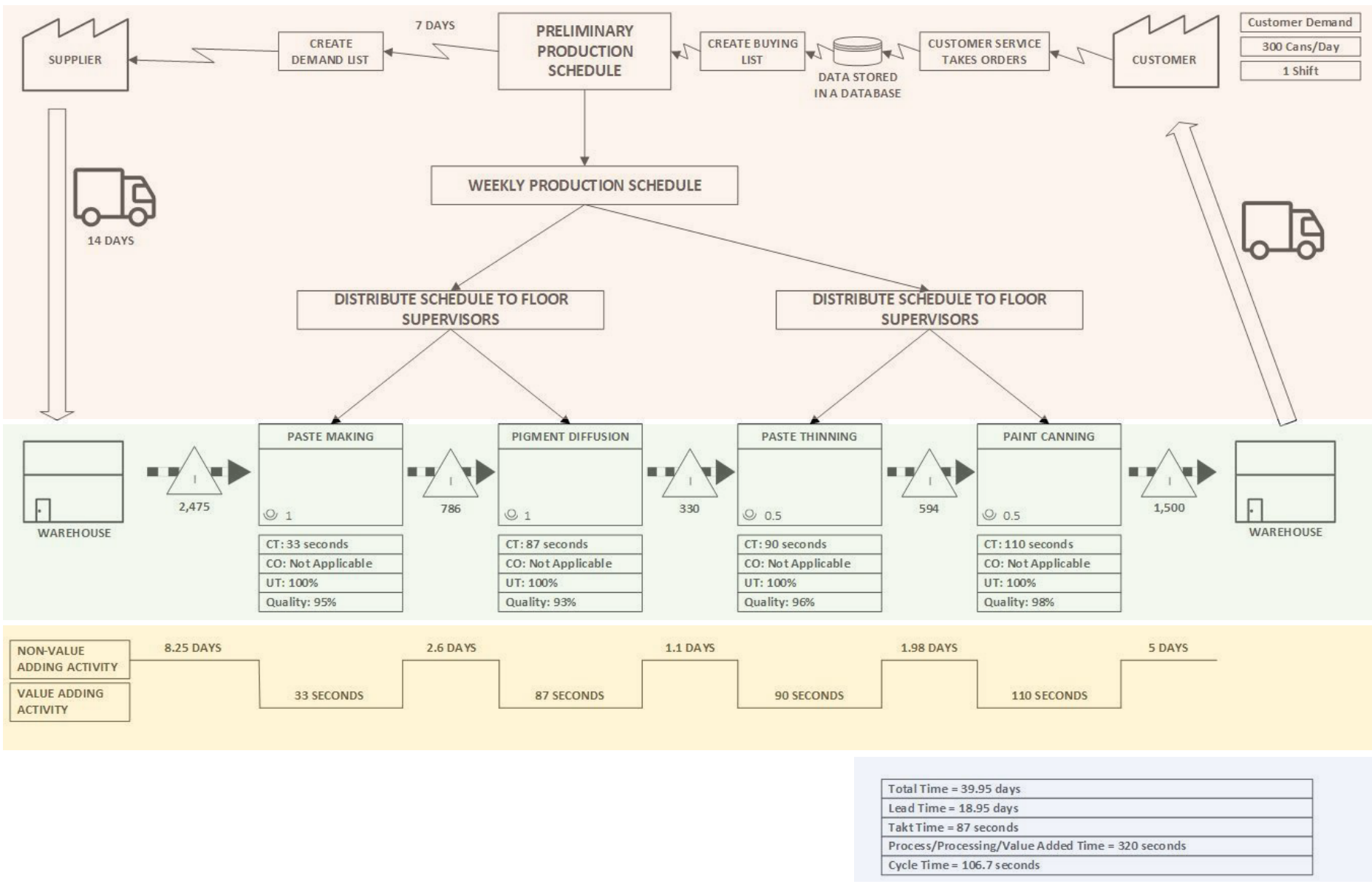


NON-VALUE ADDED ARE UNNECESSARY ACTIVITIES THAT DO NOT ADD ANY VALUE TO THE PROCESS & SHOULD BE ELIMINATED OR MINIMIZED

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COMPONENTS OF VSM

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INFORMATION FLOW

MATERIAL FLOW

TIME TAKEN AT EACH STEP

CALCULATIONS

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CURRENT STATE VS. FUTURE STATE MAP

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CURRENT STATE MAP

GRAPHIC LAYOUT OF A FINAL PRODUCT

DESCRIBES A HUGE PART OF PRODUCTION & EARNINGS

FILLED WITH INFO SUCH AS SCHEDULE, PUSH, PULL, FIFO, ETC

FUTURE STATE MAP

PRODUCED AFTER DEVELOPING CURRENT STATE MAP

IMPROVES UPON THE CURRENT SYSTEM, TWEAKS VALUE-ADDING STEPS, REMOVES NON-VALUE ADDING STEPS

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WHAT IS VALUE STREAM MAPPING SO POPULAR?



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REASONS FOR VSM'S POPULARITY

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REASONS FOR VSM'S POPULARITY

- ✓ FAST AND EASY TO LEARN AND TO FIND BOTTLENECKS
- ✓ YOUR ENTIRE WORKFORCE CAN BE INVOLVED IN CREATION OF THE VALUE STREAM MAP
- ✓ CAN SERVE AS A GUIDE TO EFFECT THE ACTUAL CHANGES DEPLOYED IN A BUSINESS PROCESS
- ✓ HIGHLY COST-EFFECTIVE
- ✓ OPEN TO SUGGESTIONS ABOUT THE PROBLEMS, AND WORKABLE SOLUTIONS FROM OTHER MEMBERS OF THE TEAM
- ✓ NOT LIMITED TO PRODUCTION AND MANUFACTURING. VSM IS INDUSTRY-INDEPENDENT
- ✓ EFFECTIVELY DEPICTS THE PRODUCTION PROCESS ALLOWING YOU TO QUICKLY SPOT WASTE IN YOUR PROCESS
- ✓ LITTLE TRAINING OF VSM ICONS NEEDED TO CEMENT VSM KNOWLEDGE
- ✓ IS A QUANTITATIVE AND QUALITATIVE TOOL
- ✓ HAS A DIRECT POSITIVE IMPACT ON CUSTOMER SATISFACTION, DEFECTS, LEAD-TIME, WASTE, NON-VALUE ADDED ACTIVITIES, EMPLOYEE PRODUCTIVITY, AND ACHIEVING COST SAVINGS

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SECTION CONCLUSION: INTRODUCTION TO VSM



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SECTION CONCLUSION

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IN THIS SECTION, YOU LEARNED:

1. WHAT IS VALUE STREAM MAPPING?
2. ITS DEFINITION
3. WHY IS VALUE STREAM MAPPING SO POPULAR?

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SECTION INTRODUCTION:
VSM TERMINOLOGIES & SYMBOLS



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VALUE STREAM MAPPING (VSM) MASTERCLASS

Introduction to Value Stream Map (VSM)

- Course Introduction
- What is Value Stream Mapping (VSM)?
- Why is Value Stream Mapping so popular?
- Section Conclusion

Terminologies and Symbols used in VSM

- Section Introduction
- Terminologies used in VSM
- Symbols used in VSM Process
- Material Flow Symbols
- Information Flow Symbols
- General Symbols
- Activity: Terminologies and Symbols used in VSM
- Activity Solution: Terminologies and Symbols used in VSM
- Section Conclusion

Steps to Create the Current State VSM

- Section and Case Study Introduction
- Step 1. Pick a Product Family
- Step 2. Choose Your VSM Team
- Step 3: Figure out Customer Demand
- Step 4: Map and Design the Process Flow
- Step 5: Map the Material Flow
- Step 6: Map the Information Flow
- Case Study Recap: Current State VSM
- Perform Calculations | Analyze Current State VSM
- Calculate Total Time | Lead Time | Takt Time
- Calculate Process Time | Cycle Time
- Activity: Steps to Create Current State VSM
- Activity Solution: Steps to Create Current State VSM
- Section Conclusion

Examine the Current State VSM

- Section Introduction
- Meeting Customer Demand with existing No. of Operators
- Meeting Customer Demand with increased No. of Operators
- Analyze the Process Balance
- Watch for Waste
- Section Conclusion

Strong Foundations for Future State VSM

- Section Introduction
- Change to Continuous Flow
- CONWIP
- Kanban
- Readjusting Your Process
- Alter the Batch Sizes
- Smoothening out Production
- Heijunka | Heijunka Box
- Cut back Changeover Time
- Improve Quality
- Putting it all Together

Key Questions to ask for Future State VSM

- Section Introduction
- What is the Takt Time?
- Are you building to a warehouse or directly to shipping?
- Where should Continuous Flow Control the Production?
- Controlling area of upstream processes
- Where will you implement the Pull System?
- How do you intend to level the production mix?
- What process improvements do you intend to achieve?
- What is the Implementation Plan?
- Closure of the Case Study
- Section Conclusion

VSM Guidelines

- Section Introduction
- How to get the most of your VSM?
- What are the Pitfalls to avoid?
- What are the advantages of VSM?
- What are the disadvantages of VSM?
- Section Conclusion

Course Conclusion

- Course Summary
- Practice Questions
- Become a Certified VSM Practitioner!

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WHAT ARE THE DIFFERENT TERMINOLOGIES USED IN VSM? PART 1



VSM TERMINOLOGIES

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BOTTLENECK

The bottleneck is the term given to the operation with the longest cycle time per unit or the one with the lowest capacity. A bottleneck operation is not only the deciding limit for the production pace, but it also determines the capacity of the production process.

BUFFER

The Buffer, which is the inventory taken between operations, creates a counterbalance in the operations cycle time.

CHANGEOVER

The changeover time or setup time is used to prepare an operation for a new product.

CUSTOMER DEMAND

This term is used to describe the probable number of products that will be ordered by customers within a given period of time.

CYCLE TIME

This is the total amount of time required to complete one cycle of a task from start to finish by an operator before he/she repeats this operation.

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VSM TERMINOLOGIES

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DOWNSTREAM

Downstream processes are the next set of operations that receive the finished or unfinished work-in-progress goods from the current operations in focus.

DOWNTIME

Downtime is the period of time when the piece of equipment or machinery is needed is NOT available for several reasons. These reasons can range from breakdown of the machinery to its scheduled maintenance.

ESSENTIAL NON-VALUE ADDED TIME

This term collectively describes all the unnecessary but uneconomical periods spent in an operational process such as unpacking deliveries, delivering parts, and even transferring tools from one hand to the next.

FIFO

FIFO is an abbreviation of the term First-In-First-Out. FIFO is a queuing system that manages products as they arrive in queue.

FLOW

A continuous production process with no buffers in the middle of operations is termed flow.

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VSM TERMINOLOGIES

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HEIJUNKA BOX

The heijunka box is a scheduling system that equalizes operations. We have a separate lecture that discusses the Heijunka box in detail.

INVENTORY

All stored products are termed as inventory. Inventory is done before, within, or after a process with the finished products.

INVENTORY LEAD TIME

This is the period of waiting for the arrival of the goods or products in inventory.

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WHAT ARE THE DIFFERENT TERMINOLOGIES USED IN VSM? PART 2



VSM TERMINOLOGIES

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KANBAN

Kanban cards are signs used to direct the customer's need downstream.

LEAD TIME

The lead time can be minutes, hours, or days. It is the amount of time that is required for the completion of an operation process. The lead time can also stand for the amount of time it takes before the next action can begin.

MANUFACTURI--NG LEAD TIME

See process time.

NON-VALUE-ADDING TIME

This is the time that the customer is NOT willing to pay for. It includes the time spent on activities such as double handling, stacking the intermediate products, or even waiting time are termed non-value-adding time because they do not add value to the product directly. It is a waste that needs to be removed entirely or minimized as much as possible.

OPERATION TIME

Calculated as the changeover time plus the processing time, the operation time is the amount of time spent on the production of a product.

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VSM TERMINOLOGIES

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PROCESS EFFICIENCY

Also known as the flow time efficiency, the process efficiency is derived from the division of the total process time by the total lead time. It is the ratio of the process or value-adding time with the lead time.

PROCESS TIME

Also known as the process lead time, the processing time is the approximate amount of time required to handle a process in a production step. The processing time includes the order preparation time, the move time, the run time, the put-away time, and the inspection time. In simple processes, the processing time and the cycle time are interchanged.

PULL

The pull principle believes that customer demands drive production. The lack of customer demands means no production process.

PUSH

The push principle comes in play when the production process operates at full capacity, notwithstanding the customer demands.

QUALITY RATE

While the quality rate can be indicated by defect products sometimes, it is well known as the ratio of all acceptable products.

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VSM TERMINOLOGIES

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SETUP TIME

Check changeover time.

TAKT TIME

Takt time is the time needed to produce one unit of a product to meet customer requirements. The formula for calculating Takt Time is: T is equal to Ta divided by Td. Here, T stands for Takt Time. Ta stands for Net Available Time to work and Td stands for total customer demand.

UPSTREAM

Upstream is a term that encompasses all the operations that are included in the first parts of the production process or the value stream.

UPTIME

The uptime is a term used to denote that the needed equipment is available for use. It is the opposite of the term DOWNTIME.

VALUE-ADDING TIME

It is time the customer is willing to pay for. Usually indicated by the processing time, the value-adding time is the total amount of time spent on activities that add value to the product.

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VSM TERMINOLOGIES

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VALUE-ADDING OPERATIONS

The operations involved in the transformation of raw materials into finished products are collectively termed value-adding operations.

WAITING TIME

Waiting time is a form of waste when both the equipment and/or the personnel do nothing and remain idle. It can occur during inventory or while waiting for the arrival of raw materials. It can also occur during unscheduled downtime of a machine.

WASTE

Waste is a collective term that describes any process, action, or inaction that does not add benefit to the production process. Waste includes overproduction, over-processing, unnecessary transportation, inventory, motion, waiting periods, and defects.

WORK-IN-PROGRESS

The work in progress, which is abbreviated to WIP, follows the product in every step of the production process. It shows the current location of the product in the production chain. Used interchangeably, it is often used before the finished goods inventory.

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WHAT ARE THE DIFFERENT SYMBOLS USED IN THE VSM PROCESS?



VSM SYMBOLS

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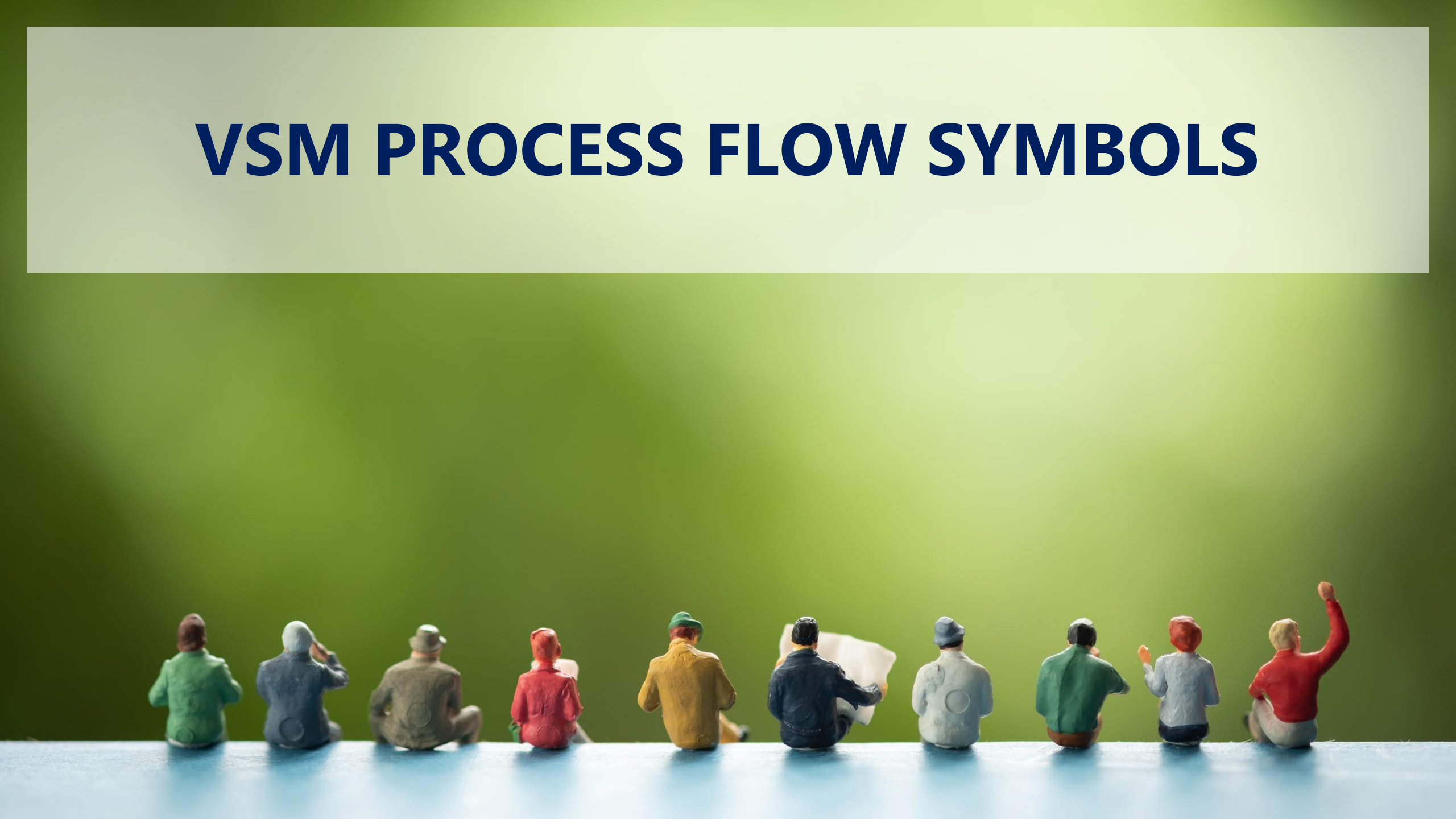
AT A HIGH-LEVEL, VSM SYMBOLS ARE CLASSIFIED INTO FOUR CATEGORIES:

- PROCESS FLOW
- MATERIAL FLOW
- INFORMATION FLOW
- GENERAL SYMBOLS



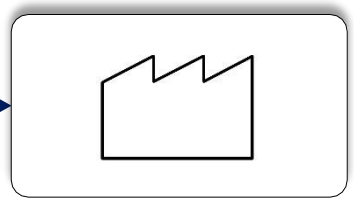
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VSM PROCESS FLOW SYMBOLS

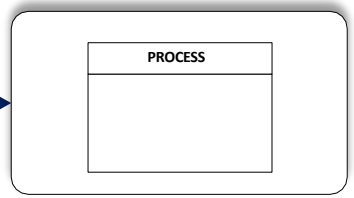


VSM PROCESS FLOW SYMBOLS

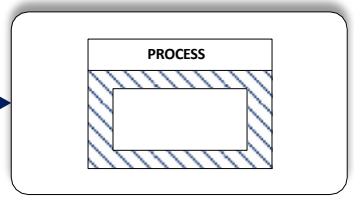
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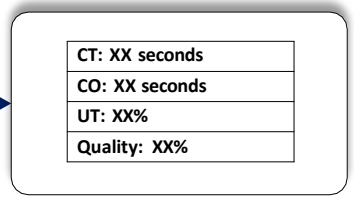
Customer/Supplier Icon: Indicative of the end point of the material, this icon represents the seller on the upper left and the customer on the upper right of the value stream map.



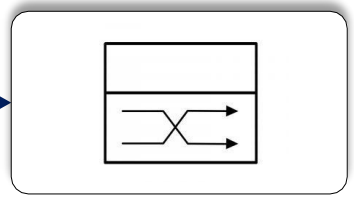
Dedicated Process Flow Icon: This icon represents a process, machine, operation or department through which material flows. It symbolizes departments with continuous and set internal flow.



Shared Process Icon: This icon represents a work center, process, machine, or operation that is used by other value stream families.



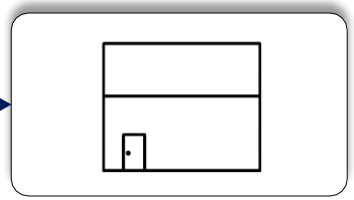
Data Box Icon: This icon is listed with other icons that have the pertinent information needed for the analysis and inspection of the system.



Cross-dock Icon: this icon represents the receiving dock of the facility that transfers the incoming materials to outbound shipping without inspection.

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VSM PROCESS FLOW SYMBOLS



Warehouse Icon: this icon represents the warehouse that stores the materials.

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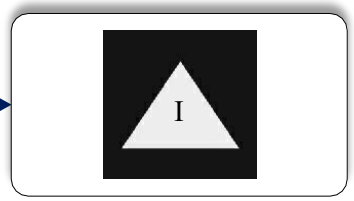
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VSM MATERIAL FLOW SYMBOLS

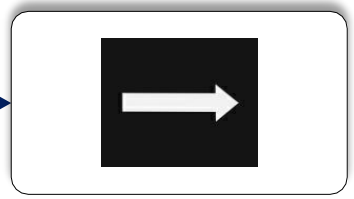


VSM MATERIAL FLOW SYMBOLS

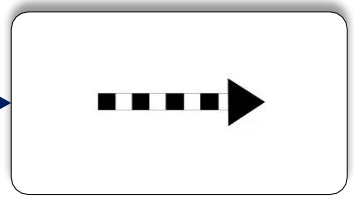
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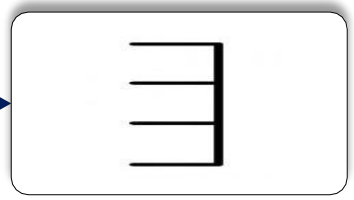
Inventory Icon: indicates the inventory in the middle of two processes.



Shipment Icon: this icon can either represent the movement of raw materials from the sellers to till it gets to the factory or the movement of the finished products from the moment it leaves the factory until it gets to the customer.



Push Arrow Icon: signifies the movement of materials from one process to the next.



Supermarket Icon: this icon represents a Kanban stock point or an inventory ‘supermarket’.

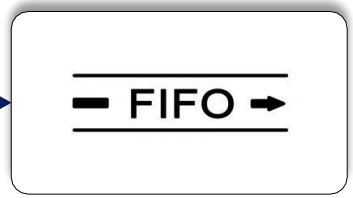


Material Pull Icon: this icon, which indicates physical removal, link the supermarkets with the downstream processes.

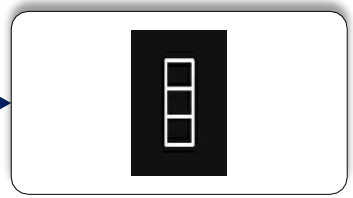
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VSM MATERIAL FLOW SYMBOLS

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FIFO Lane Icon: depicting the First-In-First-Out, this icon links processes that are attached by a FIFO system which restricts input.



Safety Stock Icon: the icon is indicative of the safety stock or the inventory edge available to protect against sudden problems such as sudden changes in customer orders, downtime or system failures.



External Shipment Icon: this portrays the movement of materials or products to sellers or customers using external transport.

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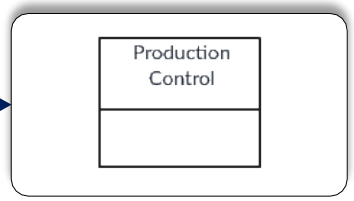
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VSM INFORMATION FLOW SYMBOLS

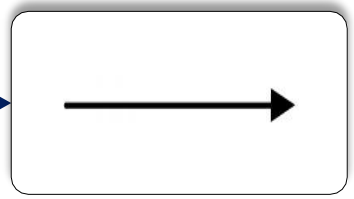


VSM INFORMATION FLOW SYMBOLS

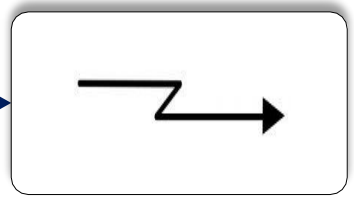
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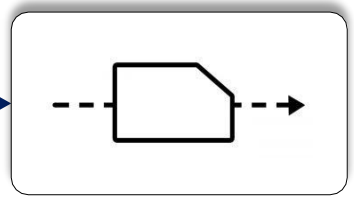
Production Control Icon: this icon can mean a person, department or operation. It signifies a key production arrangement.



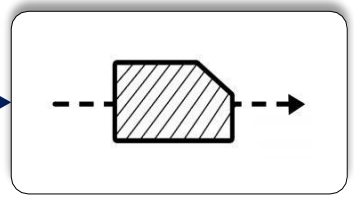
Manual Info Icon: the thin and straight arrow signifies the information flow from memos, conversations or reports.



Electronic Info Icon: this icon shows how information flows by the means of electronic devices through the internet, the intranets, the LANs (local area networks), the WANs (wide area network) or the EDIs (electronic data exchange). It also indicates the frequency of information exchange, type of media used and type of information exchanged.



Production Kanban Icon: this icon kick starts the supplying procedure to deliver parts to a downstream process. This icon signifies that the manufacturing of a set amount of parts.

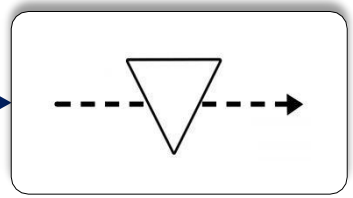


Withdrawal Kanban Icon: the presence of this card or icon informs the worker handling the materials to deliver parts from the supermarket to the receiving area.

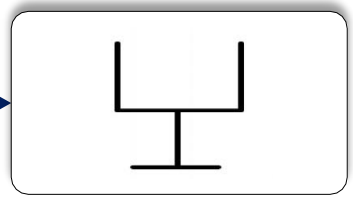
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VSM INFORMATION FLOW SYMBOLS

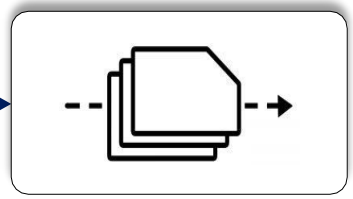
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Signal Kanban Icon: also known as the ‘one-per-batch’ Kanban, this icon is used when the inventory levels in the supermarket on-hand drops to a minimum point between two procedures.



Kanban Post Icon: used with the two-card system for the removal and construction Kanban, this icon reveals the site where the Kanban sign stay for pickup.



Kanban in Batches Icon: the presence of this icon reveals that several Kanban cards are going through the assembly process together.



Sequenced Pull Icon: the pull system that delivers instructions to the several subassembly processes to make a specified number and type of products without the supermarket is indicated with this icon.

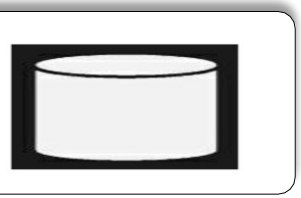


Load Leveling Icon: this icon is used to batch Kanbans. Kanbans are batched for the smoothening of the manufacture volume and mix over a particular period of time.

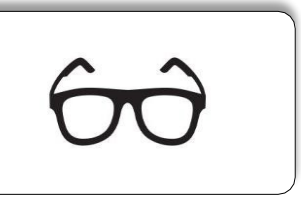
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VSM INFORMATION FLOW SYMBOLS

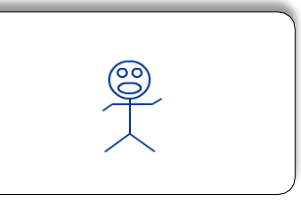
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MRP/ERP Icon: it is used to plan the usage of MRP/ERP or any other centralized system.



Go See Icon: this icon stands for the visual collection of information or information that is personally and visually obtained.



Verbal Information Icon: this icon stands for the flow of information orally and personally.

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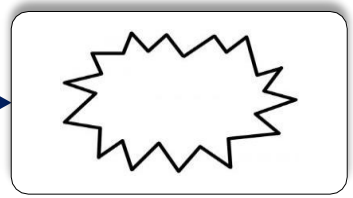
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VSM GENERAL SYMBOLS

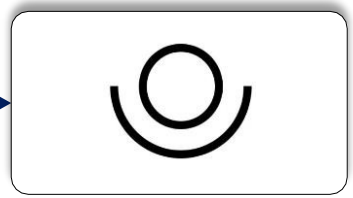


VSM GENERAL SYMBOLS

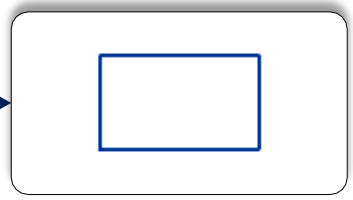
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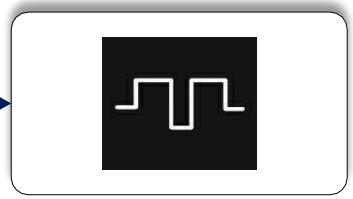
Kaizen Burst Icon: not only does this icon emphasize the need to upgrade, it also plans the future kaizen workshops at precise processes. These processes are essential in the implementation of the future state map of the value stream.



Operator Icon: portraying an operator, this icon shows the number of operators that are needed to work in a particular value stream map family at a specific workstation.



Other Icon: this icon is used to highlight any other information you deem important.



Timeline Icon: it illustrates the value added and non-value-added times. Those times are important for the calculation of the lead time and the total cycle time.

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ACTIVITY: TERMINOLOGIES AND SYMBOLS USED IN VSM



ACTIVITY

YOU ARE GIVEN A FEW QUESTIONS

BASED ON YOUR UNDERSTANDING, ANSWER THOSE QUESTIONS FROM THE GIVEN CHOICES



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<u>QUESTION 1:</u> THE NEXT SET OF OPERATIONS THAT RECEIVE THE FINISHED OR UNFINISHED WORK-IN-PROGRESS GOODS FROM THE CURRENT OPERATIONS IN FOCUS ARE TERMED AS?

1. **UPSTREAM PROCESS**
2. **DOWNSTREAM PROCESS**
3. **FIFO**
4. **INVENTORY**

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QUESTION 2: CYCLE TIME IS DEFINED AS:

1. **THE PROCESS WITH NO BUFFERS IN THE MIDDLE OF OPERATIONS.**

2. **THE APPROXIMATE AMOUNT OF TIME REQUIRED TO HANDLE A PROCESS IN A PRODUCTION STEP.**

3. **THE TIME NEEDED TO PRODUCE ONE UNIT OF A PRODUCT TO MEET CUSTOMER REQUIREMENTS.**

4. **THE TOTAL AMOUNT OF TIME REQUIRED TO COMPLETE ONE CYCLE OF TASK FROM START TO FINISH BY AN OPERATOR BEFORE HE/SHE REPEATS THIS OPERATION.**

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<u>QUESTION 3:</u> THIS PRINCIPLE COMES IN PLAY WHEN THE PRODUCTION PROCESS OPERATES AT FULL CAPACITY, NOTWITHSTANDING THE CUSTOMER DEMAND.

1. **PULL PRINCIPLE**
2. **UPSTREAM PRINCIPLE**
3. **PUSH PRINCIPLE**
4. **VALUE ADDING TIME**

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ACTIVITY

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<u>QUESTION 4:</u> WHICH OF THE FOLLOWING IS THE APPROXIMATE AMOUNT OF TIME REQUIRED TO HANDLE A PROCESS IN A PRODUCTION STEP?

1. **PROCESS TIME**
2. **CYCLE TIME**
3. **SETUP TIME**
4. **WAITING TIME**

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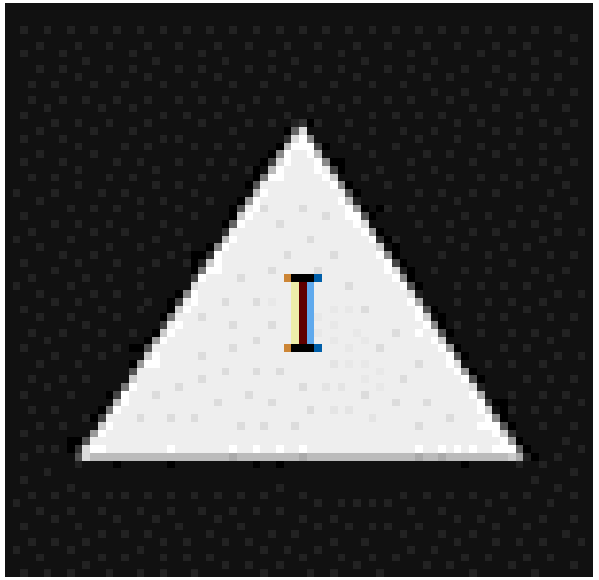
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2. MATERIAL PULL ICON: INDICATES PHYSICAL REMOVAL THAT LINK THE SUPERMARKETS WITH THE DOWNSTREAM PROCESSES.

3. SAFETY STOCK ICON: INDICATIVE OF THE SAFETY STOCK TO PROTECT AGAINST SUDDEN PROBLEMS.

4. PRODUCTION CONTROL ICON: THIS ICON CAN MEAN A PERSON, DEPARTMENT OR OPERATION. IT SIGNIFIES A KEY PRODUCTION ARRANGEMENT.



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ACTIVITY

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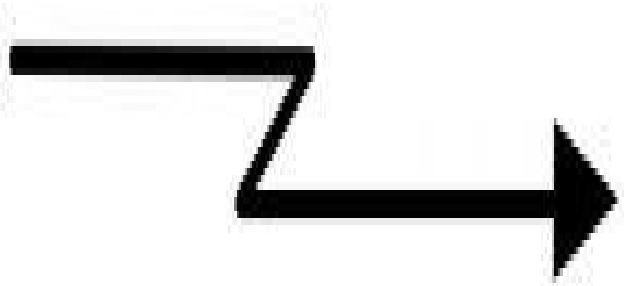
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3. IS INDICATIVE OF THE SAFETY STOCK TO PROTECT AGAINST SUDDEN PROBLEMS.

4. SHOWS HOW INFORMATION FLOWS BY THE MEANS OF ELECTRONIC DEVICES THROUGH THE INTERNET, THE INTRANETS, THE LANS, THE WANS, ETC.



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DRIVE SUCCESS TO EXCELLENCE

ACTIVITY SOLUTIONS: TERMINOLOGIES AND SYMBOLS USED IN VSM



ACTIVITY

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ACTIVITY SOLUTION

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ACTIVITY

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ACTIVITY SOLUTION

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ACTIVITY SOLUTION

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ACTIVITY SOLUTION

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ACTIVITY

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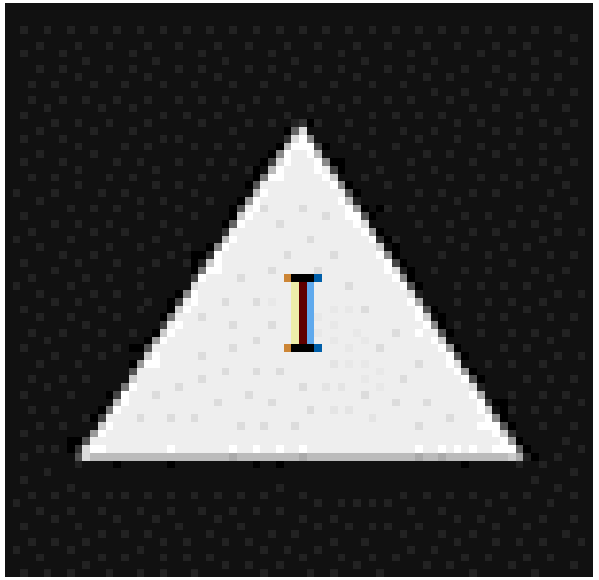
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ACTIVITY SOLUTION

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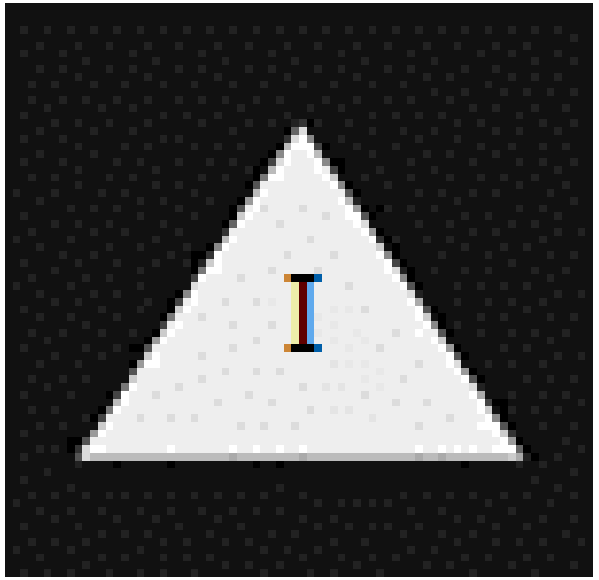
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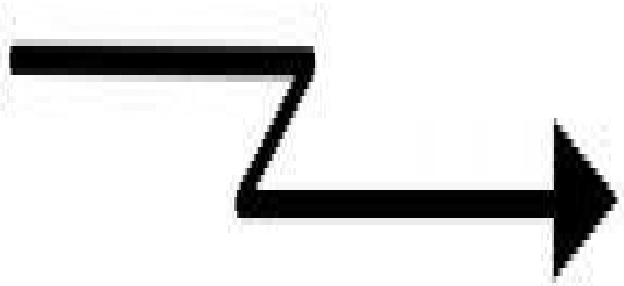
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ACTIVITY SOLUTION

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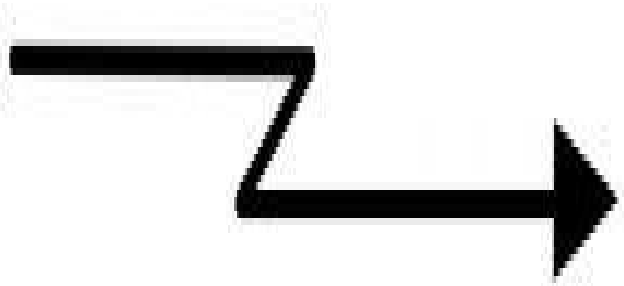
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SECTION CONCLUSION:
VSM TERMINOLOGIES & SYMBOLS



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SECTION CONCLUSION

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IN THIS SECTION, YOU LEARNED:

1. **WHAT ARE THE DIFFERENT TERMINOLOGIES USED IN VALUE STREAM MAPPING?**
2. **WHAT ARE THE DIFFERENT SYMBOLS USED IN THE VALUE STREAM MAPPING PROCESS?**
3. **WHAT ARE THE DIFFERENT INFORMATION FLOW SYMBOLS?**
4. **WHAT ARE THE MATERIAL FLOW SYMBOLS?**
5. **AND WHAT ARE THE GENERAL SYMBOLS USED IN VALUE STREAM MAPPING?**
6. **ACTIVITY AND ACTIVITY SOLUTION**

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SECTION INTRODUCTION:
STEPS TO CREATE CURRENT STATE VSM



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VALUE STREAM MAPPING (VSM) MASTERCLASS

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Introduction to Value Stream Map (VSM)

- Course Introduction
- What is Value Stream Mapping (VSM)?
- Why is Value Stream Mapping so popular?
- Section Conclusion

Terminologies and Symbols used in VSM0

- Section Introduction
- Terminologies used in VSM
- Symbols used in VSM Process
- Material Flow Symbols
- Information Flow Symbols
- General Symbols
- Activity: Terminologies and Symbols used in VSM
- Activity Solution: Terminologies and Symbols used in VSM
- Section Conclusion

Steps to Create the Current State VSM

- Section and Case Study Introduction
- Step 1. Pick a Product Family
- Step 2. Choose Your VSM Team
- Step 3: Figure out Customer Demand
- Step 4: Map and Design the Process Flow
- Step 5: Map the Material Flow
- Step 6: Map the Information Flow
- Case Study Recap: Current State VSM
- Perform Calculations | Analyze Current State VSM
- Calculate Total Time | Lead Time | Takt Time
- Calculate Process Time | Cycle Time
- Activity: Steps to Create Current State VSM
- Activity Solution: Steps to Create Current State VSM
- Section Conclusion

Examine the Current State VSM

- Section Introduction
- Meeting Customer Demand with existing No. of Operators
- Meeting Customer Demand with increased No. of Operators
- Analyze the Process Balance
- Watch for Waste
- Section Conclusion

Strong Foundations for Future State VSM

- Section Introduction
- Change to Continuous Flow
- CONWIP
- Kanban
- Readjusting Your Process
- Alter the Batch Sizes
- Smoothening out Production
- Heijunka | Heijunka Box
- Cut back Changeover Time
- Improve Quality
- Putting it all Together

Key Questions to ask for Future State VSM

- Section Introduction
- What is the Takt Time?
- Are you building to a warehouse or directly to shipping?
- Where should Continuous Flow Control the Production?
- Controlling area of upstream processes
- Where will you implement the Pull System?
- How do you intend to level the production mix?
- What process improvements do you intend to achieve?
- What is the Implementation Plan?
- Closure of the Case Study
- Section Conclusion

VSM Guidelines

- Section Introduction
- How to get the most of your VSM?
- What are the Pitfalls to avoid?
- What are the advantages of VSM?
- What are the disadvantages of VSM
- Section Conclusion

Course Conclusion

- Course Summary
- Practice Questions
- Become a Certified VSM Practitioner!

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INTRODUCTION TO THE VSM
CASE STUDY



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CASE STUDY INTRODUCTION

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DYNAMIC LTD. MANUFACTURES PAINTS

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CASE STUDY INTRODUCTION

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PAINTS ARE COATING MATERIALS THAT ARE APPLIED OVER MATERIAL SURFACES, WHICH CAN EITHER BE METALLIC OR NON-METALLIC

THEY ARE USED FOR EITHER PROTECTIVE OR DECORATIVE PURPOSES OR BOTH

GOOD PAINT POSSESSES SEVERAL QUALITIES, WHICH ARE COLOR, GOOD HIDING POWER, QUICK AND EASY APPLICATION, HIGH RESISTANCE TO ATMOSPHERIC CONDITIONS, AND GOOD PAINT MUST BE ECONOMICAL, AND OF LOW COST



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CASE STUDY INTRODUCTION

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FOR THIS CASE STUDY, YOU ARE THE PROCESS IMPROVEMENT EXPERT FOR DYNAMIC LTD.

BASED ON THE SENIOR MANAGEMENT DISCUSSIONS, YOU'VE DECIDED TO EXECUTE THE VALUE STREAM MAPPING EXERCISE FOR THE PRODUCT THAT IS CURRENTLY THEIR BEST SELLER.

YOUR COMPANY'S DIRECT COMPETITOR CAN PRODUCE THIS PRODUCT FASTER, AND THAT IS SOMETHING YOU WANT TO CURB.

INCREASING THE REVENUE AND REDUCING COST IS NOT THE ONLY FOCUS OF THIS EXERCISE, BUT YOU MUST ALSO REDUCE THE TIME TAKEN TO DELIVER THIS PRODUCT TO ITS CUSTOMERS.



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WHAT ARE THE STEPS TO CREATE A VALUE STREAM MAP?



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KEY POINTS

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THE BEST WAY TO CREATE A VSM IS BY HAND

VSM MADE BY A COMPUTER GIVES THE IDEA OF A FINISHED PRODUCT THAT LEAVES NO ROOM FOR SUGGESTIONS OR CORRECTIONS



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STEPS FOR VSM

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1. PICK A PRODUCT FAMILY
2. CHOOSE YOUR VSM TEAM
3. FIGURE OUT CUSTOMER DEMAND
4. MAP & DESIGN PROCESS FLOW
5. MAP THE MATERIAL FLOW
6. MAP THE INFORMATION FLOW
7. PERFORM CALCULATIONS

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STEP 1. PICK A PRODUCT FAMILY



STEP 1: PICK A PRODUCT FAMILY

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TIP # 1

- SELECT A LARGE-QUANTITY AND HIGH-REVENUE PRODUCT FAMILY.
- FOCUS ON ELIMINATING WASTE, REDUCING THE CYCLE TIME, AND IMPROVING THE PRODUCT'S QUALITY THAT IS YOUR BEST SELLER AND HELPS YOU GENERATE HIGH REVENUE.
- YOU CAN PICK THIS PRODUCT FAMILY DUE TO ITS EXPECTED GAINS.

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STEP 1: PICK A PRODUCT FAMILY

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TIP # 2

- **RECOGNIZE THE VALUE STREAM FROM CONSUMER'S PERSPECTIVE.**
- **ESSENTIAL FACTORS TO UNDERSTAND CONSUMER'S PERSPECTIVE:**
  - **SPEED OF DELIVERY**
  - **COST**
  - **SERVICE**
  - **CUSTOMER SATISFACTION**
  - **RISK**
  - **SAFETY**

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STEP 1: PICK A PRODUCT FAMILY

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TIP # 3

- **IF YOU HAVE SELECTED MULTIPLE PRODUCTS IN A PRODUCT FAMILY, ENSURE THAT THE SELECTED PRODUCTS GO THROUGH SIMILAR OPERATIONS OR PROCESSES.**

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STEP 1: PICK A PRODUCT FAMILY

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TIP # 4

- **THE ANALYSIS PERFORMED IS WITHIN AN INTRA-COMPANY FOCUS.**

- **THIS MEANS THAT YOUR VSM MUST FOCUS NOT JUST ON ONE DEPARTMENT BUT ALSO ON ALL THE DEPARTMENTS THAT COME IN THE PURVIEW OF BUILDING THAT PRODUCT.**

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STEP 1: PICK A PRODUCT FAMILY

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TIP # 5

- KNOW ALL THE PRODUCTS IN THAT PRODUCT FAMILY.
- IF YOU ARE A MANAGER HANDLING MULTIPLE PROCESSES, KNOWING ALL THE PRODUCTS MAY BE COMPLICATED.
- THIS IS WHERE YOU SHOULD LEVERAGE YOUR TEAM MEMBERS.
- YOUR TEAM NEEDS TO KNOW ALL THE PRODUCTS IN THAT PRODUCT.

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STEP 1: PICK A PRODUCT FAMILY

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SUMMARY

- BEGIN BY SELECTING A LARGE-QUANTITY AND HIGH-REVENUE PRODUCT FAMILY
- RECOGNIZE THE VALUE STREAM FROM THE CUSTOMER PERSPECTIVE
- SELECTED PRODUCTS SHOULD GO THROUGH SIMILAR PROCESSES
- THE ANALYSIS PERFORMED IS WITHIN AN INTRA-COMPANY FOCUS
- KNOW ALL THE PRODUCTS IN THAT PRODUCT FAMILY

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STEP 1: PICK A PRODUCT FAMILY

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FOR OUR DYNAMIC LTD. CASE STUDY:

- **WE HAVE SELECTED A SINGLE PRODUCT.**
- **HENCE, SELECTING THE CORRECT PRODUCT FAMILY IS NOT A REQUIREMENT.**



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STEP 2. CHOOSE YOUR VSM TEAM



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STEP 2. CHOOSE YOUR VSM TEAM

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CREATING A VALUE STREAM MAP IS NOT A ONE-PERSON ACTIVITY. IT IS A TEAM EFFORT. AND YOU ARE IN CHARGE OF BUILDING YOUR TEAM.



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STEP 2: CHOOSE YOUR VSM TEAM

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GUIDELINE # 1

- **GET THE RIGHT SUBJECT MATTER EXPERTS (SME'S) FROM ALL DEPARTMENTS.**

- **HELPS IN UNDERSTANDING THE PERSPECTIVE OF UPSTREAM, DOWNSTREAM AS WELL AS PARALLEL PROCESSES.**



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STEP 2: CHOOSE YOUR VSM TEAM

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GUIDELINE # 2

- **PICK MEMBERS THAT ARE QUITE FAMILIAR AND WELL ACQUAINTED WITH THE PRODUCT.**

- **IT WOULD BE BEST IF YOU HAD SME'S WHO HAVE NECESSARY KNOWLEDGE OF THE PRODUCT.**



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STEP 2: CHOOSE YOUR VSM TEAM

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GUIDELINE # 3

- **MAKE SURE THE SELECTED MEMBERS HAVE A THOROUGH UNDERSTANDING OF VALUE STREAM MAPPING AND ITS USE.**

- **YOUR TEAM SHOULD KNOW:**

WHAT IS VALUE STREAM MAPPING ALL ABOUT?	**WHY IS THE ORGANIZATION DOING THIS EXERCISE?**	**WHAT IS THEIR ROLE IN CREATING THE VALUE STREAM MAP?**	**WHAT ARE THE STEPS INVOLVED?**



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STEP 2: CHOOSE YOUR VSM TEAM

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GUIDELINE # 4: ASSIGN A TEAM LEADER

- **THE TEAM LEADER WILL BE RESPONSIBLE FOR:**

PROVIDING INPUTS TO SELECT THE RIGHT SUBJECT MATTER EXPERTS	COORDINATING WITH TEAM MEMBERS OF DIFFERENT DEPARTMENTS TO FIX THE VENUE OF VALUE STREAM MAP BRAINSTORMING	SET UP THE DISCUSSION ROOM WITH NECESSARY STATIONERY AND MATERIAL
COLLECT DATA	HELP IN CONSTRUCTING THE VALUE STREAM MAP ON A COMPUTER	IMPLEMENTATION PLANNING AND EXECUTION



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STEP 2: CHOOSE YOUR VSM TEAM

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GUIDELINE SUMMARY:

GUIDELINE # 1

CREATE A CROSS-FUNCTIONAL TEAM



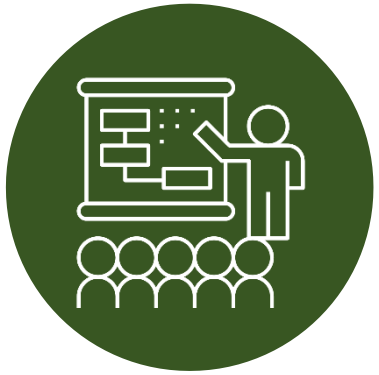
GUIDELINE # 2

PICK MEMBERS THAT ARE FAMILIAR AND ACQUAINTED WITH THE PRODUCT



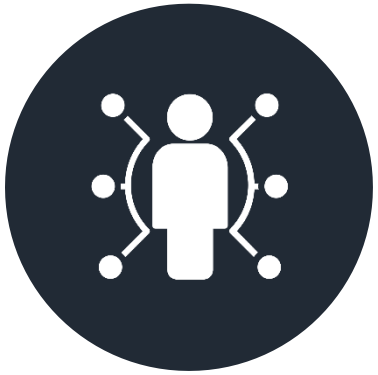
GUIDELINE # 3

MAKE SURE THE SELECTED MEMBERS HAVE A THOROUGH UNDERSTANDING OF VSM AND ITS USE



GUIDELINE # 4

ASSIGN A TEAM LEADER



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STEP 2: CHOOSE YOUR VSM TEAM

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FOR OUR DYNAMIC LTD. CASE STUDY:

- **WE ARE ASSUMING THAT THIS STEP IS COVERED, AND YOU ARE READY FOR THE NEXT STEPS.**



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DRIVE SUCCESS TO EXCELLENCE

STEP 3. FIGURE OUT CUSTOMER DEMAND



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STEP 3. FIGURE OUT CUSTOMER DEMAND

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SOME ORGANIZATIONS ROBUSTLY TRACK THEIR CUSTOMER DEMAND, WHILE OTHERS DO NOT DO A GREAT JOB.



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STEP 3. FIGURE OUT CUSTOMER DEMAND

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01



CHECK THE PRODUCTION HISTORY

REVIEW THE SALES RECORDS



02

03



HAVE WEEKLY AND MONTHLY FORECASTS

PREPARE FOR ALL PARTS AND NOT JUST A FEW



04

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STEP 3. FIGURE OUT CUSTOMER DEMAND

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FOR OUR DYNAMIC LTD. CASE STUDY:

- **CUSTOMER DEMAND IS 300 PAINT CANS PER DAY.**

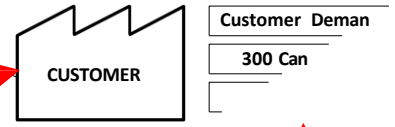


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YOUR CUSTOMER WILL BE LISTED ON THE TOP RIGHT CORNER OF YOUR VALUE STREAM MAP



YOU WILL THEN LIST THE CUSTOMER DEMAND TO THE RIGHT OF THE CUSTOMER ICON.

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STEP 4. MAP AND DESIGN THE PROCESS FLOW

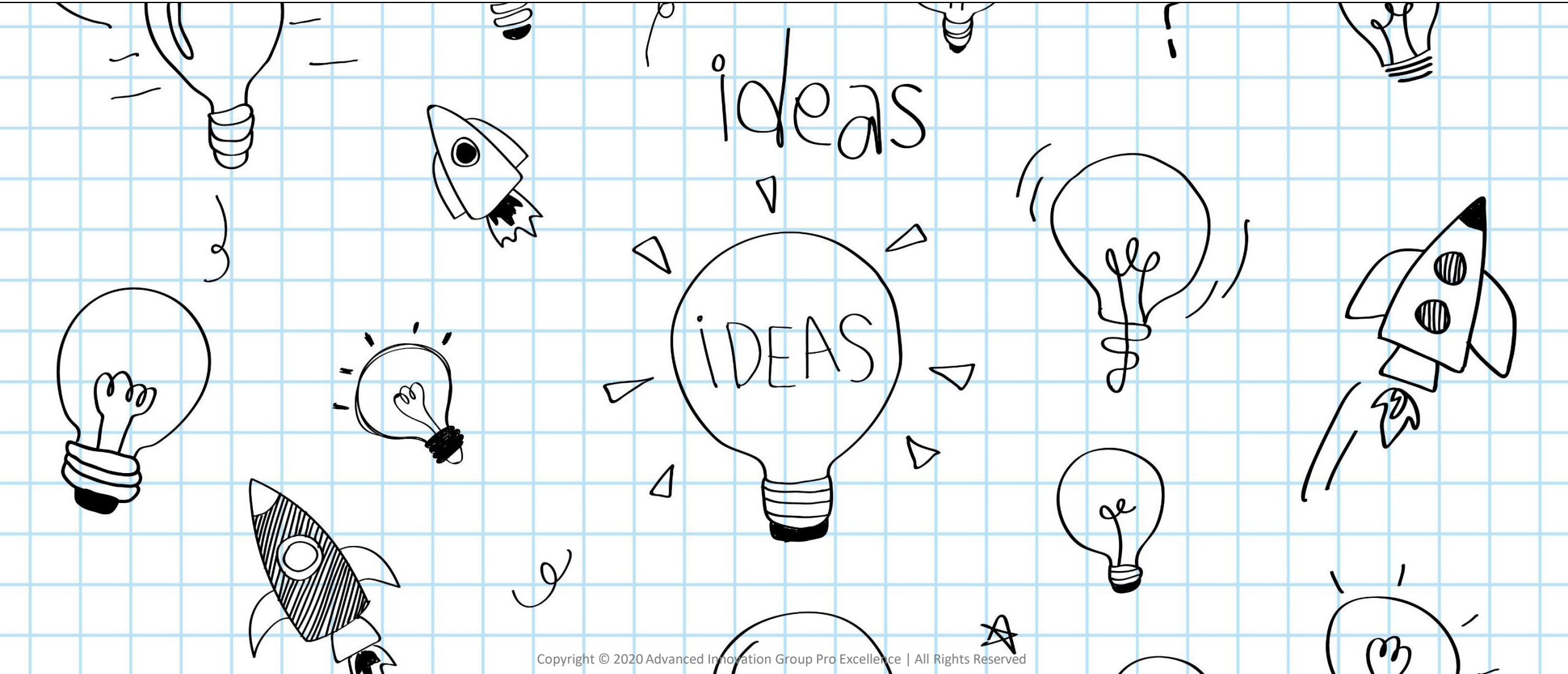


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STEP 4. MAP AND DESIGN THE PROCESS FLOW

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THIS STEP IS WHERE YOU START SKETCHING YOUR VALUE STREAM MAP.



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STEP 4: MAP AND DESIGN THE PROCESS FLOW

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TIP # 1: DEFINE THE START AND END POINTS OF THE PROCESS

- IDENTIFYING THE START AND ENDPOINTS AND DEFINING THEM IS CRITICAL BECAUSE UNCLEAR BOUNDARIES CAN LEAD TO SOMETHING CALLED "SCOPE CREEP."

- SCOPE CREEP IS A TERM USED WHEN YOUR PROJECT GOES OVER AND BEYOND THE INITIALLY DECIDED SCOPE.

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STEP 4: MAP AND DESIGN THE PROCESS FLOW

AIGPE

TIP # 2: GET TO THE FLOOR TO MAP AND DESIGN THE PROCESS FLOW

- **GET YOUR TEAM ON THE SHOP FLOOR, IDENTIFY ALL THE STEPS EXECUTED BY THE OPERATIONAL AND FUNCTIONAL TEAMS TO BRING THE PRODUCT TO LIFE, AND MAP THEM.**

- **GOOD PRACTICE TO:**

START MAPPING AT THE SHIPPING END AND THEN TO MOVE UPSTREAM FROM THERE

MAP THE PROCESS ON A PIECE OF PAPER RATHER THAN A COMPUTER

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STEP 4: MAP AND DESIGN THE PROCESS FLOW

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FOR OUR DYNAMIC LTD. CASE STUDY:

- **LET'S MAP THE PROCESS FROM THE SHIPPING END**



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THE RAW MATERIALS ARE KEPT IN THE WAREHOUSE AS THEY ARE RECEIVED FROM THE SUPPLIERS

BEFORE SHIPPING THE PAINT CANS, THEY ARE KEPT IN A WAREHOUSE



THERE ARE FOUR KEY PROCESSES IN THE PRODUCTION PROCESS. THE LAST ONE IS PAINT CANNING. THE ONE BEFORE IS PASTE THINNING. THE ONE BEFORE THAT IS PIGMENT DIFFUSION, AND THE FIRST ONE IS PASTE MAKING.

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DRIVE SUCCESS TO EXCELLENCE

STEP 5. MAP THE MATERIAL FLOW

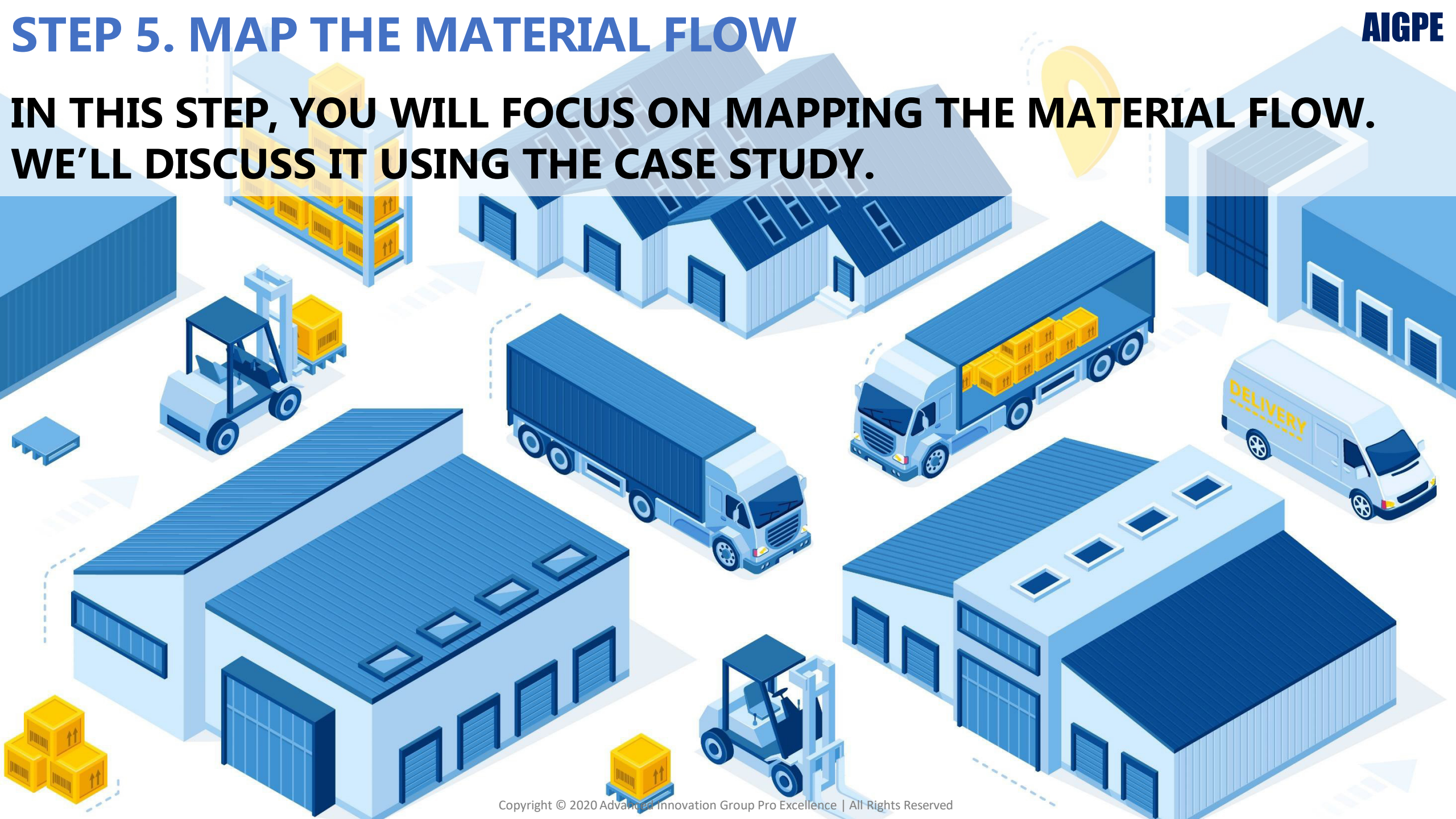


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STEP 5. MAP THE MATERIAL FLOW

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IN THIS STEP, YOU WILL FOCUS ON MAPPING THE MATERIAL FLOW. WE'LL DISCUSS IT USING THE CASE STUDY.



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STEP 5. MAP THE MATERIAL FLOW

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FOR OUR DYNAMIC LTD. CASE STUDY:

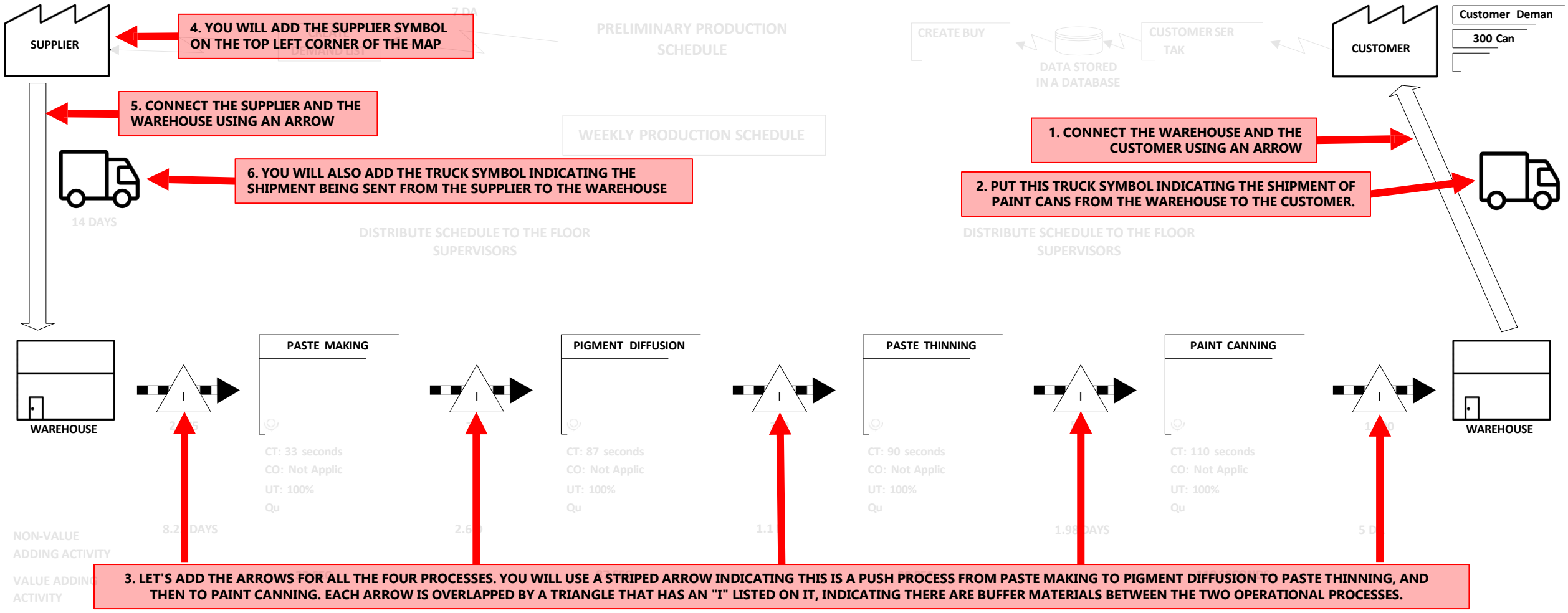
- **LET'S MAP THE MATERIAL FLOW**
- **ALTHOUGH THE MATERIAL FLOW IS FROM LEFT TO RIGHT, YOU CAN TRACK THE PRODUCTION FROM RIGHT TO LEFT, I.E., FROM THE CUSTOMER TO THE SUPPLIER.**



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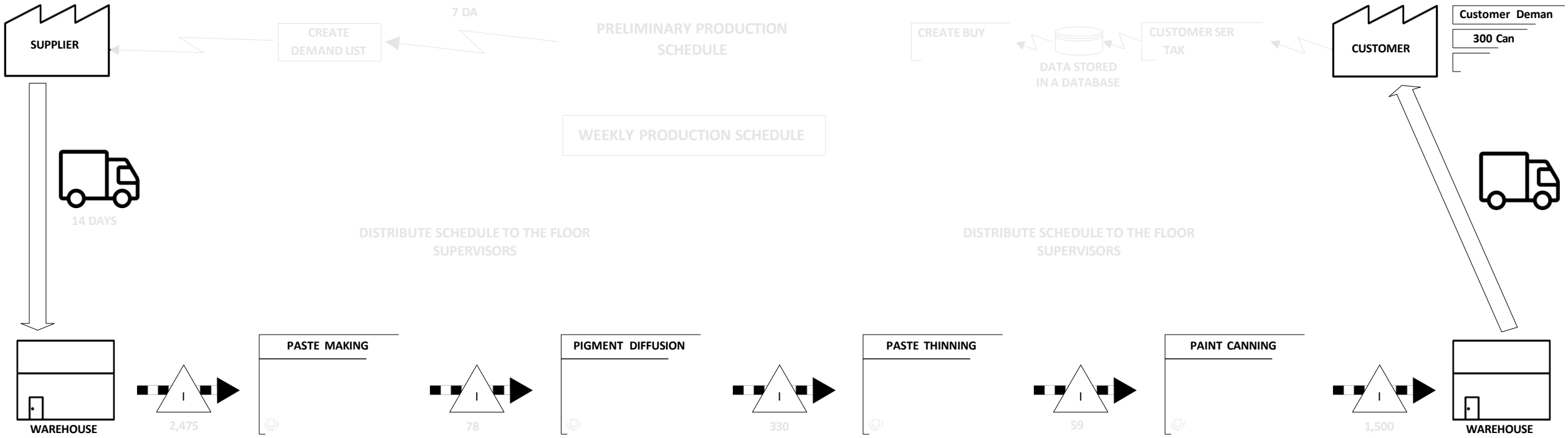
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GLIMPSE OF THE CURRENT STATE VSM AFTER STEP 4. MAPPING THE MATERIAL FLOW

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STEP 5. MAP THE MATERIAL FLOW (CONTINUED)



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STEP 5: MAP THE MATERIAL FLOW (CONTINUED)

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FOR OUR DYNAMIC LTD. CASE STUDY:

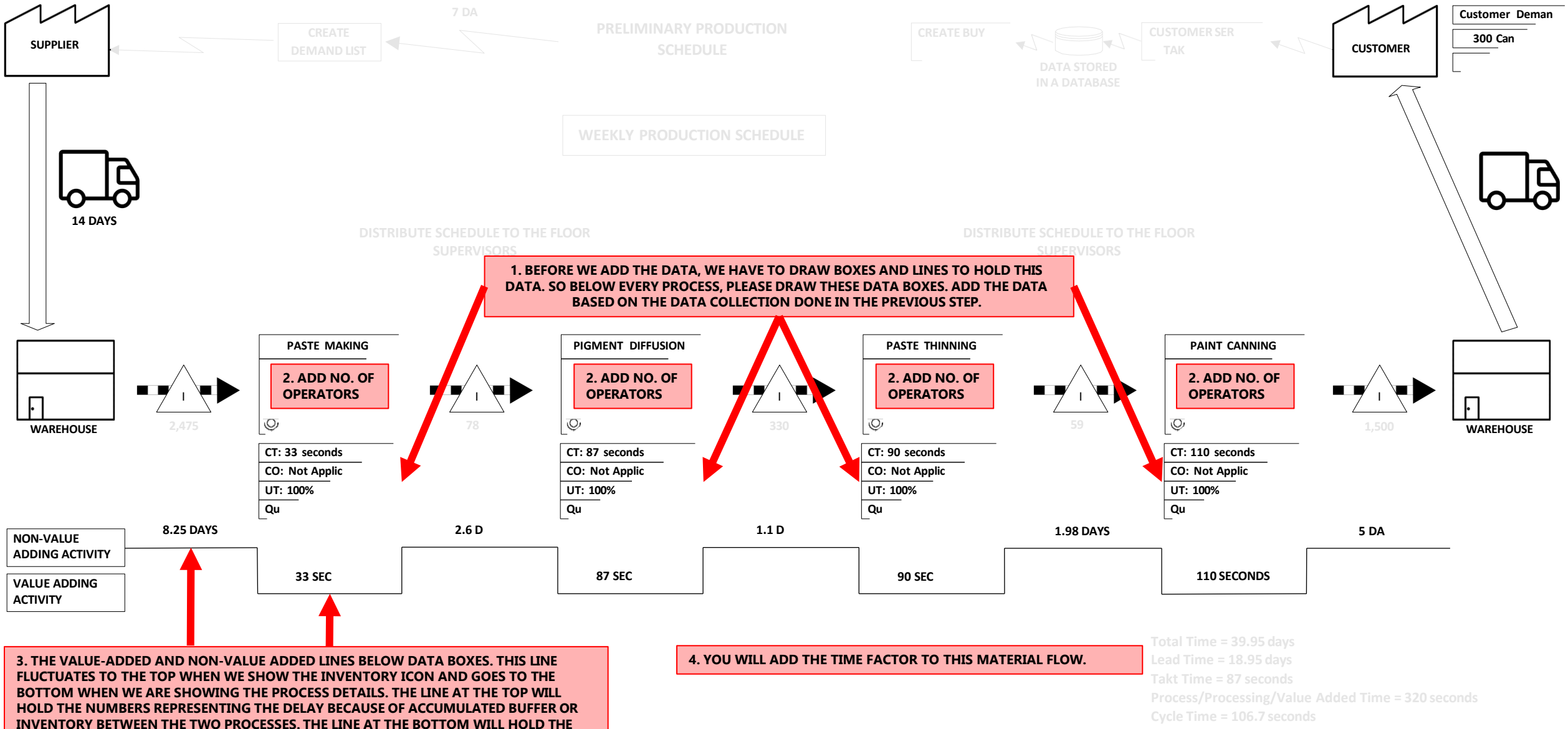
- YOU WILL NOW ADD THE COLLECTED DATA TO EACH OF THESE STEPS IN THE MATERIAL FLOW.
- AS WE ARE DEALING WITH A SINGLE PAINT PRODUCT, THERE IS NO CHANGEOVER TIME NEEDED TO BE ADDED FOR THESE PROCESSES.
- SECOND, THE UPTIME IS 100% IN ALL OPERATIONAL PROCESSES AS THE MACHINERY IN THIS MANUFACTURING PLANT IS STILL NEW.



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DYNAMIC LTD. CASE STUDY

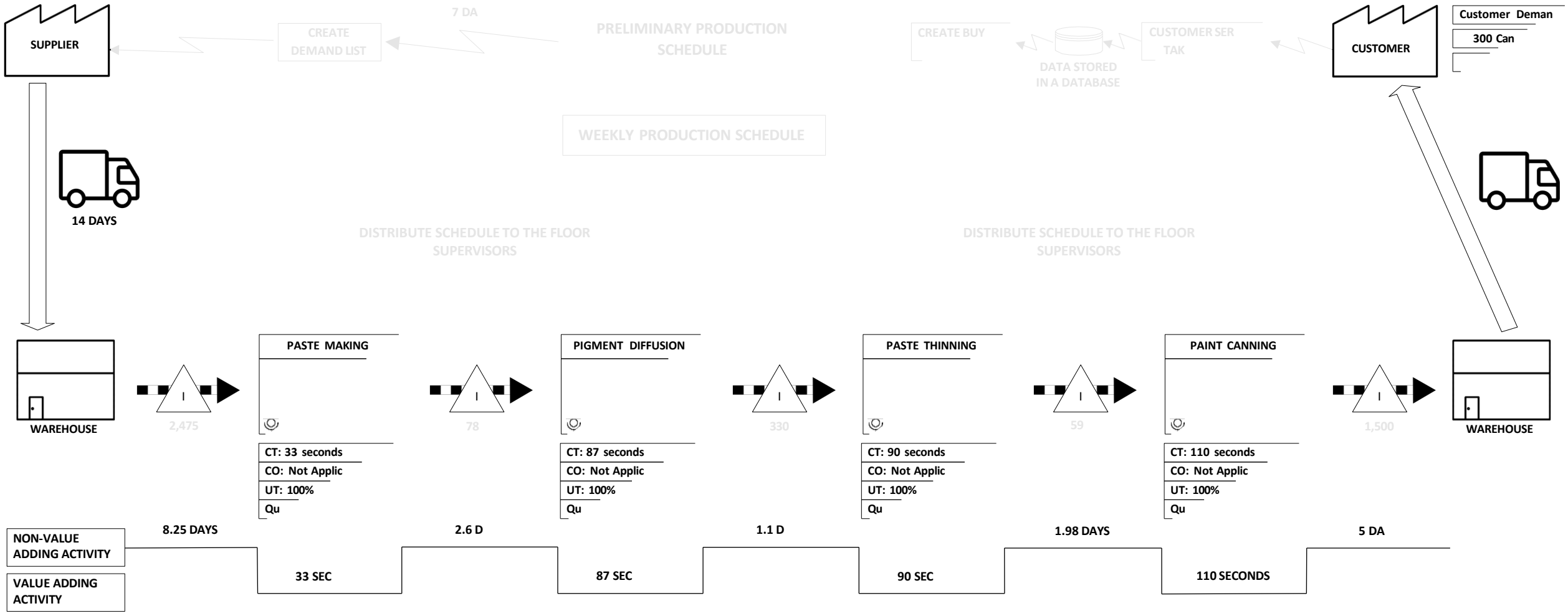
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GLIMPSE OF THE CURRENT STATE VSM AFTER STEP 4. MAPPING THE MATERIAL FLOW & ADDING THE COLLECTED DATA TO THIS FLOW

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STEP 6. MAP THE INFORMATION FLOW



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STEP 6: MAP THE INFORMATION FLOW

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FOR OUR DYNAMIC LTD. CASE STUDY:

RECALL THAT:

- **THE MATERIAL FLOW GOES FROM LEFT TO RIGHT ON YOUR VALUE STREAM MAP.**

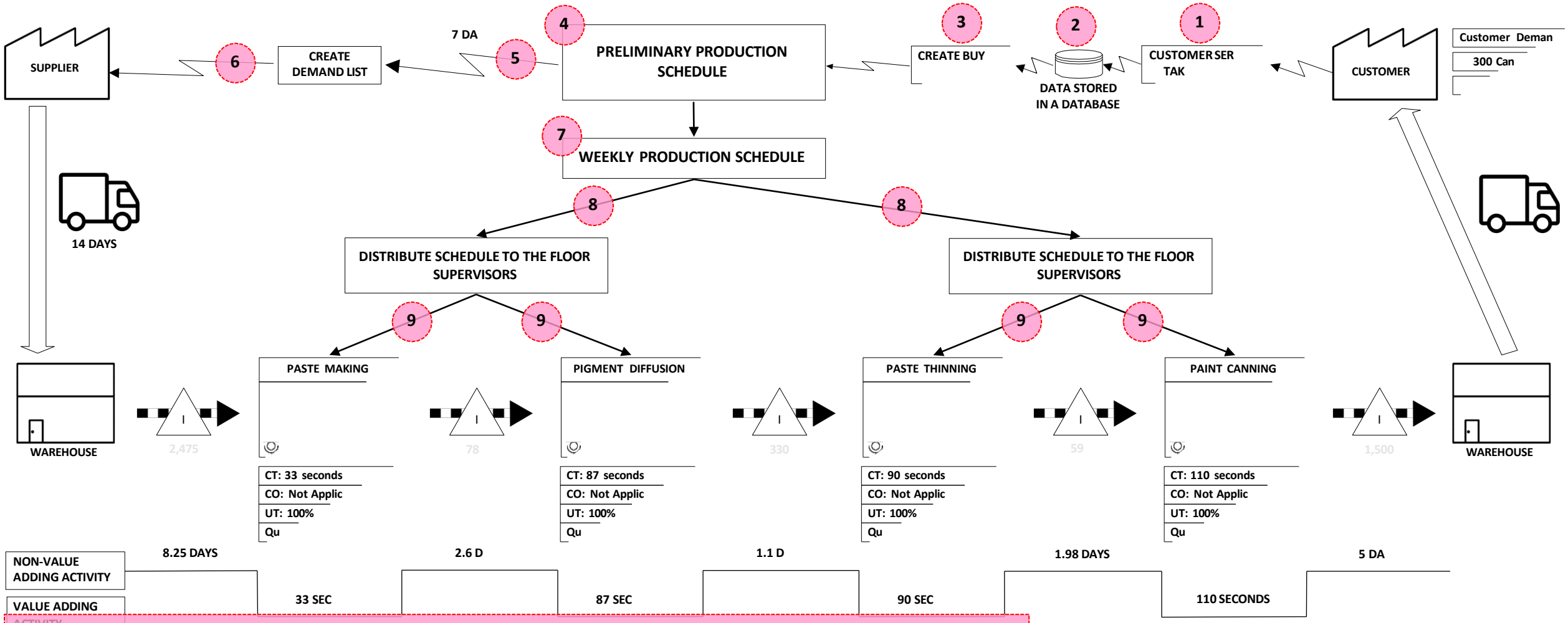
- **THE INFORMATION FLOW FROM RIGHT TO LEFT.**



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DYNAMIC LTD. CASE STUDY

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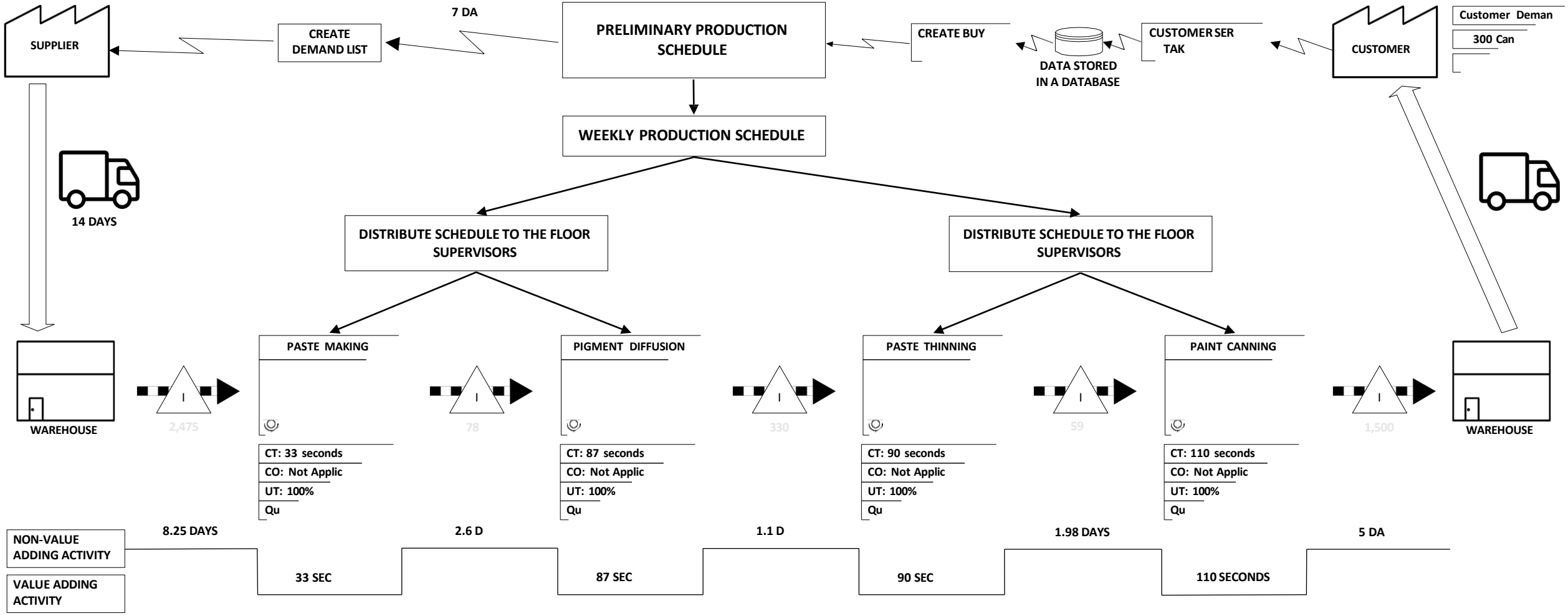
1. THE CUSTOMER SERVICE TEAM TAKES THE CALL AND ENTERS THE CUSTOMER ORDER DETAILS IN A SYSTEM.
2. THE DATA IS STORED IN A DATA STORAGE AND RETRIEVAL SERVER.
3. THE CUSTOMER SERVICE TEAM CREATES A BUYING LIST OF RAW MATERIALS BASED ON THE ORDERS AND AVAILABILITY OF THE EXISTING INVENTORY.
4. BASED ON THE CUSTOMER ORDERS AND THE BUYING LIST, A PRELIMINARY PRODUCTION SCHEDULE IS CREATED.
5. THE PROCUREMENT TEAM WAITS FOR ANOTHER WEEK AND CREATES A DEMAND LIST.
6. THE DEMAND LIST IS SENT TO THE SUPPLIER FOR PROCURING THE RAW MATERIALS.
7. AS THE RAW MATERIALS ARE RECEIVED FROM THE SUPPLIER, THE PRODUCTION MANAGER CREATES A WEEKLY PRODUCTION SCHEDULE.
8. THE WEEKLY PRODUCTION SCHEDULE IS DISTRIBUTED TO THE FLOOR SUPERVISORS.
9. THE FLOOR SUPERVISORS PASS THE WEEKLY PRODUCTION SCHEDULE TO EACH OF THE OPERATIONAL TEAMS.

Total Time = 39.95 days
Lead Time = 18.95 days
Takt Time = 87 seconds
Process/Processing/Value Added Time = 320 seconds
Cycle Time = 106.7 seconds

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DYNAMIC LTD. CASE STUDY

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GLIMPSE OF THE CURRENT STATE VSM AFTER STEP 5. MAPPING THE INFORMATION FLOW

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RECAP OF THE CASE STUDY: CURRENT STATE VALUE STREAM MAP



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RECAP OF THE CASE STUDY

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FOR OUR DYNAMIC LTD. CASE STUDY:

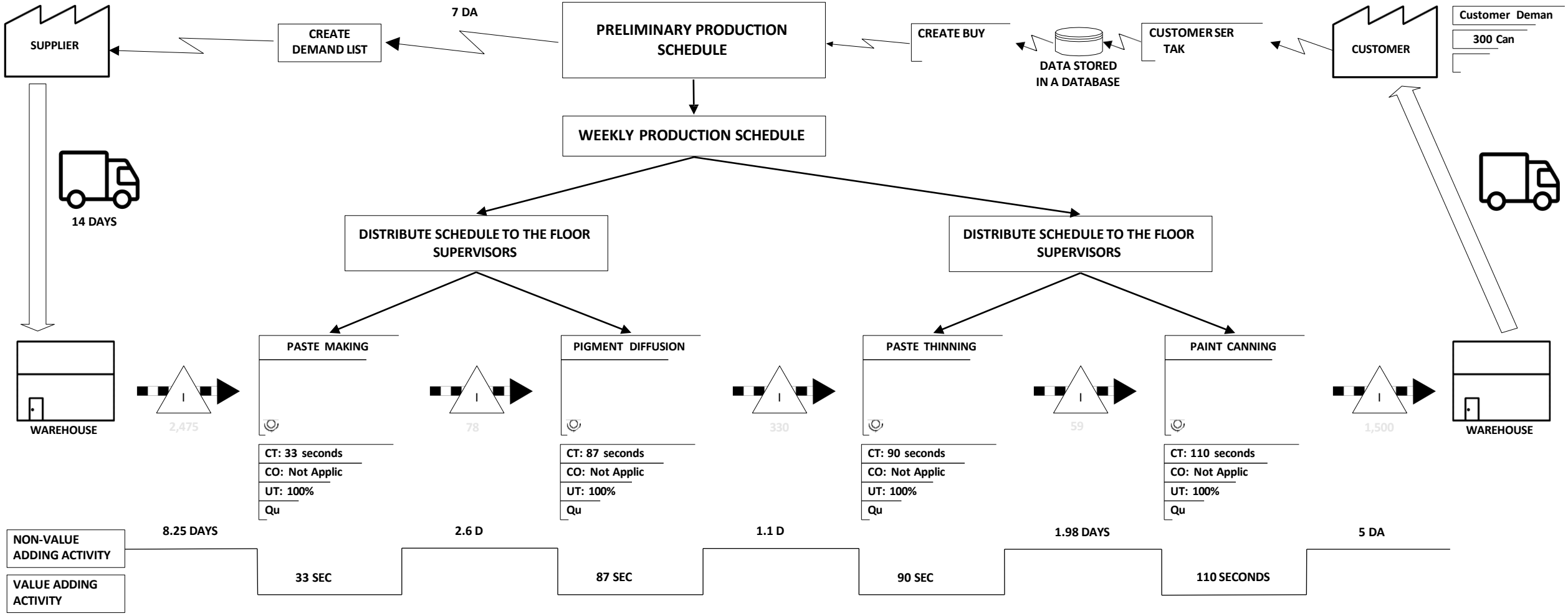
BEFORE WE MOVE ON TO THIS NEXT STEP, IT WOULD BE GREAT TO LOOK AT THE COMPLETE CURRENT STATE VSM



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GLIMPSE OF THE CURRENT STATE VSM AFTER STEP 5. MAPPING THE INFORMATION FLOW

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PERFORM CALCULATIONS AND ANALYZE THE CURRENT STATE VSM



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PERFORM CALCULATIONS

THIS STEP INVOLVES THE CALCULATION OF THE FOLLOWING:

- **LEAD TIME OF THE PROCESS**
- **STORAGE TIME OF THE PROCESS**
- **TAKT TIME**
- **PROCESS/PROCESSING/ VALUE ADDED TIME**
- **CYCLE TIME OF THE PROCESS**



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CALCULATE LEAD TIME



WHAT IS LEAD TIME?

LEAD TIME IS ALL THE TIME TAKEN COLLECTIVELY FROM ORDER TO SHIPPING.



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PERFORM CALCULATIONS

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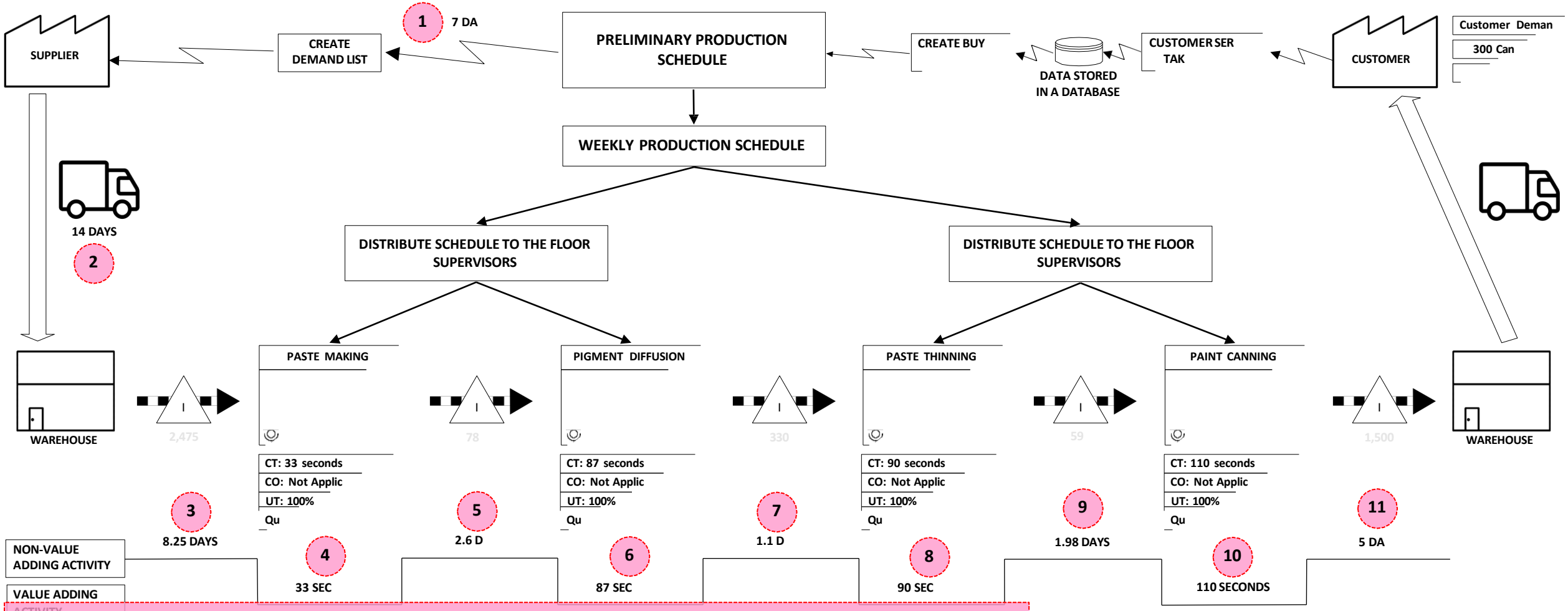
LET'S DO THE CALCULATION OF THE LEAD TIME:



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1. THE PROCUREMENT TEAM WAITS FOR ONE MORE WEEK TO CREATE THE DEMAND LIST – SO THAT'S 7 ADDITIONAL DAYS
2. SUPPLIER TAKES 14 DAYS TO DELIVER GOODS TO THE WAREHOUSE
3. THE AGEING OF RAW MATERIALS AT THE WAREHOUSE (BEFORE THE PASTE MAKING PROCESS) IS 8.25 DAYS
4. THE PASTE MAKING PROCESS TAKES 33 SECONDS
5. THE AGEING OF THE WORK-IN-PROGRESS GOODS BETWEEN PASTE MAKING AND PIGMENT DIFFUSION PROCESS IS 2.6 DAYS
6. THE PIGMENT DIFFUSION PROCESS TAKES 87 SECONDS
7. THE AGEING OF THE WORK-IN-PROGRESS GOODS BETWEEN PIGMENT DIFFUSION AND PASTE THINNING IS 1.1 DAYS
8. THE PASTE THINNING PROCESS TAKES 90 SECONDS
9. THE AGEING OF WORK-IN-PROGRESS GOODS BETWEEN PASTE THINNING AND PAINT CANNING IS 1.98 DAYS
10. THE PAINT CANNING PROCESS TAKES 110 SECONDS
11. AND THE FINISHED GOODS ARE KEPT IN THE WAREHOUSE FOR 5 DAYS BEFORE BEING SHIPPED TO THE CUSTOMER

Lead Time = 39.93 days
Storage Time = 18.95 days
Takt Time = 87 seconds
Process/Processing/Value Added Time = 320 seconds
Cycle Time = 106.7 seconds

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LEAD TIME OF THE PROCESS

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		# OF DAYS	SECONDS
1.	THE PROCUREMENT TEAM WAITS FOR ONE MORE WEEK TO CREATE THE DEMAND LIST – SO THAT'S 7 ADDITIONAL DAYS	7	
2.	SUPPLIER TAKES 14 DAYS TO DELIVER GOODS TO THE WAREHOUSE	14	
3.	THE AGEING OF RAW MATERIALS AT THE WAREHOUSE (BEFORE THE PASTE MAKING PROCESS) IS 8.25 DAYS	8.25	
4.	THE PASTE MAKING PROCESS TAKES 33 SECONDS		33
5.	THE AGEING OF THE WORK-IN-PROGRESS GOODS BETWEEN PASTE MAKING AND PIGMENT DIFFUSION PROCESS IS 2.6 DAYS	2.6	
6.	THE PIGMENT DIFFUSION PROCESS TAKES 87 SECONDS		87
7.	THE AGEING OF THE WORK-IN-PROGRESS GOODS BETWEEN PIGMENT DIFFUSION AND PASTE THINNING IS 1.1 DAYS	1.1	
8.	THE PASTE THINNING PROCESS TAKES 90 SECONDS		90
9.	THE AGEING OF WORK-IN-PROGRESS GOODS BETWEEN PASTE THINNING AND PAINT CANNING IS 1.98 DAYS	1.98	
10.	THE PAINT CANNING PROCESS TAKES 110 SECONDS		110
11.	AND THE FINISHED GOODS ARE KEPT IN THE WAREHOUSE FOR 5 DAYS BEFORE BEING SHIPPED TO THE CUSTOMER	5	
		39.93 DAYS	320 SECONDS

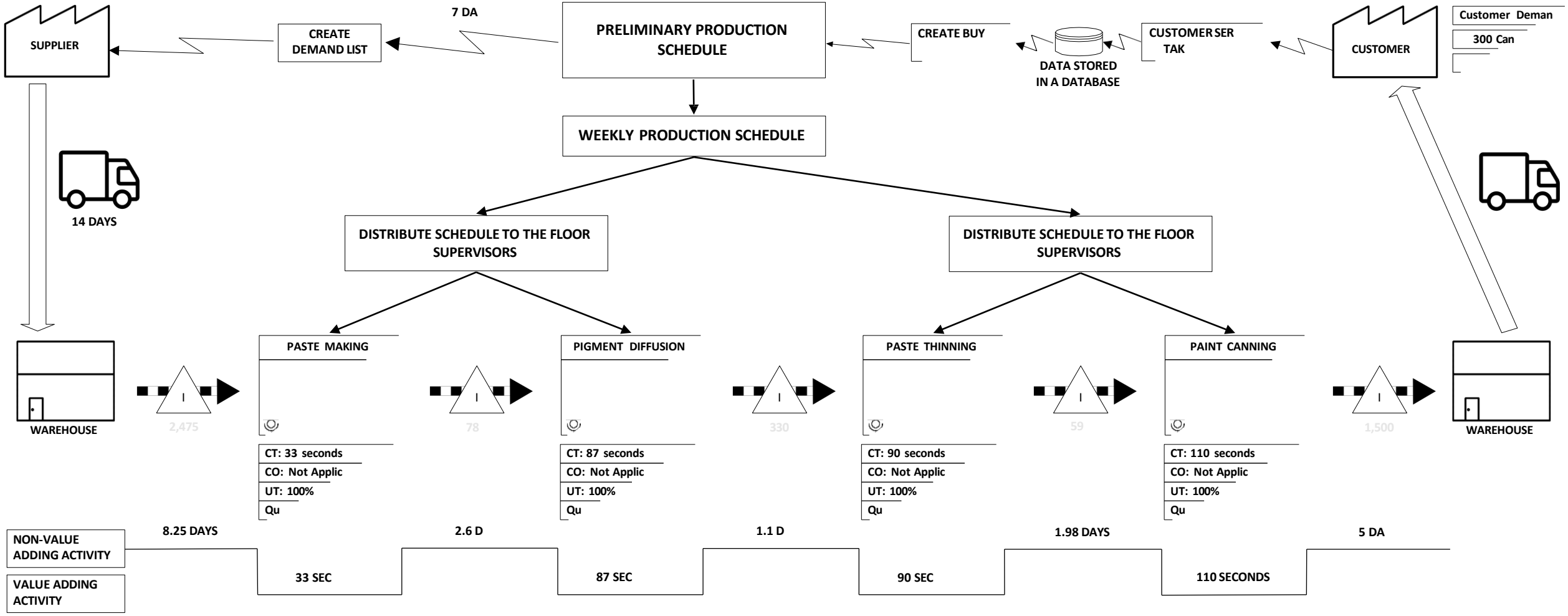
SINCE 320 SECONDS IS INFINITESIMALLY SMALLER THAN 39.95 DAYS, WE CAN IGNORE THIS NUMBER.

SO, THE LEAD TIME IS 39.93 DAYS FROM ORDER TO DELIVERY.

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DYNAMIC LTD. CASE STUDY

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ON THE BOTTOM RIGHT CORNER OF YOUR CURRENT STATE VALUE STREAM MAP, YOU CAN CREATE A FEW BOXES AND NAME THEM – LEAD TIME, TAKT TIME, PROCESS TIME AS WELL AS THE CYCLE TIME. SINCE YOU HAVE CALCULATED THE LEAD TIME, LET'S UPDATE THAT NUMBER IN THE FIRST BOX.

Lead Time = 39.93 days
Storage Time =
Takt Time =
Process/Processing/Value Added Time =
Cycle Time =

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HOW TO CALCULATE STORAGE TIME?



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RECALL: STORAGE TIME

STORAGE TIME CAN BE MINUTES, HOURS, OR DAYS.

IT IS THE AMOUNT OF TIME THAT IS REQUIRED FOR THE RAW MATERIAL THAT MAY BE SITTING IN THE WAREHOUSE TO GO THROUGH THE DIFFERENT STAGES OF OPERATIONS ALL THE WAY UNTIL THE STORAGE OF FINISHED PRODUCTS BEFORE GETTING SHIPPED TO THE CUSTOMER.



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CALCULATE STORAGE TIME

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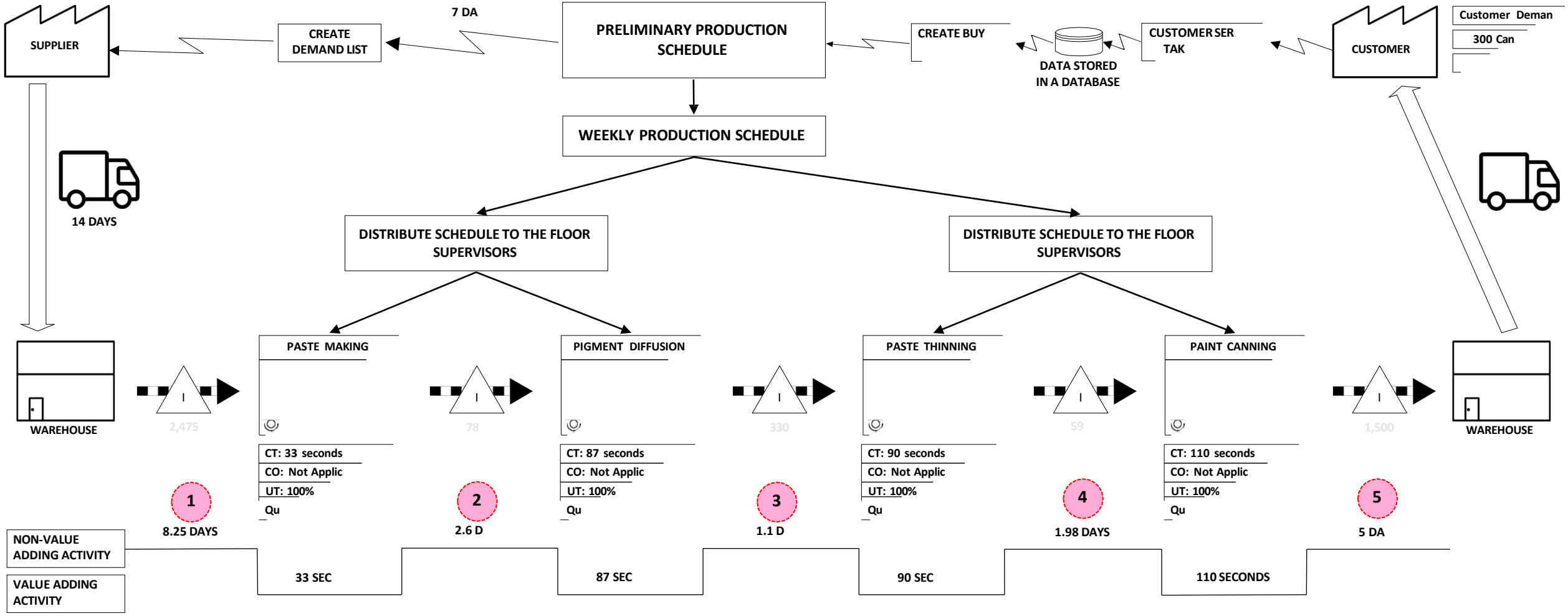
LET'S DO THE CALCULATION OF THE STORAGE TIME:



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1. THE AGEING OF RAW MATERIALS AT THE WAREHOUSE (BEFORE THE PASTE MAKING PROCESS) IS 8.25 DAYS
2. THE AGEING OF THE WORK-IN-PROGRESS GOODS BETWEEN PASTE MAKING AND PIGMENT DIFFUSION PROCESS IS 2.6 DAYS
3. THE AGEING OF THE WORK-IN-PROGRESS GOODS BETWEEN PIGMENT DIFFUSION AND PASTE THINNING IS 1.1 DAYS
4. THE AGEING OF WORK-IN-PROGRESS GOODS BETWEEN PASTE THINNING AND PAINT CANNING IS 1.98 DAYS
5. AND THE FINISHED GOODS ARE KEPT IN THE WAREHOUSE FOR 5 DAYS BEFORE BEING SHIPPED TO THE CUSTOMER

Lead Time = 39.95 days
Storage Time = 18.95 days
Takt Time = 87 seconds
Process/Processing/Value Added Time = 320 seconds
Cycle Time = 106.7 seconds

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STORAGE TIME OF THE PROCESS

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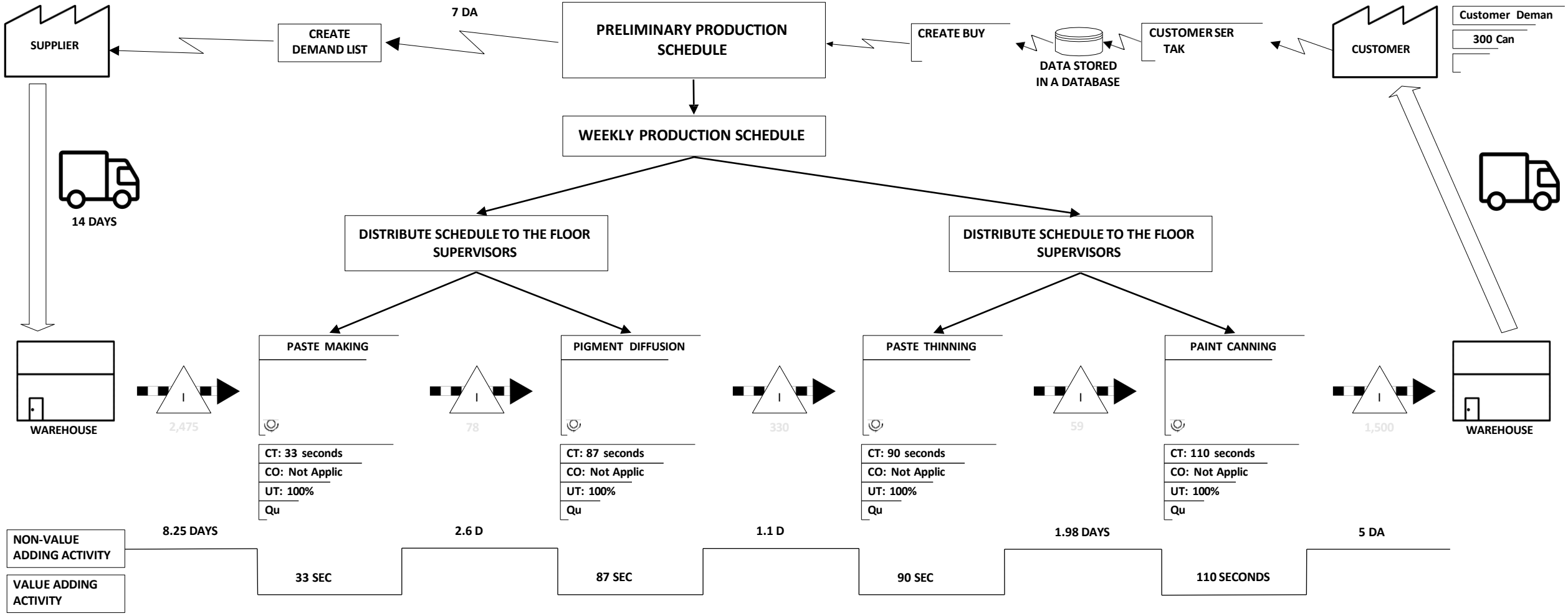
		# OF DAYS	SECONDS
1.	THE AGEING OF RAW MATERIALS AT THE WAREHOUSE (BEFORE THE PASTE MAKING PROCESS) IS 8.25 DAYS	8.25	
2.	THE AGEING OF THE WORK-IN-PROGRESS GOODS BETWEEN PASTE MAKING AND PIGMENT DIFFUSION PROCESS IS 2.6 DAYS	2.6	
3.	THE AGEING OF THE WORK-IN-PROGRESS GOODS BETWEEN PIGMENT DIFFUSION AND PASTE THINNING IS 1.1 DAYS	1.1	
4.	THE AGEING OF WORK-IN-PROGRESS GOODS BETWEEN PASTE THINNING AND PAINT CANNING IS 1.98 DAYS	1.98	
5.	AND THE FINISHED GOODS ARE KEPT IN THE WAREHOUSE FOR 5 DAYS BEFORE BEING SHIPPED TO THE CUSTOMER	5	
		18.93 DAYS	0 SECONDS

SO, THE STORAGE TIME IS 18.93 DAYS.

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ON THE BOTTOM RIGHT CORNER OF YOUR CURRENT STATE VALUE STREAM MAP, YOU CAN UPDATE THE STORAGE TIME AS 18.93 DAYS

Lead Time = 39.93 days
Storage Time = 18.93 days
Takt Time =
Process/Processing/Value Added Time =
Cycle Time =

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HOW TO CALCULATE THE TAKT TIME?



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RECALL: TAKT TIME



TAKT TIME IS THE TIME NEEDED TO PRODUCE ONE UNIT OF A PRODUCT TO MEET CUSTOMER REQUIREMENTS.

$$T = \frac{T_a}{T_d}$$

T = TAKT TIME

T_a = TOTAL AVAILABLE TIME

T_d = CUSTOMER DEMAND

TAKT IS A GERMAN WORD FOR "DRUM BEAT." IT ESTABLISHES THE PACE OF THE PROCESSES NEEDED TO MEET THE CUSTOMER DEMAND. IT SHOWS WHERE THE FLOW IS UNEVEN AND IS RESULTING IN WASTE.

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CALCULATE TAKT TIME

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FOR OUR DYNAMIC LTD. CASE STUDY:

LET'S DO THE CALCULATION OF THE TAKT TIME:



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TAKT TIME OF THE PROCESS

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CALCULATE TOTAL AVAILABLE TIME:

- **TOTAL AVAILABLE TIME = 8 HOURS WITH A BREAK OF 15 MINUTES (AT LEAST 3 TIMES)**
- **TOTAL AVAILABLE TIME = (8 X 60 MINUTES X 60 SECONDS) – [(15 MINUTES + 15 MINUTES + 15 MINUTES) X 60 SECONDS]**
- **TOTAL AVAILABLE TIME = (28,800 – 2,700)**
- **TOTAL AVAILABLE TIME = 26,100 SECONDS**
- **T_a = 26,100 SECONDS**

IDENTIFY CUSTOMER DEMAND:

- **CUSTOMER DEMAND = 300 CANS PER DAY**
- **T_d = 300**

TAKT TIME:

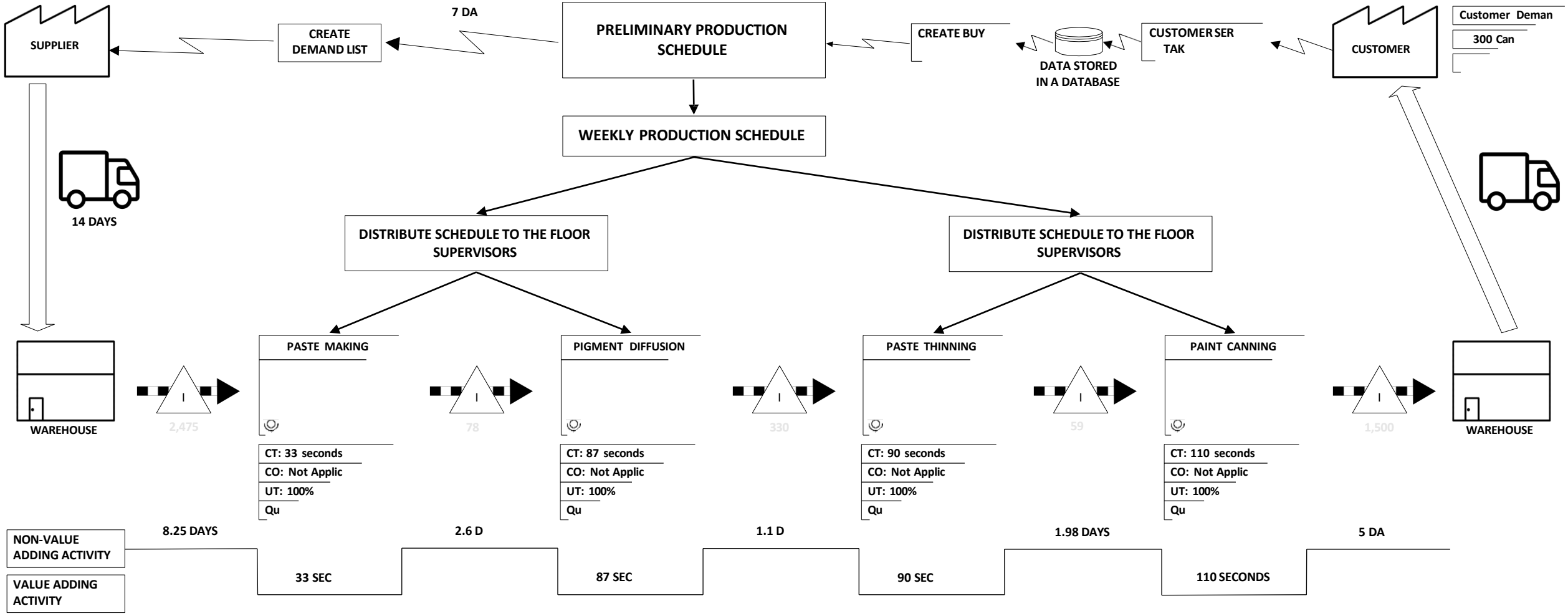
- $T = \frac{T_a}{T_d} = \frac{26{,}100}{300}$ **= 87 SECONDS**

THIS INDICATES THAT IT SHOULD TAKE NO MORE THAN 87 SECONDS TO PRODUCE ONE UNIT OF THE PAINT CAN TO MEET THE CUSTOMER REQUIREMENT OF 300 CANS PER DAY.

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YOU CAN UPDATE THE TAKT TIME OF 87 SECONDS IN THIS BOX ON YOUR CURRENT STATE VALUE STREAM MAP.

Lead Time = 39.93 days
Storage Time = 18.93 days
Takt Time = 87 seconds
Process/Processing/Value Added Time =
Cycle Time =

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HOW TO CALCULATE THE PROCESS TIME?



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RECALL: PROCESS TIME

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THE PROCESS TIME IS THE APPROXIMATE AMOUNT OF TIME REQUIRED TO HANDLE A PROCESS IN A PRODUCTION STEP.

THE PROCESSING TIME INCLUDES THE ORDER PREPARATION TIME, THE MOVE TIME, THE RUN TIME, THE PUT-AWAY TIME, AND THE INSPECTION TIME.



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CALCULATE PROCESS TIME

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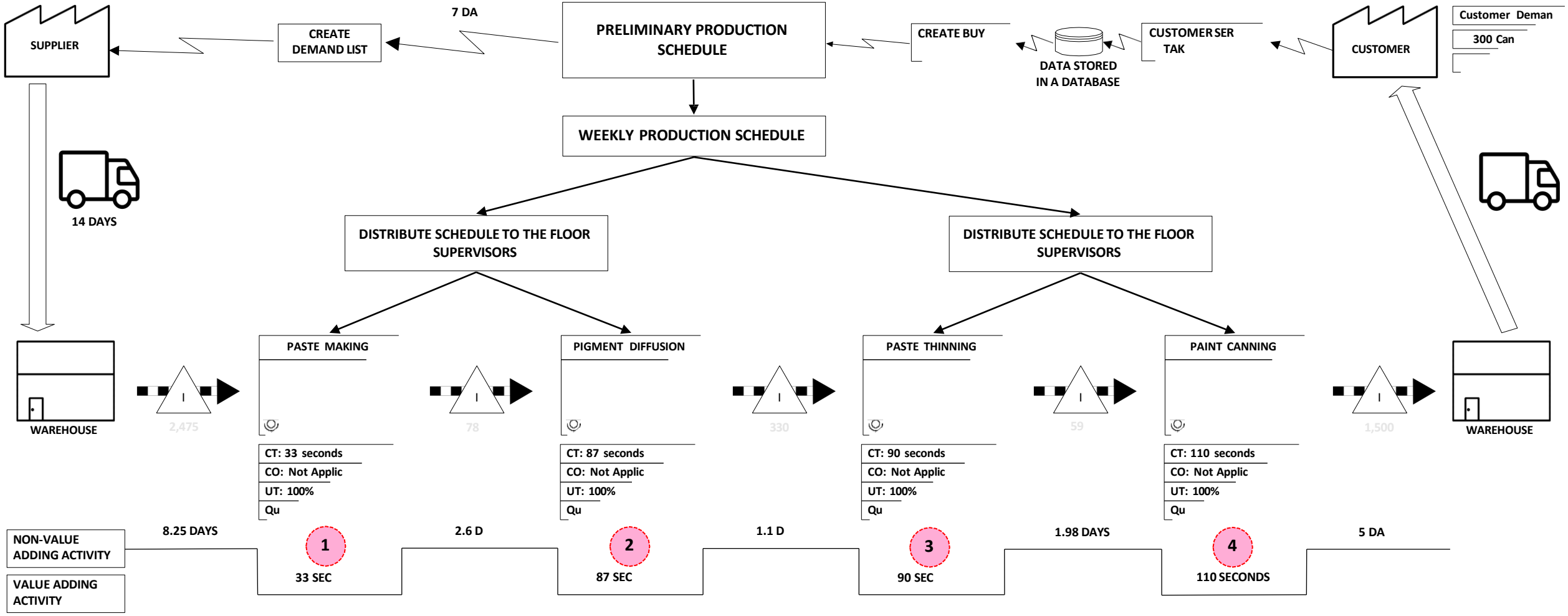
LET'S DO THE CALCULATION OF THE PROCESS TIME:



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1. THE PASTE MAKING PROCESS TAKES 33 SECONDS
2. THE PIGMENT DIFFUSION PROCESS TAKES 87 SECONDS
3. THE PASTE THINNING PROCESS TAKES 90 SECONDS
4. THE PAINT CANNING PROCESS TAKES 110 SECONDS

Lead Time = 39.95 days
Storage Time = 18.95 days
Takt Time = 87 seconds
Process/Processing/Value Added Time = 320 seconds
Cycle Time = 106.7 seconds

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PROCESS TIME

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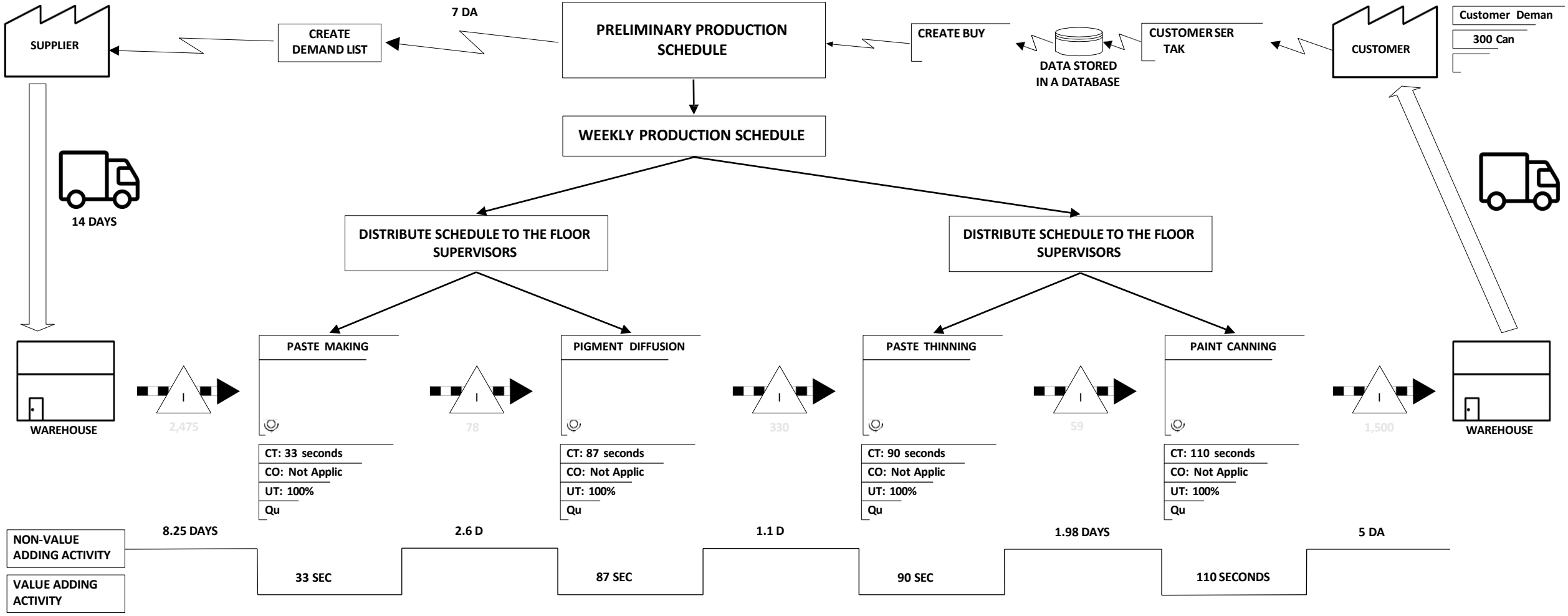
		# OF DAYS	SECONDS
1.	THE PASTE MAKING PROCESS TAKES 33 SECONDS	0	33
2.	THE PIGMENT DIFFUSION PROCESS TAKES 87 SECONDS	0	87
3.	THE PASTE THINNING PROCESS TAKES 90 SECONDS	0	90
4.	THE PAINT CANNING PROCESS TAKES 110 SECONDS	0	110
		0 DAYS	320 SECONDS

THE PROCESS TIME OR PROCESSING TIME OR THE VALUE ADDED TIME IS 320 SECONDS

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DYNAMIC LTD. CASE STUDY

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YOU CAN UPDATE THE PROCESS TIME OF 320 SECONDS IN THIS BOX ON YOUR CURRENT STATE VALUE STREAM MAP.

Lead Time = 39.93 days
Storage Time = 18.93 days
Takt Time = 87 seconds
Process/Processing/Value Added Time = 320 seconds
Cycle Time =

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HOW TO CALCULATE THE CYCLE TIME?



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RECALL: CYCLE TIME

IT IS THE TOTAL AMOUNT OF TIME REQUIRED TO COMPLETE ONE CYCLE OF TASK FROM START TO FINISH BY AN OPERATOR BEFORE HE/SHE REPEATS THIS OPERATION.

USE THE FOLLOWING FORMULA:

$$\text{CYCLE TIME} = \frac{PROCESS\ TIME}{NUMBER\ OF\ OPERATORS}$$



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CALCULATE CYCLE TIME

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FOR OUR DYNAMIC LTD. CASE STUDY:

LET'S DO THE CALCULATION OF THE CYCLE TIME:

IN THE CONTEXT OF OUR CASE STUDY, CYCLE TIME IS THE TOTAL TIME NEEDED TO PRODUCE ONE CAN OF PAINT BY ONE OPERATOR FROM START OF THE PRODUCTION TO FINISH.



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CYCLE TIME

AIGPE

CALCULATE PROCESS TIME:

- **PROCESS TIME = 33 SECONDS + 87 SECONDS + 90 SECONDS + 110 SECONDS (CALCULATED PREVIOUSLY)**
- **PROCESS TIME = 320 SECONDS**

IDENTIFY THE NUMBER OF OPERATORS:

- **NUMBER OF OPERATORS INVOLVED = 3**

CYCLE TIME:

$$\text{CYCLE TIME} = \frac{PROCESS\ TIME}{NUMBER\ OF\ OPERATORS} = \frac{320}{3}$$

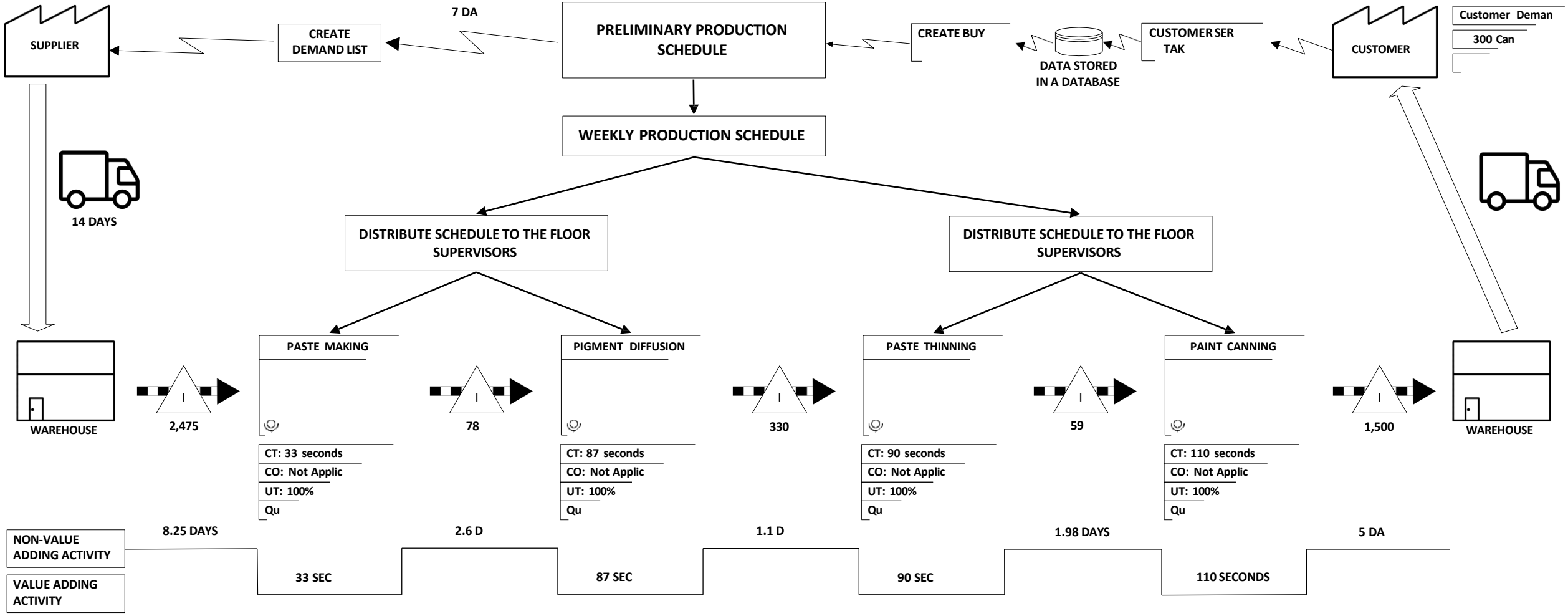
CYCLE TIME = 106.7 SECONDS

IT TAKES 106.7 SECONDS TO COMPLETE ONE CYCLE OF A TASK FROM START TO FINISH I.E. IT TAKES 106.7 SECONDS TO PRODUCE ONE PAINT CAN FROM THE FIRST TO THE LAST PROCESS.

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DYNAMIC LTD. CASE STUDY

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Lead Time = 39.93 days
Storage Time = 18.93 days
Takt Time = 87 seconds
Process/Processing/Value Added Time = 320 seconds
Cycle Time = 106.7 seconds

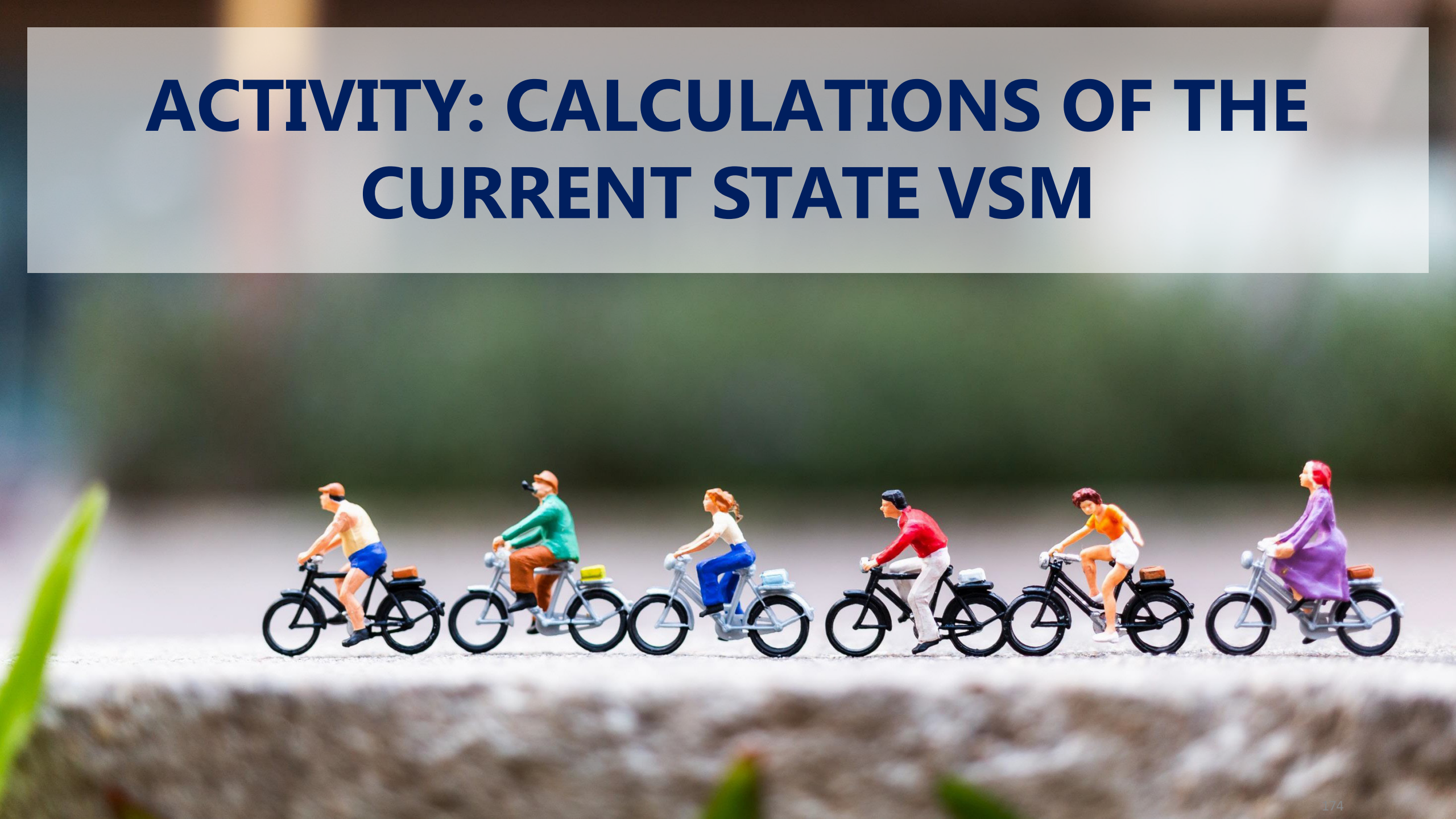
YOU CAN UPDATE THE CYCLE TIME OF 106.7 SECONDS IN THIS BOX ON YOUR CURRENT STATE VALUE STREAM MAP.

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ACTIVITY: CALCULATIONS OF THE CURRENT STATE VSM



ACTIVITY

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YOU ARE GIVEN A FEW QUESTIONS AND PROBLEMS TO SOLVE.

BASED ON YOUR UNDERSTANDING, ANSWER THOSE QUESTIONS.



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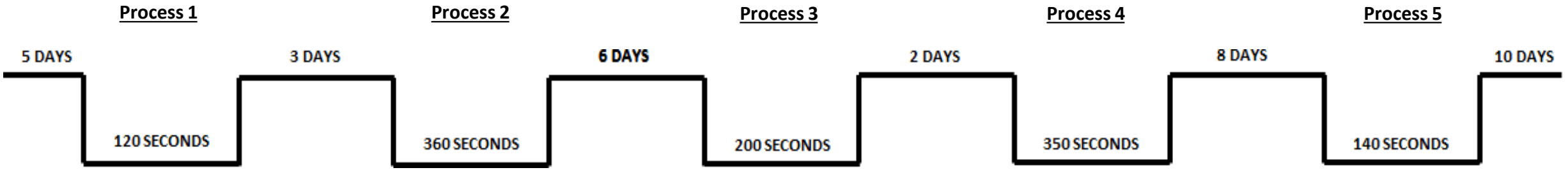
ACTIVITY

AIGPE

QUESTION 1:

YOU WORK AS A MANAGER OF A TOY FIGURE MANUFACTURING ASSEMBLY PROCESS. YOUR OPERATIONS RUN IN 1 SHIFT OF 9 HOURS. EMPLOYEES ARE ALLOWED TO TAKE 3 BREAKS OF 10 MINUTES EACH. NUMBER OF OPERATORS = 2 (IMPORTANT FOR CYCLE TIME CALCULATION)

CUSTOMER DEMAND IS 30 TOY FIGURES PER DAY. THERE ARE FIVE DEPARTMENTS. REQUIRED DETAILS ARE PROVIDED IN THE TIMELINE BELOW. BASED ON THIS DATA, CALCULATE THE TAKT TIME, PROCESS TIME, AND CYCLE TIME.



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ACTIVITY

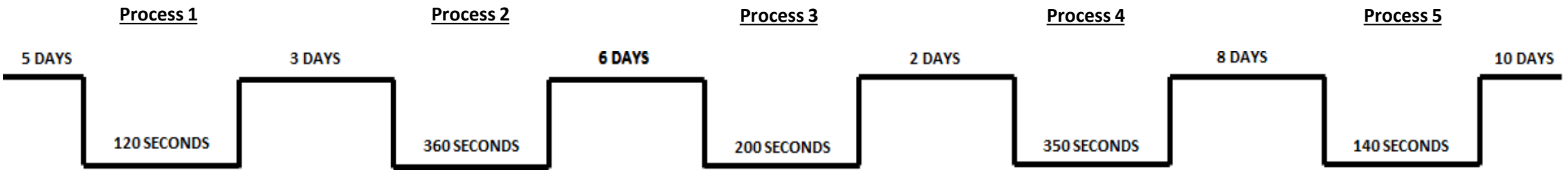
AIGPE

QUESTION 2:

IT TAKES 10 DAYS FOR THE SUPPLIER TO DELIVER THE RAW MATERIALS TO THE WAREHOUSE.

THE TIMELINE OF YOUR NEWLY CREATED CURRENT STATE VALUE STREAM MAP IS AS OUTLINED.

CALCULATE THE LEAD TIME AND THE STORAGE TIME.



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ACTIVITY SOLUTION: CALCULATIONS OF THE CURRENT STATE VSM



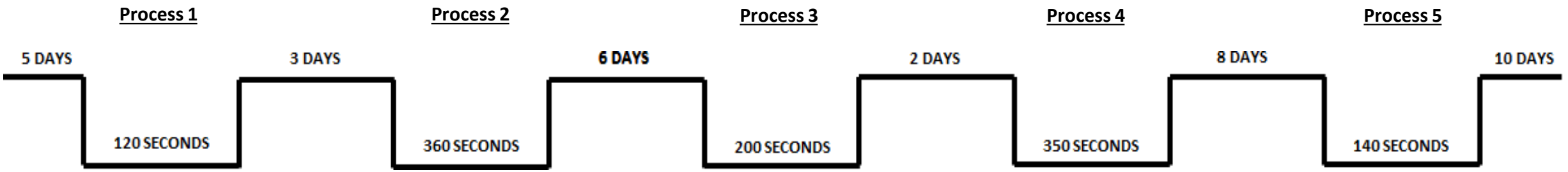
ACTIVITY

AIGPE

QUESTION 1:

YOU WORK AS A MANAGER OF A TOY FIGURE MANUFACTURING ASSEMBLY PROCESS. YOUR OPERATIONS RUN IN 1 SHIFT OF 9 HOURS. EMPLOYEES ARE ALLOWED TO TAKE 3 BREAKS OF 10 MINUTES EACH. NUMBER OF OPERATORS = 2 (IMPORTANT FOR CYCLE TIME CALCULATION)

CUSTOMER DEMAND IS 30 TOY FIGURES PER DAY. THERE ARE FIVE DEPARTMENTS. REQUIRED DETAILS ARE PROVIDED IN THE TIMELINE BELOW. BASED ON THIS DATA, CALCULATE THE TAKT TIME, PROCESS TIME, AND CYCLE TIME.



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TAKT TIME

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LET'S CALCULATE THE TAKT TIME FIRST:

CALCULATE TOTAL AVAILABLE TIME:

- **TOTAL AVAILABLE TIME = 9 HOURS WITH A BREAK OF 10 MINUTES (AT LEAST 3 TIMES)**
- **TOTAL AVAILABLE TIME = (9 X 60 MINUTES X 60 SECONDS) – [(10 MINUTES + 10 MINUTES + 10 MINUTES) X 60 SECONDS]**
- **TOTAL AVAILABLE TIME = (32,400 – 1,800)**
- **TOTAL AVAILABLE TIME = 30,600 SECONDS**
- **T_a = 30,600 SECONDS**

IDENTIFY CUSTOMER DEMAND:

- **CUSTOMER DEMAND = 30 UNITS PER DAY**
- **T_d = 30**

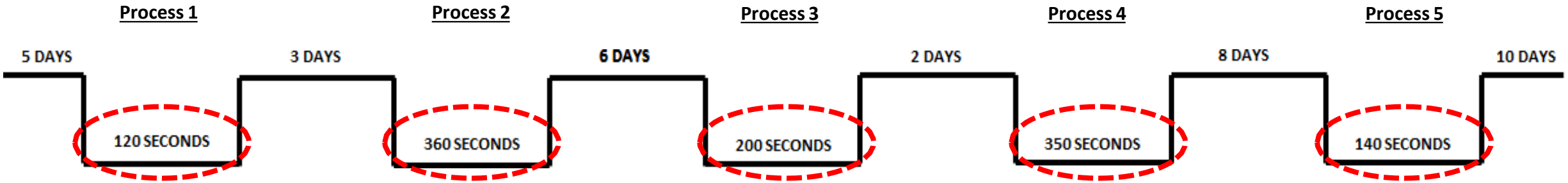
TAKT TIME:

- $T = \frac{T_a}{T_d} = \frac{30{,}600}{30}$ **= 1,020 SECONDS**

THIS INDICATES THAT IT SHOULD TAKE NO MORE THAN 1,020 SECONDS TO PRODUCE ONE UNIT OF THE PRODUCT TO MEET THE CUSTOMER REQUIREMENT OF 30 UNITS PER DAY.

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PROCESS TIME



RECALL THAT THE PROCESS TIME IS THE APPROXIMATE AMOUNT OF TIME REQUIRED TO HANDLE A PROCESS IN A PRODUCTION STEP. HERE, THE PRODUCTION STEPS ARE PROCESS 1, 2, 3, 4 AND 5.

	# OF DAYS	SECONDS
1. PROCESS 1 TAKES 120 SECONDS	0	120
2. PROCESS 2 TAKES 360 SECONDS	0	360
3. PROCESS 3 TAKES 200 SECONDS	0	200
4. PROCESS 4 TAKES 350 SECONDS	0	350
5. PROCESS 5 TAKES 140 SECONDS	0	140
	0 DAYS	1,070 SECONDS

THE PROCESS TIME OR PROCESSING TIME OR THE VALUE-ADDED TIME IS 1,070 SECONDS

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CYCLE TIME

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RECALL THE DEFINITION OF CYCLE TIME: IT IS THE TOTAL AMOUNT OF TIME REQUIRED TO COMPLETE ONE CYCLE OF TASK FROM START TO FINISH BY AN OPERATOR BEFORE HE/SHE REPEATS THIS OPERATION.

CALCULATE PROCESS TIME:

- **PROCESS TIME = 120 SECONDS + 360 SECONDS + 200 SECONDS + 350 SECONDS + 140 SECONDS**
- **PROCESS TIME = 1,070 SECONDS**

IDENTIFY THE NUMBER OF OPERATORS:

- **NUMBER OF OPERATORS INVOLVED = 2**

CYCLE TIME:

$$\text{CYCLE TIME} = \frac{PROCESS\ TIME}{NUMBER\ OF\ OPERATORS} = \frac{1{,}070}{2}$$

CYCLE TIME = 585 SECONDS

IT TAKES 585 SECONDS TO COMPLETE ONE CYCLE OF A TASK FROM START TO FINISH.

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ACTIVITY

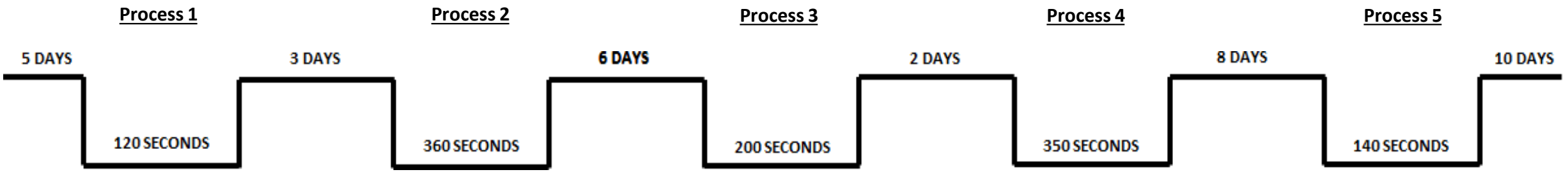
AIGPE

QUESTION 2:

IT TAKES 10 DAYS FOR THE SUPPLIER TO DELIVER THE RAW MATERIALS TO THE WAREHOUSE.

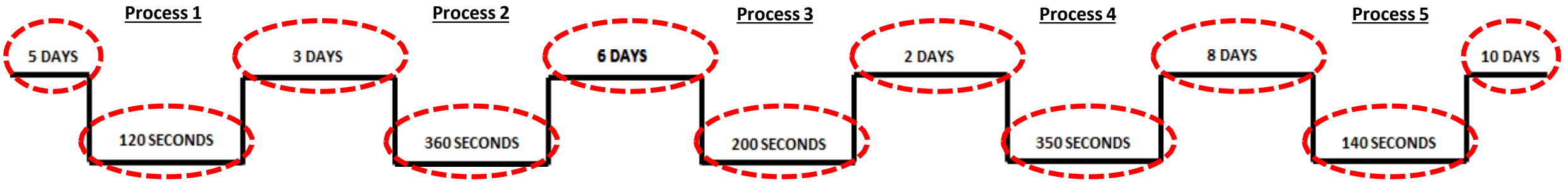
THE TIMELINE OF YOUR NEWLY CREATED CURRENT STATE VALUE STREAM MAP IS AS OUTLINED.

CALCULATE THE LEAD TIME AND THE STORAGE TIME.



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LEAD TIME



THE LEAD TIME TAKES INTO ACCOUNT ALL OF THE TIME TAKEN IN THE ENTIRE PROCESS. THE QUESTION STATES THAT THE IT TAKES 10 DAYS FOR THE SUPPLIER TO DELIVER THE RAW MATERIALS TO THE WAREHOUSE. IN ADDITION, WE ARE PROVIDED THE TIMELINE OF THE CURRENT STATE VALUE STREAM MAP. SO, TO CALCULATE THE LEAD TIME, WE HAVE TO ADD THE TIME GIVEN ON THIS TIMELINE WITH 10 DAYS THAT THE SUPPLIER TAKES.

WE'LL PERFORM THE CALCULATIONS ON THE NEXT PAGE.

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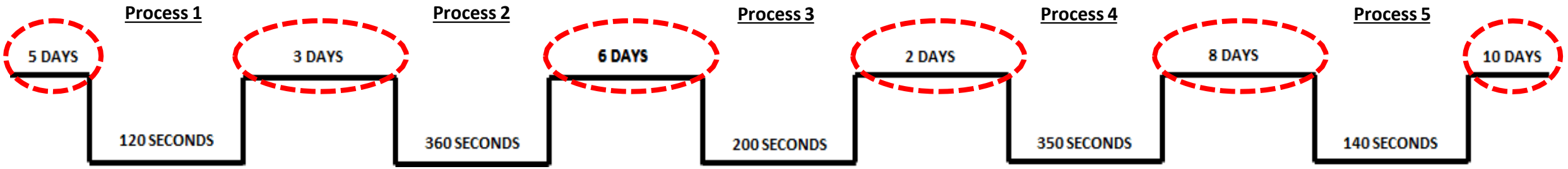
TOTAL TIME

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		# OF DAYS	SECONDS
1.	IT TAKES 10 DAYS FOR THE SUPPLIER TO DELIVER THE RAW MATERIAL TO THE WAREHOUSE	10	
2.	THE AGEING OF RAW MATERIALS AT THE WAREHOUSE IS 5 DAYS	5	
3.	PROCESS 1 TAKES 120 SECONDS		120
4.	THE AGEING OF THE WORK-IN-PROGRESS GOODS BETWEEN PROCESS 1 AND 2 IS 3 DAYS	3	
5.	PROCESS 2 TAKES 360 SECONDS		360
6.	THE AGEING OF THE WORK-IN-PROGRESS GOODS BETWEEN PROCESS 2 AND 3 IS 6 DAYS	6	
7.	PROCESS 3 TAKES 200 SECONDS		200
8.	THE AGEING OF THE WORK-IN-PROGRESS GOODS BETWEEN PROCESS 3 AND 4 IS 2 DAYS	2	
9.	PROCESS 4 TAKES 350 SECONDS		350
10.	THE AGEING OF THE WORK-IN-PROGRESS GOODS BETWEEN PROCESS 4 AND 5 IS 8 DAYS	8	
11.	PROCESS 5 TAKES 140 SECONDS		140
12.	FINISHED GOODS ARE KEPT IN THE WAREHOUSE FOR 10 DAYS BEFORE BEING SHIPPED TO THE CUSTOMER	10	
		44 DAYS	1,070 SECONDS

SINCE THE TIME TAKEN IN SECONDS IN INFINITESIMALLY SMALL COMPARED TO THE NUMBER OF DAYS, YOU CAN IGNORE THAT NUMBER. AS A RESULT, THE LEAD TIME IS 44 DAYS.

STORAGE TIME



THE STORAGE TIME CAN BE MINUTES, HOURS, OR DAYS. IT IS THE AMOUNT OF TIME THAT IS REQUIRED FOR THE RAW MATERIAL THAT MAY BE SITTING IN THE WAREHOUSE TO GO THROUGH THE DIFFERENT STAGES OF OPERATIONS ALL THE WAY UNTIL THE STORAGE OF FINISHED PRODUCTS BEFORE GETTING SHIPPED TO THE CUSTOMER.

WE'LL PERFORM THE CALCULATIONS ON THE NEXT PAGE.

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STORAGE TIME OF THE PROCESS

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		# OF DAYS	SECONDS
1.	THE AGEING OF RAW MATERIALS AT THE WAREHOUSE IS 5 DAYS	5	
2.	THE AGEING OF THE WORK-IN-PROGRESS GOODS BETWEEN PROCESS 1 AND 2 IS 3 DAYS	3	
3.	THE AGEING OF THE WORK-IN-PROGRESS GOODS BETWEEN PROCESS 2 AND 3 IS 6 DAYS	6	
4.	THE AGEING OF THE WORK-IN-PROGRESS GOODS BETWEEN PROCESS 3 AND 4 IS 2 DAYS	2	
5.	THE AGEING OF THE WORK-IN-PROGRESS GOODS BETWEEN PROCESS 4 AND 5 IS 8 DAYS	8	
6.	FINISHED GOODS ARE KEPT IN THE WAREHOUSE FOR 10 DAYS BEFORE BEING SHIPPED TO THE CUSTOMER	10	
		34 DAYS	0 SECONDS

SO, THE STORAGE TIME IS 34 DAYS.

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SECTION CONCLUSION:
STEPS TO CREATE CURRENT STATE VSM



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SECTION CONCLUSION

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IN THIS SECTION, YOU LEARNED HOW TO CALCULATE:

1. **THE LEAD TIME OF THE PROCESS**
2. **STORAGE TIME**
3. **TAKT TIME**
4. **PROCESS/PROCESSING/VALUE-ADDED TIME**
5. **CYCLE TIME**

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SECTION INTRODUCTION:
HOW TO EXAMINE THE CURRENT STATE VSM?



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VALUE STREAM MAPPING (VSM) MASTERCLASS

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Introduction to Value Stream Map (VSM)

- Course Introduction
- What is Value Stream Mapping (VSM)?
- Why is Value Stream Mapping so popular?
- Section Conclusion

Terminologies and Symbols used in VSM0

- Section Introduction
- Terminologies used in VSM
- Symbols used in VSM Process
- Material Flow Symbols
- Information Flow Symbols
- General Symbols
- Activity: Terminologies and Symbols used in VSM
- Activity Solution: Terminologies and Symbols used in VSM
- Section Conclusion

Steps to Create the Current State VSM

- Section and Case Study Introduction
- Step 1. Pick a Product Family
- Step 2. Choose Your VSM Team
- Step 3: Figure out Customer Demand
- Step 4: Map and Design the Process Flow
- Step 5: Map the Material Flow
- Step 6: Map the Information Flow
- Case Study Recap: Current State VSM
- Perform Calculations | Analyze Current State VSM
- Calculate Total Time | Lead Time | Takt Time
- Calculate Process Time | Cycle Time
- Activity: Steps to Create Current State VSM
- Activity Solution: Steps to Create Current State VSM
- Section Conclusion

Examine the Current State VSM

- Section Introduction
- Meeting Customer Demand with existing No. of Operators
- Meeting Customer Demand with increased No. of Operators
- Analyze the Process Balance
- Watch for Waste
- Section Conclusion

Strong Foundations for Future State VSM

- Section Introduction
- Change to Continuous Flow
- CONWIP
- Kanban
- Readjusting Your Process
- Alter the Batch Sizes
- Smoothening out Production
- Heijunka | Heijunka Box
- Cut back Changeover Time
- Improve Quality
- Putting it all Together

Key Questions to ask for Future State VSM

- Section Introduction
- What is the Takt Time?
- Are you building to a warehouse or directly to shipping?
- Where should Continuous Flow Control the Production?
- Controlling area of upstream processes
- Where will you implement the Pull System?
- How do you intend to level the production mix?
- What process improvements do you intend to achieve?
- What is the Implementation Plan?
- Closure of the Case Study
- Section Conclusion

VSM Guidelines

- Section Introduction
- How to get the most of your VSM?
- What are the Pitfalls to avoid?
- What are the advantages of VSM?
- What are the disadvantages of VSM
- Section Conclusion

Course Conclusion

- Course Summary
- Practice Questions
- Become a Certified VSM Practitioner!

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CAN YOUR PROCESS MEET CUSTOMER DEMAND WITH EXISTING NO. OF OPERATORS?



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CAN YOUR PROCESS MEET DEMAND WITH EXISTING NO. OF OPERATORS?

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FOR OUR DYNAMIC LTD. CASE STUDY:

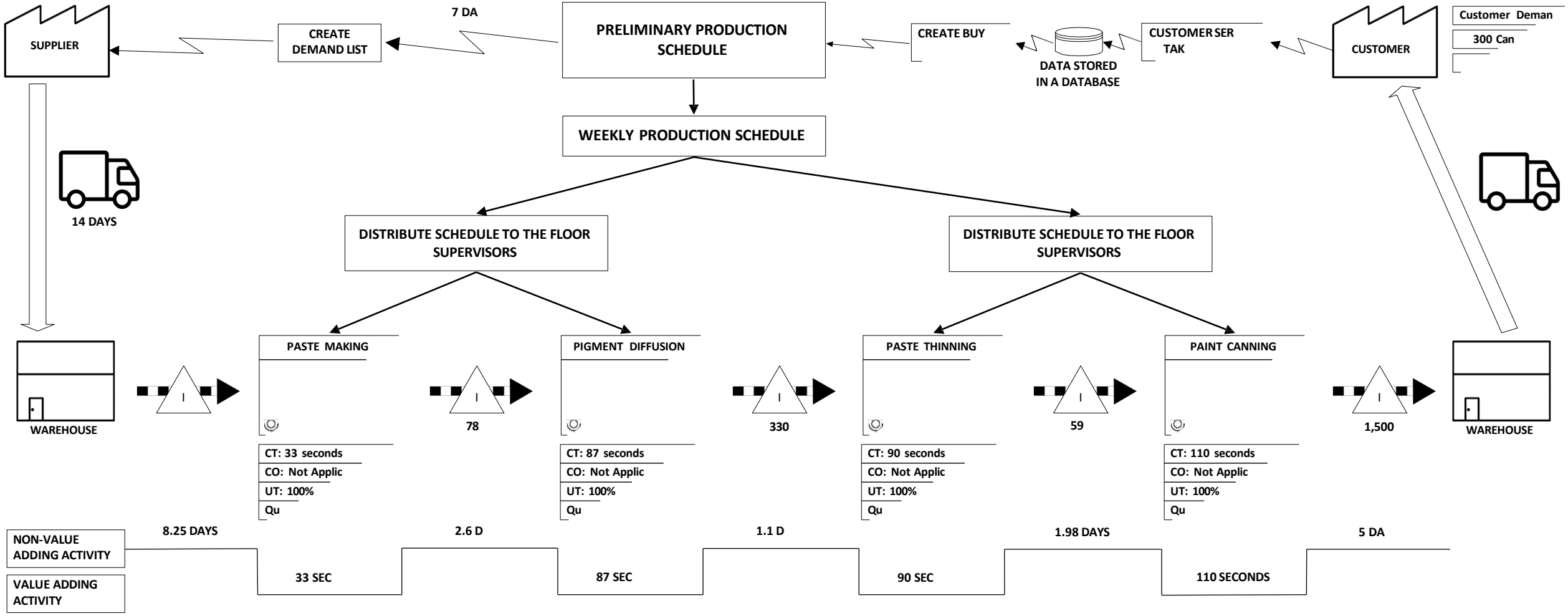
LET'S DO THE MATH:



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DYNAMIC LTD. CASE STUDY

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TAKT TIME INDICATES THAT YOUR PROCESS SHOULD NOT TAKE MORE THAN 87 SECONDS TO PRODUCT ONE UNIT OF THE PRODUCT TO MEET THE DAILY CUSTOMER DEMAND OF 300 CANS/DAY

CYCLE TIME INDICATES THAT WITH 3 RESOURCES, THE PROCESS TAKES 106.7 SECONDS TO PRODUCT ONE UNIT OF THE PAINT CAN

Total Time = 39.93 days
Lead Time = 18.93 days
Takt Time = 87 seconds
Process/Processing/Value Added Time = 320 seconds
Cycle Time = 106.7 seconds

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CAN YOUR PROCESS MEET DEMAND WITH EXISTING NO. OF OPERATORS?

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TAKT TIME	87 seconds	<	CYCLE TIME	106.7 seconds

TAKT TIME:

- **CALCULATED AS 87 SECONDS.**
- **SO IT MEANS THAT IT SHOULD TAKE NO MORE THAN 87 SECONDS TO PRODUCE ONE UNIT OF THE PAINT CAN TO MEET THE CUSTOMER DEMAND.**

CYCLE TIME:

- **CALCULATED AS 106.7 SECONDS**
- **WITH THREE RESOURCES, THE PROCESS TAKES 106.7 SECONDS TO PRODUCE ONE UNIT OF THE PAINT CAN.**

AS YOU COMPARE THE TAKT TIME OF 87 VS. THE CYCLE TIME OF 106.7, YOU FIND THAT THE THREE EMPLOYEES ARE UNABLE TO MEET THE CUSTOMER DEMAND OF 300 CANS PER DAY.

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CAN YOUR PROCESS MEET CUSTOMER DEMAND WITH INCREASED NO. OF OPERATORS?



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CAN YOUR PROCESS MEET DEMAND WITH INCREASED NO. OF OPERATORS?

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FOR OUR DYNAMIC LTD. CASE STUDY:

LET'S DO THE MATH:



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CAN YOUR PROCESS MEET DEMAND WITH INCREASED NO. OF OPERATORS?

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CYCLE TIME WITH 3 OPERATORS	106.7 seconds	CYCLE TIME WITH 4 OPERATORS	? seconds

LET'S CALCULATE THE CYCLE TIME WITH 4 OPERATORS:

FIRST, CALCULATE PROCESS TIME:

- PROCESS TIME = 33 SECONDS + 87 SECONDS + 90 SECONDS + 110 SECONDS (CALCULATED PREVIOUSLY)
- PROCESS TIME = 320 SECONDS

SECONDS, IDENTIFY THE NUMBER OF OPERATORS:

- NUMBER OF OPERATORS INVOLVED = 4

THIRD, CALCULATE CYCLE TIME:

$$\text{CYCLE TIME} = \frac{PROCESS\ TIME}{NUMBER\ OF\ OPERATORS} = \frac{320}{4}$$

CYCLE TIME WITH 4 OPERATORS = 80 SECONDS

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CAN YOUR PROCESS MEET DEMAND WITH INCREASED NO. OF OPERATORS?

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CYCLE TIME WITH 3 OPERATORS	106.7 seconds	>	CYCLE TIME WITH 4 OPERATORS	80 seconds

CYCLE TIME WITH 3 OPERATORS VS. WITH 4 OPERATORS:

- **THE CYCLE TIME WITH THREE OPERATORS IS 106.7 SECONDS, AND THAT WITH FOUR OPERATORS, IT GETS REDUCED TO 80 SECONDS. IT IS MUCH LESSER THAN 106.7 SECONDS.**

CYCLE TIME WITH 4 OPERATORS VS. TAKT TIME:

- **THE TAKT TIME IS 87 SECONDS. AND THE CYCLE TIME WITH 4 OPERATORS IS 80 SECONDS. THUS, THIS CYCLE TIME IS LESSER THAN THE TAKT TIME TOO.**

CONCLUSION:

- **THEREFORE, IT INDICATES THAT IF WE ADD ONE MORE OPERATOR, THE TEAM CAN MEET THE DAILY CUSTOMER DEMAND.**

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DRIVE SUCCESS TO EXCELLENCE

ANALYZE THE PROCESS BALANCE



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ANALYZE THE PROCESS BALANCE

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UNLESS A BALANCE CAN BE STRUCK BETWEEN THE CUSTOMER DEMANDS AND THE PROCESSING CAPACITY, YOU WILL BE UNABLE TO MEET THE CUSTOMER DEMAND.

EFFECTIVE PRODUCTION REQUIRES THE TAKT TIME TO BE LONGER THAN THE CYCLE TIME.



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ANALYZE THE PROCESS BALANCE

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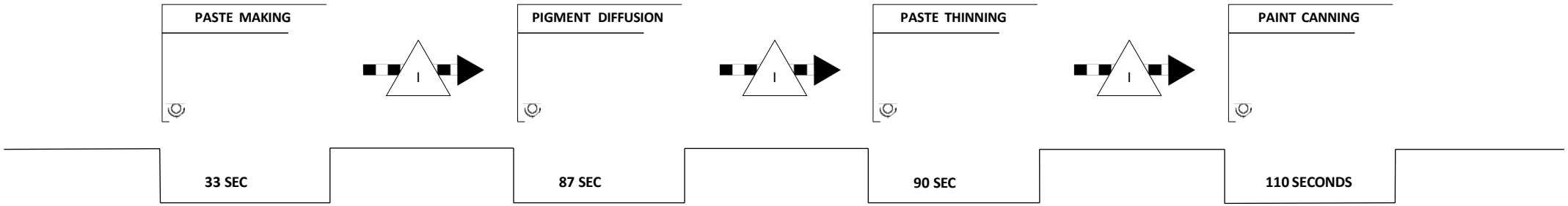
FOR OUR DYNAMIC LTD. CASE STUDY:

LET'S DO A CHART:



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ANALYZE THE PROCESS BALANCE

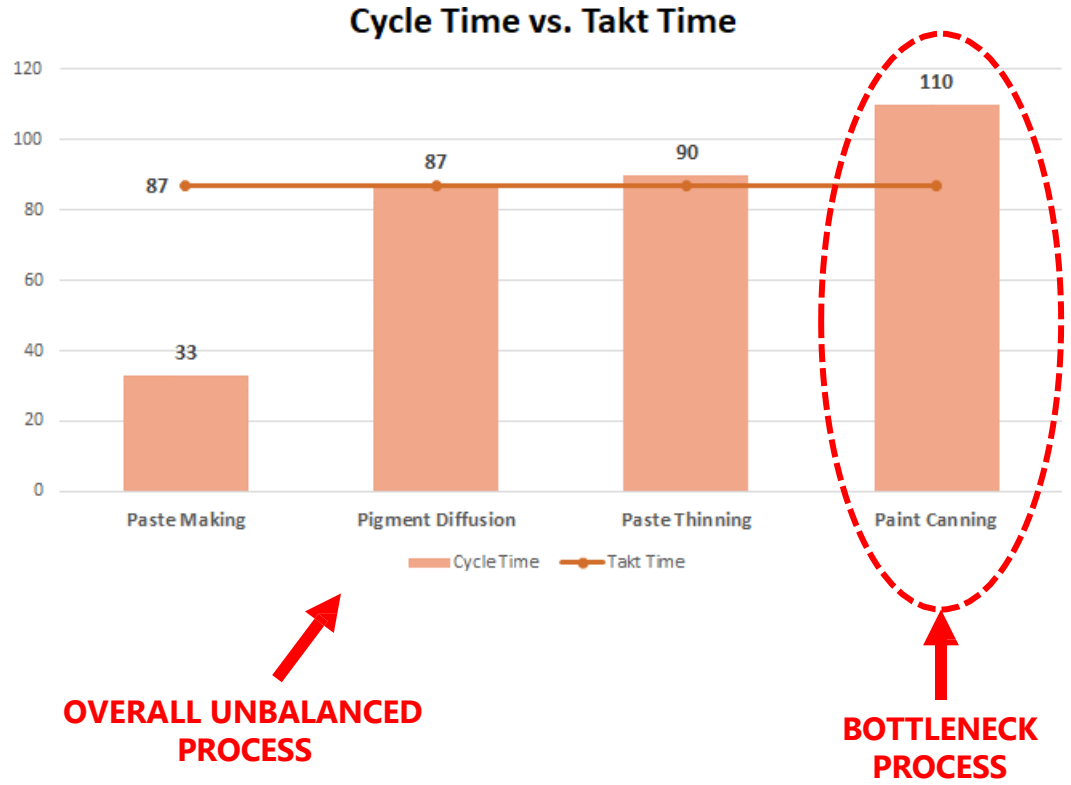


IN A BAR AND LINE CHART, PLOT THE CYCLE TIME OF EACH PROCESS AGAINST THE TAKT TIME. FOR DYNAMIC LTD., THE OBSERVATIONS ARE:

1. **CYCLE TIME OF PASTE MAKING (33 SECONDS) < 87 SECONDS**
2. **CYCLE TIME FOR PIGMENT DIFFUSION (87 SECONDS) = 87 SECONDS**
3. **CYCLE TIME FOR PASTE THINNING (90 SECONDS) > 87 SECONDS**
4. **CYCLE TIME FOR PAINT CANNING (110 SECONDS) > 87 SECONDS**

<u>THIS INDICATES TWO THINGS:</u>

1. **<u>PAINT CANNING PROCESS IS CREATING A BOTTLENECK</u>**
2. **<u>THE OVERALL PROCESS IS UNBALANCED</u>**



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ANALYZE THE PROCESS BALANCE

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KEY POINTS TO REMEMBER:

- **BY MEASURING AND COMPARING THE CYCLE TIMES IN THE MIDDLE OF OPERATIONS WITHIN THE PRODUCTION PROCESS, NOT ONLY ARE YOU EVALUATING BALANCE, BUT YOU ARE ALSO ABLE TO IDENTIFY THE BOTTLENECKS OF THE PRODUCTION PROCESS QUICKLY.**

- **BY MEASUREMENT AND COMPARISON, YOU CAN DISCOVER THE HUGE LAG BETWEEN CYCLE TIMES IN BETWEEN OPERATIONS AND CORRECT APPROPRIATELY. THIS CORRECTION IS NECESSARY TO AVOID UNDERUTILIZING RESOURCES DURING OPERATIONS WITH THE SHORTER CYCLE TIMES.**

- **IT IS INEQUITABLE, COSTLY, AND UNPRODUCTIVE TO HAVE SOME OPERATIONS RUNNING AT HALF TIME WHILE OTHERS RUN AT FULL TIME.**



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WATCH FOR WASTE



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WATCH FOR WASTE

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THERE ARE SEVEN ACCEPTED FORMS OF WASTE ACCORDING TO THE TOYOTA PRODUCTION SYSTEM, TPS.

THEY ARE:

- OVERPRODUCTION
- OVERPROCESSING
- WAITING
- TRANSPORT
- MOTION
- INVENTORY
- DEFECTS





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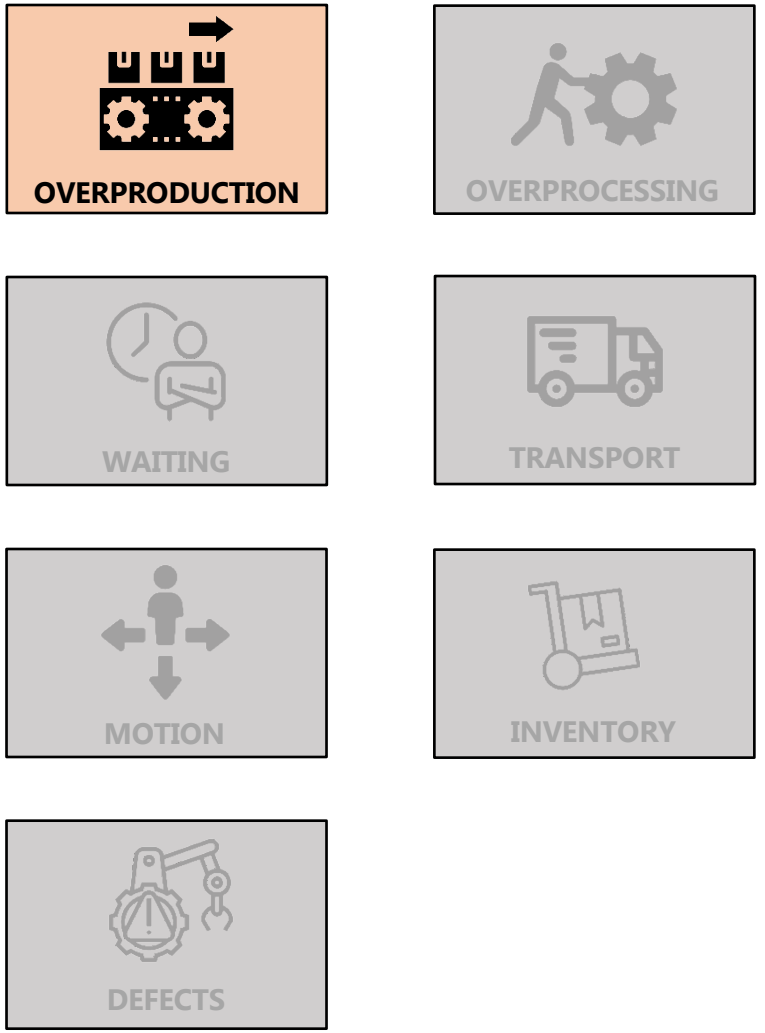
OVERPRODUCTION

AIGPE

OVERPRODUCTION IS PRODUCING TOO MUCH OR TOO SOON.

BY PRODUCING MORE THAN THE CUSTOMER'S DEMAND, YOU CREATE AN INCREASE IN INVENTORY.

THE INCREASED INVENTORY LEADS TO A LONGER WAITING PERIOD AS WELL AS INCREASED TRANSPORTATION OF GOODS OR PRODUCTS.



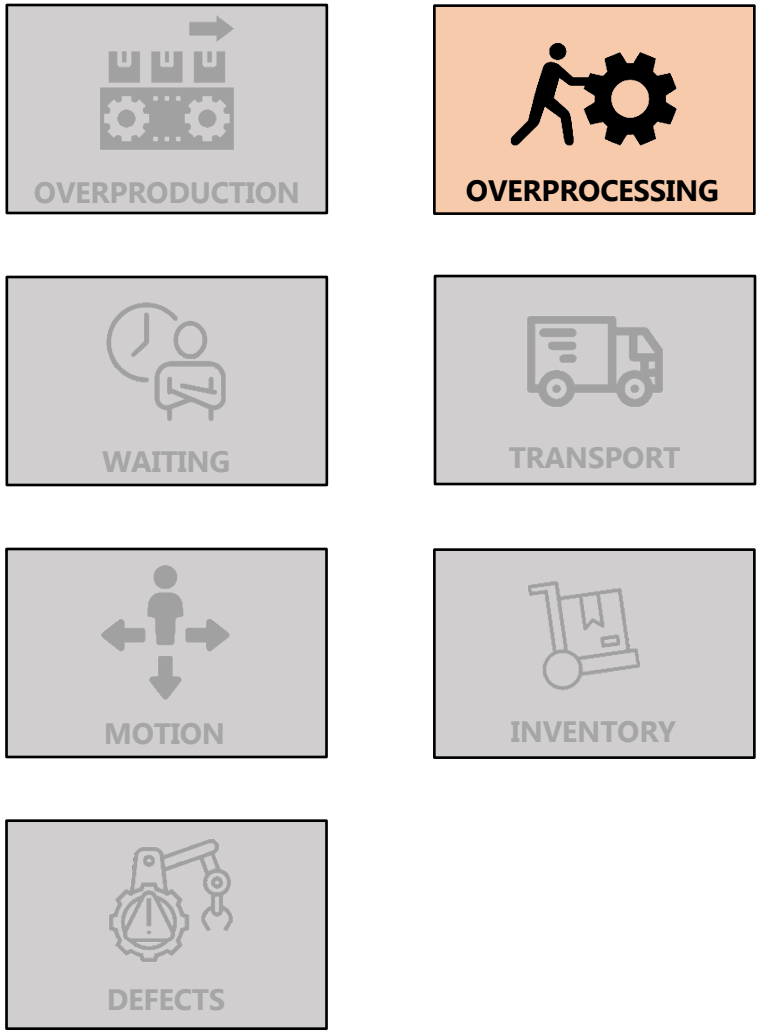
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OVERPROCESSING

AIGPE

OVERPROCESSING IS ADDITIONAL PROCESSING OF TRANSACTIONS.

OCCURS WHEN COMPLEX SOLUTIONS ARE APPLIED IN RESPONSE TO SIMPLE PROBLEMS.



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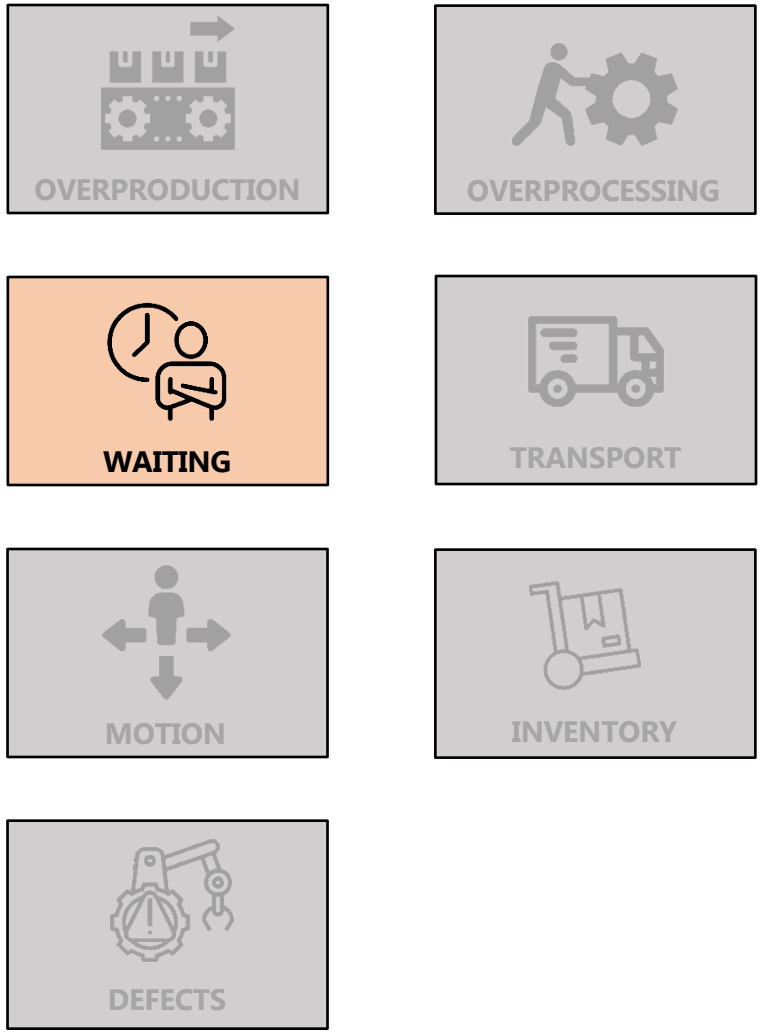
WAITING

AIGPE

WASTE OF WAITING OCCURS WHENEVER WORK HAS TO STOP FOR SOME REASON.

THAT MAY BE DUE TO:

- **THE NEXT PERSON IN LINE IS OVERWHELMED**
- **SOMETHING BROKE DOWN**
- **YOU'RE WAITING FOR APPROVAL OR**
- **YOU'VE RUN OUT OF SOMETHING**

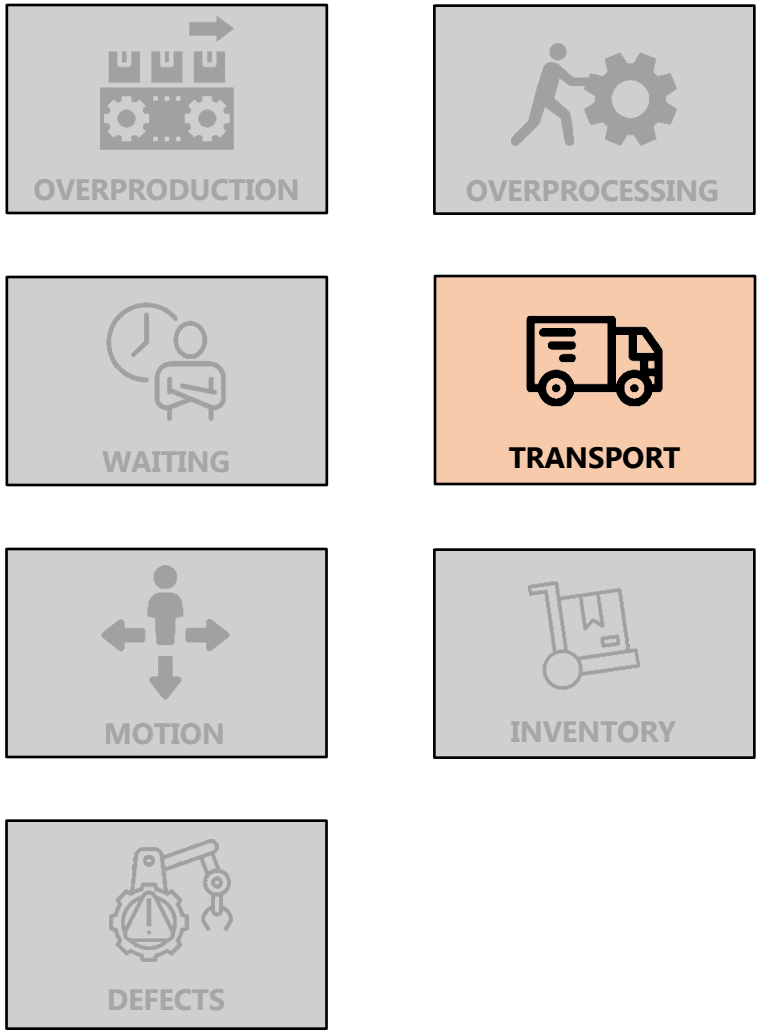


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TRANSPORT

AIGPE

THIS IS THE WASTE FROM MOVING THINGS AROUND OR WORK TRANSFERRING ACROSS PLATFORMS OR TEAMS, MOSTLY NON-ESSENTIAL TRANSPORTATION.



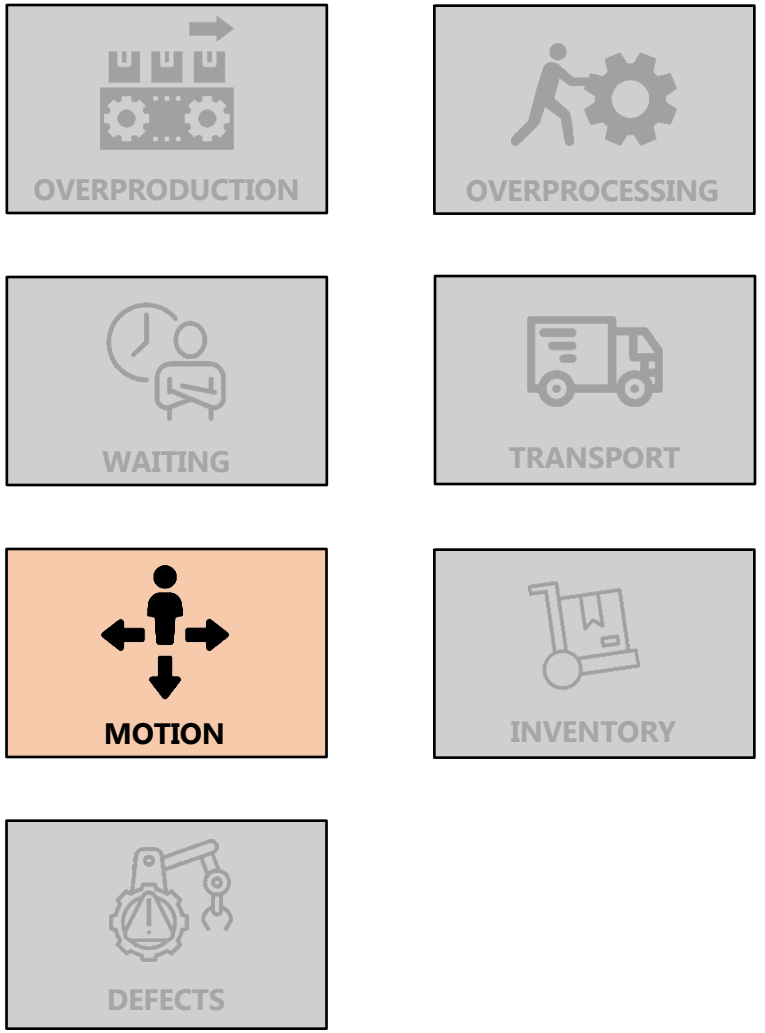
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MOTION

AIGPE

IT IS INEFFICIENT PLACEMENT OF RESOURCES CREATING MOTION.

UNNECESSARY MOVEMENT CAN BE BECAUSE OF THE CONSTANT STRETCHING AND BENDING TO PICK UP THE MATERIALS THAT ARE NEEDED FOR PRODUCTION.



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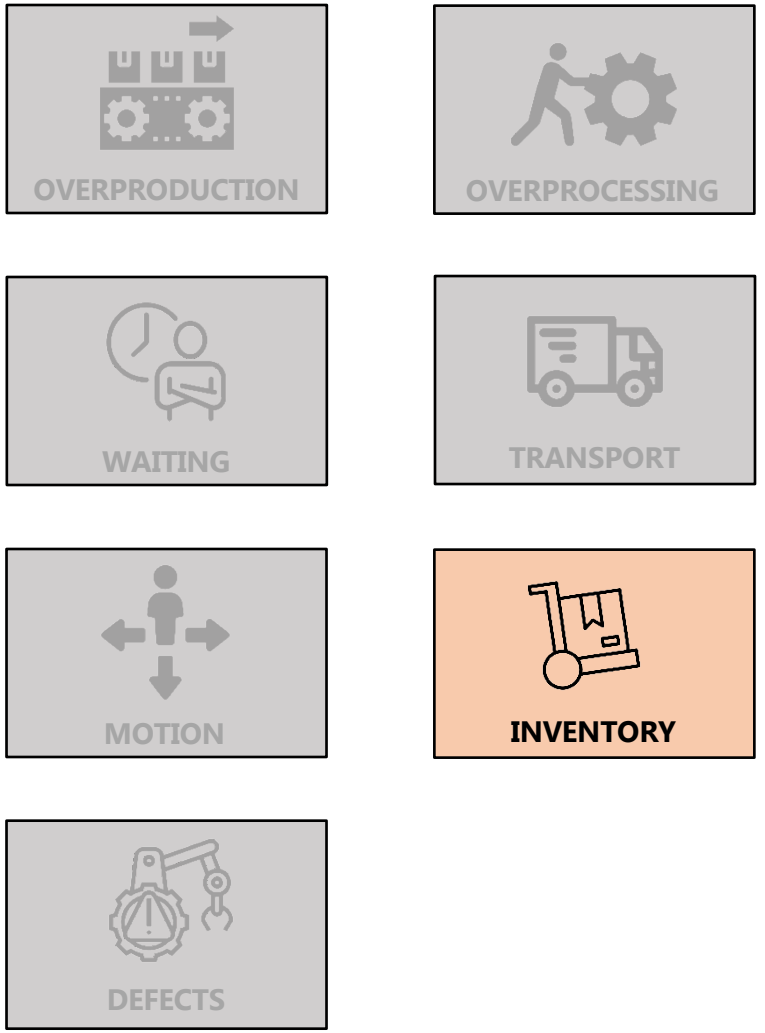
INVENTORY

AIGPE

WORK STUCK IN EMAIL IN-BOXES NOT BEING PROCESSED, IDLE FINANCIAL OR FIXED ASSETS.

WITH UNNECESSARY INVENTORY COMES INCREASED LEAD TIME, WHICH PREVENTS THE QUICK IDENTIFICATION OF PROBLEMS WITHIN THE PRODUCTION PROCESS.

THE INCREASED USE OF SPACE AND POOR COMMUNICATION AMONG STAFF IS ALSO HIDDEN BY UNNECESSARY INVENTORY.



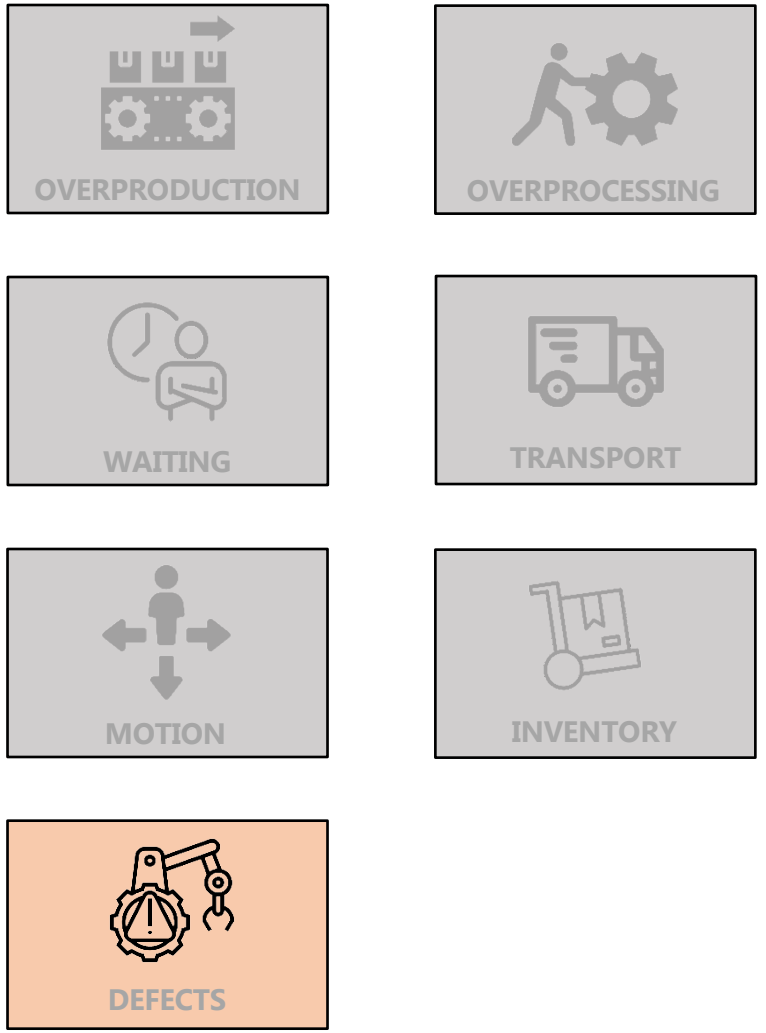
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DEFECTS

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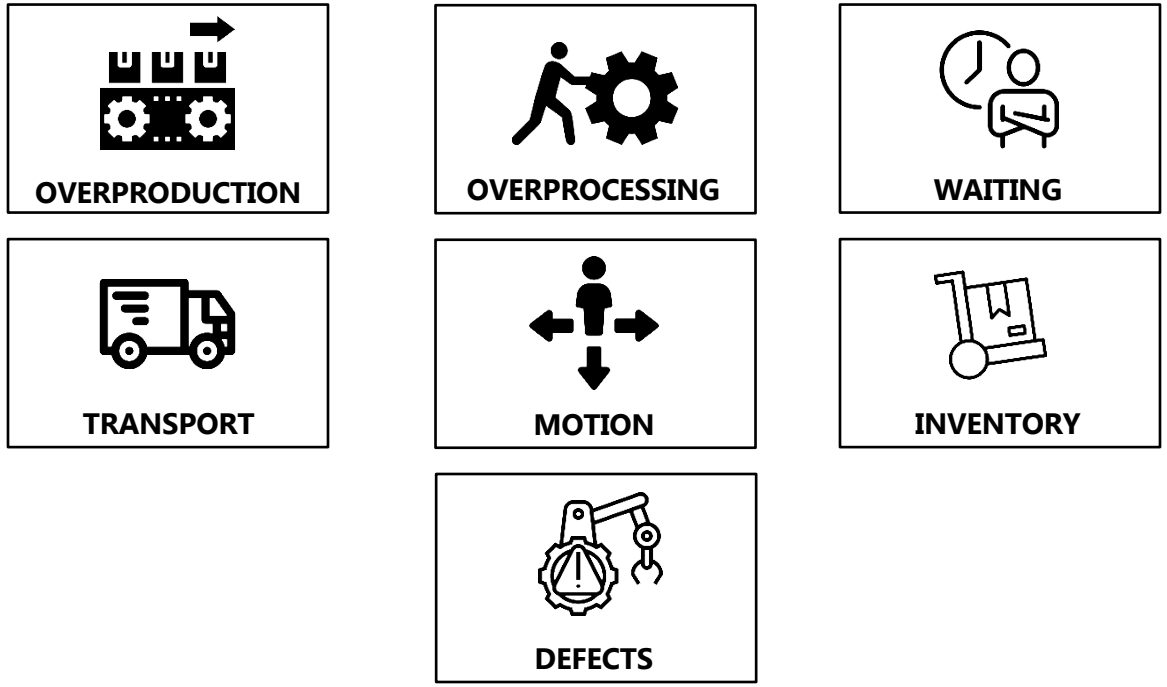
DEFECTS ARE REJECTS THAT REQUIRE ADDITIONAL TIME, RESOURCES AND MONEY TO FIX.

DEFECTS ARE ALSO CALLED REWORK AS THEY LEAD TO MORE WORK.



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WATCH FOR WASTE



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DRIVE SUCCESS TO EXCELLENCE

ACTIVITY: HOW TO EXAMINE THE CURRENT STATE VSM



ACTIVITY

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YOU ARE GIVEN A FEW QUESTIONS AND PROBLEMS TO SOLVE.

BASED ON YOUR UNDERSTANDING, ANSWER THOSE QUESTIONS.



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ACTIVITY

AIGPE

QUESTION 1:

THE TAKT TIME OF YOUR PROCESS IS 6 HOURS AND THE PROCESS CYCLE TIME IS 9.2 HOURS. THERE ARE 15 EMPLOYEES IN YOUR PROCESS. ARE YOU ABLE TO MEET THE CUSTOMER DEMAND OF 30 UNITS PER DAY?

1. **YES**

2. **NO**

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ACTIVITY

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QUESTION 2:

THE TAKT TIME OF YOUR BUSINESS PROCESS IS 20 MINUTES. THE PROCESSING TIME IS 400 MINUTES AND THE TOTAL NUMBER OF OPERATORS IS 10. ANSWER THE FOLLOWING QUESTIONS:

1. **CALCULATE THE CYCLE TIME AND DETERMINE IF THE PROCESS IS CURRENTLY ABLE TO MEET THE EXISTING CUSTOMER DEMAND?**

2. **IF THE NUMBER OF EMPLOYEES IS INCREASED FROM 10 TO 15, WILL THAT HELP IN ACHIEVING THE TAKT TIME TARGET OF 20 MINUTES?**

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ACTIVITY

AIGPE

QUESTION 3:

IF THE CUSTOMER DEMAND IS 10 PRODUCTS PER DAY AND YOU PRODUCE 15 PRODUCTS PER DAY, THIS SITUATION RESULTS IN WHICH TYPE OF WASTE?

1. **OVERPROCESSING**
2. **OVERPRODUCTION**
3. **DEFECTS**
4. **MOTION**

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ACTIVITY

AIGPE

QUESTION 4:

WHICH OF THE FOLLOWING WASTES IS CAUSED BY UNNECESSARY BENDING AND STRETCHING OF EMPLOYEES WHILE PERFORMING THEIR OFFICIAL TASKS AT WORK?

1. **DEFECTS**
2. **INVENTORY**
3. **WAITING**
4. **MOTION**

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DRIVE SUCCESS TO EXCELLENCE

ACTIVITY SOLUTION: HOW TO EXAMINE THE CURRENT STATE VSM



ACTIVITY

AIGPE

QUESTION 1:

THE TAKT TIME OF YOUR PROCESS IS 6 HOURS AND THE PROCESS/PROCESSING TIME IS 9.2 HOURS. THERE ARE 15 EMPLOYEES IN YOUR PROCESS. ARE YOU ABLE TO MEET THE CUSTOMER DEMAND OF 30 UNITS PER DAY?

1. **YES**

2. **NO**

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ACTIVITY SOLUTION

AIGPE

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ACTIVITY

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QUESTION 2: CALCULATE CYCLE TIME WITH 10 EMPLOYEES

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GIVEN TAKT TIME: 20 MINUTES

GIVEN PROCESS TIME: 400 MINUTES

GIVEN NUMBER OF OPERATORS: 10

CALCULATING THE CYCLE TIME:

$$\text{CYCLE TIME} = \frac{PROCESS\ TIME}{NUMBER\ OF\ OPERATORS} = \frac{400}{10}$$

CYCLE TIME = 40 MINUTES

THE CYCLE TIME OF 40 MINUTES INDICATES THAT ONE UNIT OF THE PRODUCT TAKES 40 MINUTES OF TIME TO PRODUCE WITH THE CURRENT WORKFORCE OF 10 EMPLOYEES.

WHEREAS, THE TAKT TIME INDICATES THAT ONE UNIT OF THE PRODUCT SHOULD TAKE 20 MINUTES OF TIME TO PRODUCE.

HENCE, GIVEN THE CALCULATIONS, THE PROCESS IS CURRENTLY UNABLE TO MEET THE EXISTING CUSTOMER DEMAND, AND THAT ANSWERS THE FIRST OF THE TWO QUESTIONS.

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QUESTION 2: CALCULATE CYCLE TIME WITH 15 EMPLOYEES

AIGPE

GIVEN TAKT TIME: 20 MINUTES

GIVEN PROCESS TIME: 400 MINUTES

GIVEN NUMBER OF OPERATORS: 15

CALCULATING THE CYCLE TIME:

$$\text{CYCLE TIME} = \frac{PROCESS\ TIME}{NUMBER\ OF\ OPERATORS} = \frac{400}{15}$$

CYCLE TIME = 26.66 MINUTES

IF THE NUMBER OF EMPLOYEES IS INCREASED FROM 10 TO 15, THE CYCLE TIME WILL BE 26.66 MINUTES.

THIS IS STILL HIGHER THAN THE TAKT TIME OF 20 MINUTES.

HENCE, EVEN IF WE INCREASE THE NUMBER OF EMPLOYEES FROM 10 TO 15, THE PROCESS WILL STILL NOT BE ABLE TO ACHIEVE THE CUSTOMER DEMAND.

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ACTIVITY

AIGPE

QUESTION 3:

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1. **OVERPROCESSING**
2. **OVERPRODUCTION**
3. **DEFECTS**
4. **MOTION**

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ACTIVITY SOLUTION

AIGPE

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ACTIVITY

AIGPE

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WHICH OF THE FOLLOWING WASTES IS CAUSED BY UNNECESSARY BENDING AND STRETCHING OF EMPLOYEES WHILE PERFORMING THEIR OFFICIAL TASKS AT WORK?

1. **DEFECTS**
2. **INVENTORY**
3. **WAITING**
4. **MOTION**

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ACTIVITY

AIGPE

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2. **INVENTORY**
3. **WAITING**
4. **MOTION**

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DRIVE SUCCESS TO EXCELLENCE

SECTION CONCLUSION:
HOW TO EXAMINE THE CURRENT STATE VSM?



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SECTION CONCLUSION

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IN THIS SECTION, YOU LEARNED HOW TO CALCULATE:

1. **CAN YOUR PROCESS MEET CUSTOMER DEMAND WITH EXISTING NO. OF OPERATORS?**
2. **CAN YOU MEET CUSTOMER DEMAND IF YOU INCREASE THE NO. OF OPERATORS?**
3. **HOW TO ANALYZE THE PROCESS BALANCE?**
4. **HOW TO WATCH FOR WASTE IN THE BUSINESS PROCESS?**

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SECTION INTRODUCTION:

HOW TO LAY A STRONG FOUNDATION TO BUILD YOUR FUTURE STATE VSM?



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VALUE STREAM MAPPING (VSM) MASTERCLASS

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Introduction to Value Stream Map (VSM)

- Course Introduction
- What is Value Stream Mapping (VSM)?
- Why is Value Stream Mapping so popular?
- Section Conclusion

Terminologies and Symbols used in VSM0

- Section Introduction
- Terminologies used in VSM
- Symbols used in VSM Process
- Material Flow Symbols
- Information Flow Symbols
- General Symbols
- Activity: Terminologies and Symbols used in VSM
- Activity Solution: Terminologies and Symbols used in VSM
- Section Conclusion

Steps to Create the Current State VSM

- Section and Case Study Introduction
- Step 1. Pick a Product Family
- Step 2. Choose Your VSM Team
- Step 3: Figure out Customer Demand
- Step 4: Map and Design the Process Flow
- Step 5: Map the Material Flow
- Step 6: Map the Information Flow
- Case Study Recap: Current State VSM
- Perform Calculations | Analyze Current State VSM
- Calculate Total Time | Lead Time | Takt Time
- Calculate Process Time | Cycle Time
- Activity: Steps to Create Current State VSM
- Activity Solution: Steps to Create Current State VSM
- Section Conclusion

Examine the Current State VSM

- Section Introduction
- Meeting Customer Demand with existing No. of Operators
- Meeting Customer Demand with increased No. of Operators
- Analyze the Process Balance
- Watch for Waste
- Section Conclusion

Strong Foundations for Future State VSM

- Section Introduction
- Change to Continuous Flow
- CONWIP
- Kanban
- Readjusting Your Process
- Alter the Batch Sizes
- Smoothening out Production
- Heijunka | Heijunka Box
- Cut back Changeover Time
- Improve Quality
- Putting it all Together

Key Questions to ask for Future State VSM

- Section Introduction
- What is the Takt Time?
- Are you building to a warehouse or directly to shipping?
- Where should Continuous Flow Control the Production?
- Controlling area of upstream processes
- Where will you implement the Pull System?
- How do you intend to level the production mix?
- What process improvements do you intend to achieve?
- What is the Implementation Plan?
- Closure of the Case Study
- Section Conclusion

VSM Guidelines

- Section Introduction
- How to get the most of your VSM?
- What are the Pitfalls to avoid?
- What are the advantages of VSM?
- What are the disadvantages of VSM
- Section Conclusion

Course Conclusion

- Course Summary
- Practice Questions
- Become a Certified VSM Practitioner!

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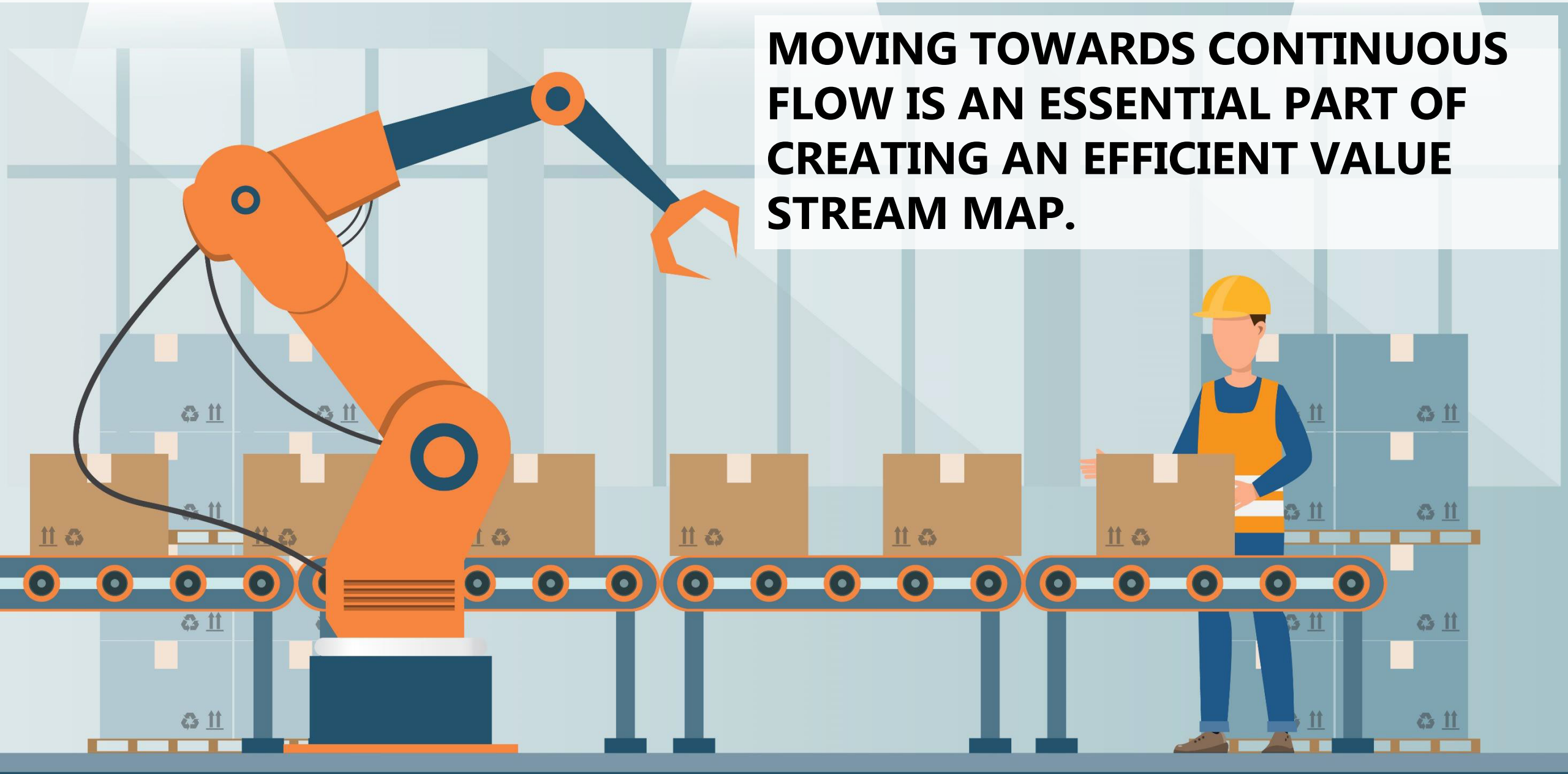
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HOW CAN YOU CHANGE YOUR EXISTING PROCESS TO CONTINUOUS FLOW?



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CHANGE TO CONTINUOUS FLOW



MOVING TOWARDS CONTINUOUS FLOW IS AN ESSENTIAL PART OF CREATING AN EFFICIENT VALUE STREAM MAP.

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WHAT IS VALUE STREAM MAP (VSM)?

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WE'LL DISCUSS THREE METHODS TO CREATE CONTINUOUS FLOW:

1. PREPARE FOR PRODUCTION TAKT VS. CONSUMER TAKT
2. CONWIP
3. KANBAN



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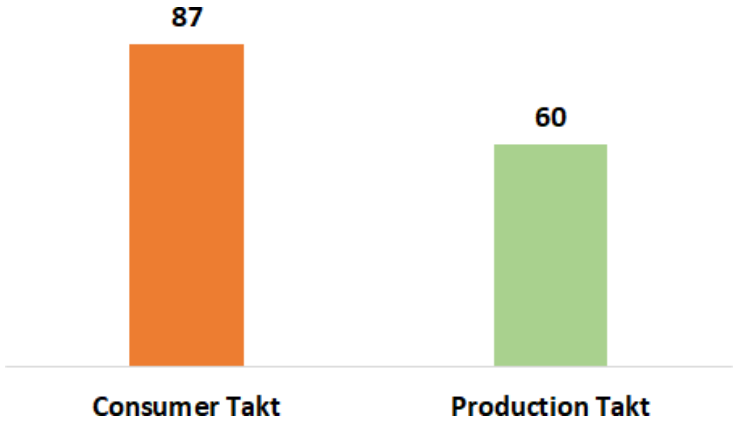
CONSUMER TAKT VS. PRODUCTION TAKT

AIGPE

RECALL THAT THE TAKT TIME IS THE TIME NEEDED TO PRODUCE ONE UNIT OF A PRODUCT TO MEET CUSTOMER REQUIREMENTS.

CONSUMER TAKT

- CONSUMER TAKT TIME IS THE TAKT TIME BASED ON CUSTOMER DEMAND.
- CONSUMER TAKT IS ALSO REFERRED TO AS IDEAL TAKT.
- IF CONSUMER TAKT TIME IS NOT MET, THE PRODUCTION TEAM MAY NOT BE ABLE TO CREATE THE PRODUCT WITHIN GIVEN TIMELINE.



PRODUCTION TAKT

- IN PRACTICAL LIFE, A BUSINESS MAY NOT RUN AT 100% EFFICIENCY. YOUR TEAMS MAY HAVE UNSCHEDULED BREAKS, URGENT MEETINGS, ABRUPT SYSTEM SHUTDOWNS, AMONG OTHERS.
- TO PREPARE FOR THIS, YOU MAY WANT TO SET UP YOUR PROCESS AT A PACE THAT IS SIGNIFICANTLY LESSER THAN THE CONSUMER TAKT TIME.
- THIS TAKT TIME IS REFERRED TO AS "PRODUCTION TAKT TIME."

- PRODUCTION TAKT TIME CAN BE DETERMINED BY USING HISTORICAL DATA, SUBJECT MATTER KNOWLEDGE, ETC.
- OPTIMAL PRODUCTION TAKT TIME ALLOWS FOR CONTINUOUS FLOW, WHICH SAFEGUARDS PRODUCTION AND AVOIDS FORMS OF WASTE LIKE OVERPRODUCTION

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WHAT IS CONWIP?



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WHAT IS CONWIP?



ABBREVIATED TERM OF “CONSTANT WORK-IN-PROGRESS”

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WHAT IS CONWIP?

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CONWIP IS ANOTHER SIMPLE WAY TO ACHIEVE CONTINUOUS FLOW. IT IS AN EXAMPLE OF A SEQUENCED PULL.



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WHAT IS CONWIP? DYNAMIC LTD. CASE STUDY

AIGPE

FOR OUR DYNAMIC LTD. CASE STUDY:

LET'S UNDERSTAND CONWIP USING THE DYNAMIC LTD. CASE STUDY

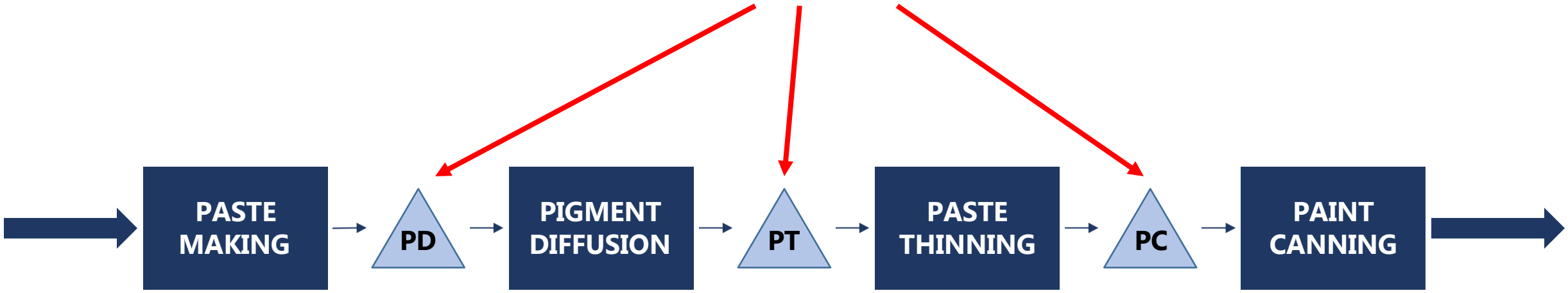


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WHAT IS CONWIP?

AIGPE

THERE ARE FOUR PROCESSES - PASTE MAKING, PIGMENT DIFFUSION, PASTE THINNING, AND PAINT CANNING. BETWEEN THESE PROCESSES, THERE ARE WORK-IN-PROGRESS JOBS. LET'S DESIGNATE THEM AS PD, PT, AND PC.





- THIS SYMBOL REPRESENTS A FREE CARD THAT DOES NOT HAVE A JOB ATTACHED TO IT.



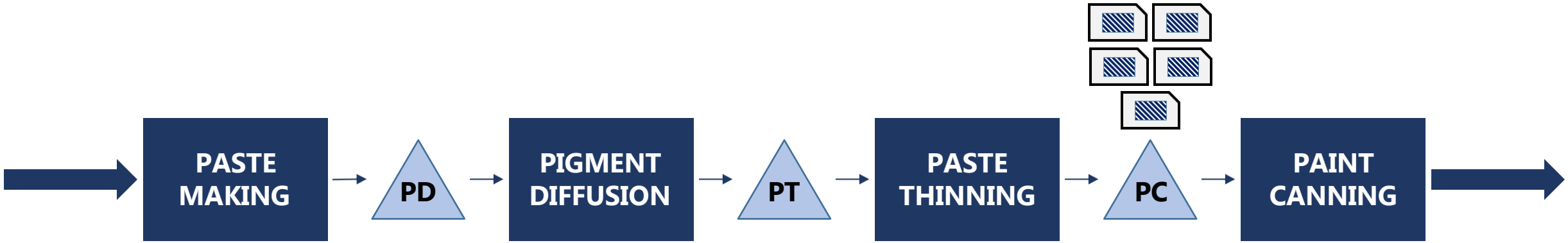
- THIS SYMBOL REPRESENTS THE CONWIP CARD ATTACHED TO A JOB.

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WHAT IS CONWIP?

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THE PAINT CANNING PROCESS TAKES THE LONGEST TIME TO FINISH. SO, WORK-IN-PROGRESS JOBS ARE THE HIGHEST HERE. LET'S ASSUME THERE ARE FIVE WORK-IN-PROGRESS JOBS. IN THE CONWIP SYSTEM, EACH OF THOSE JOBS WILL HAVE A PHYSICAL CARD ATTACHED.



TWO RULES IN THE CONWIP SYSTEM THAT YOU MUST REMEMBER:

1. **A SINGLE SET OF PHYSICAL CARDS ENFORCES ONE WORK-IN-PROGRESS LIMIT ACROSS THE WHOLE SYSTEM. IN OUR EXAMPLE, THERE IS A SINGLE SET OF PHYSICAL CARDS. THIS SET OF PHYSICAL CARDS WILL ENFORCE ONE WORK-IN-PROGRESS LIMIT ACROSS THE WHOLE PROCESS.**

2. **THE COUNT OF CARDS IS FIXED. IN OUR EXAMPLE, THE COUNT OF CARDS IS LIMITED TO FIVE.**

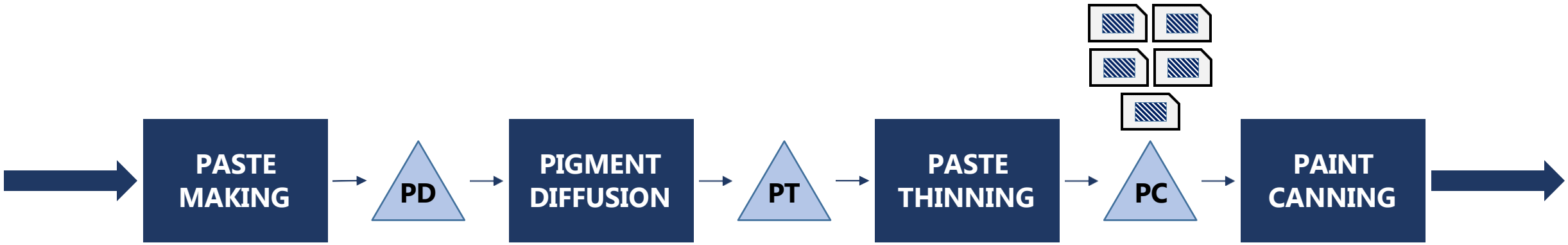
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WHAT IS CONWIP?

AIGPE

HOW DOES THE WORK FLOW THROUGH THE SYSTEM?

WHEN WORK ENTERS THE SYSTEM, A PHYSICAL CARD IS ATTACHED TO THE WORK. IF NO CARD IS AVAILABLE, THE WORK OR RAW MATERIAL MUST WAIT OUTSIDE THE SYSTEM. WHEN WORK EXITS THE SYSTEM, THE CARD IS DETACHED FROM THE WORK. THIS PHYSICAL CARD IS CARRIED BACK TO THE START OF THE SYSTEM FOR THE NEXT JOB TO ENTER.



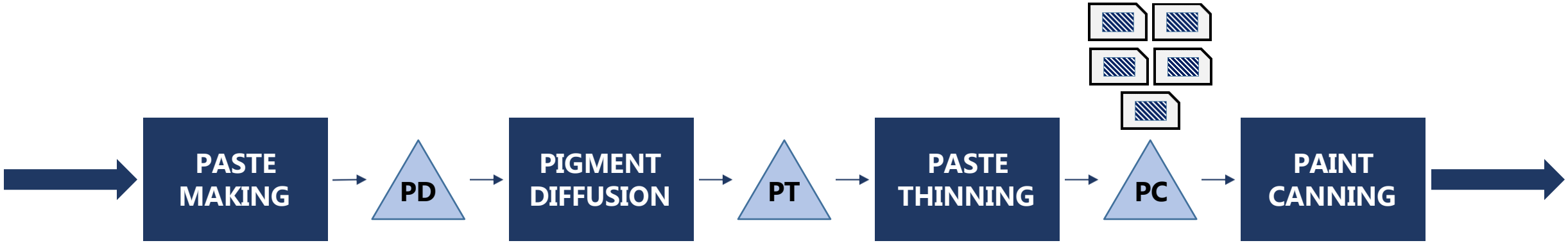
LET'S PUT THIS IN ACTION THROUGH THE EXAMPLE ON THE NEXT PAGE.

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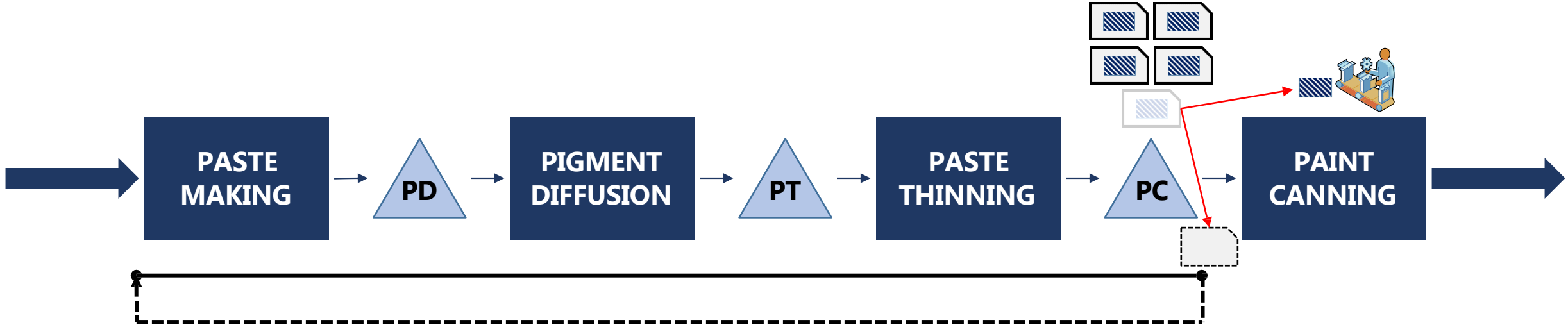
WHAT IS CONWIP?

AIGPE

1. **THE ENTIRE FLOW IS CURRENTLY IN ITS RESTING STATE, AND THERE ARE FIVE WORK-IN-PROGRESS JOBS WITH ATTACHED CONWIP CARDS AT PC.**



2. **THE ACTION STARTS WHEN THE OPERATOR IN THE PAINT CANNING PROCESS FINISHES HIS JOB. THE OPERATOR PICKS UP ONE OF THE WORK-IN-PROGRESS JOBS AT PC. THIS FREES THE PHYSICAL CONWIP CARD**

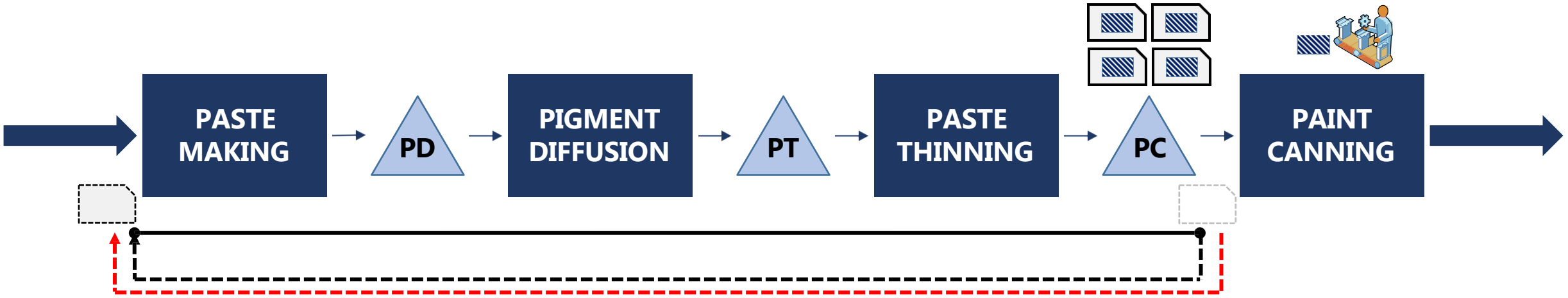


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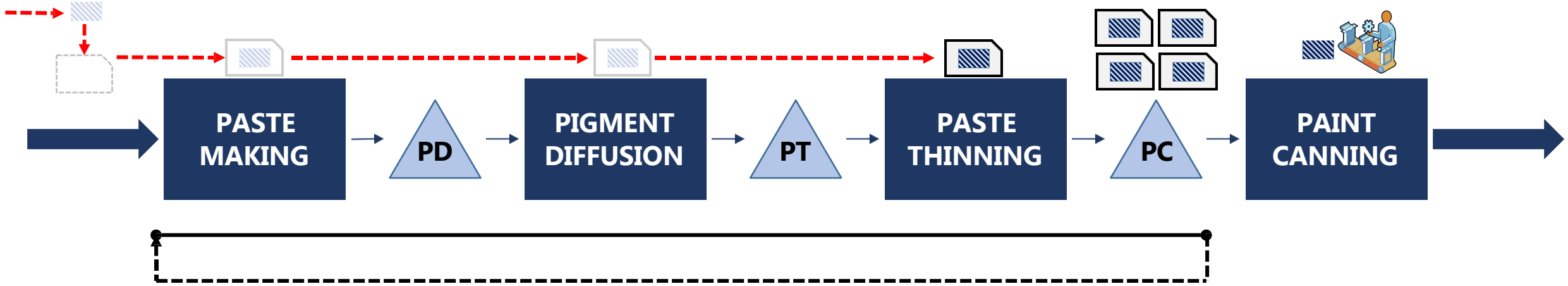
WHAT IS CONWIP?

AIGPE

3. THE FREE CONWIP CARD TRAVELS BACK TO THE SYSTEM'S ENTRY POINT



4. A NEW JOB GETS ATTACHED TO THIS FREE CONWIP CARD AND PROGRESSES THROUGH ALL THE PROCESSES OF PASTE MAKING, PIGMENT DIFFUSION, AND PASTE THINNING.

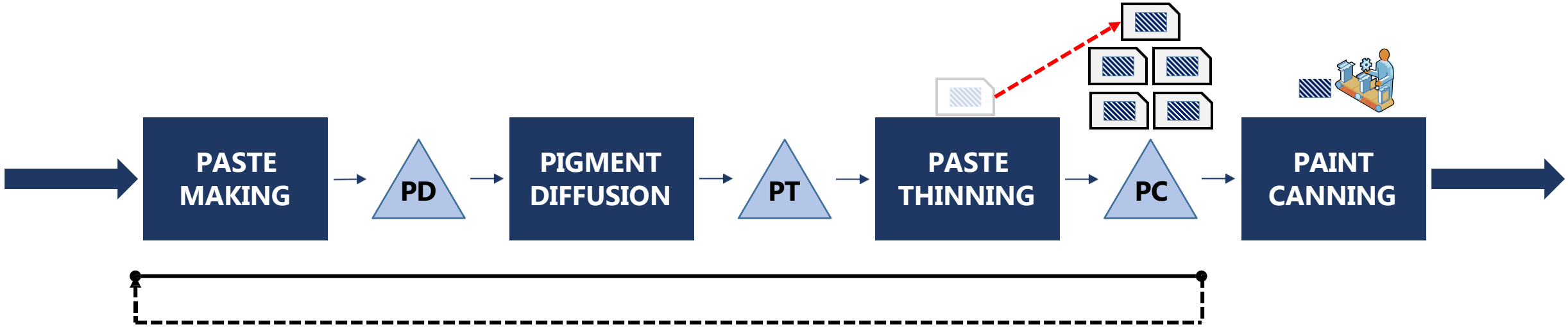


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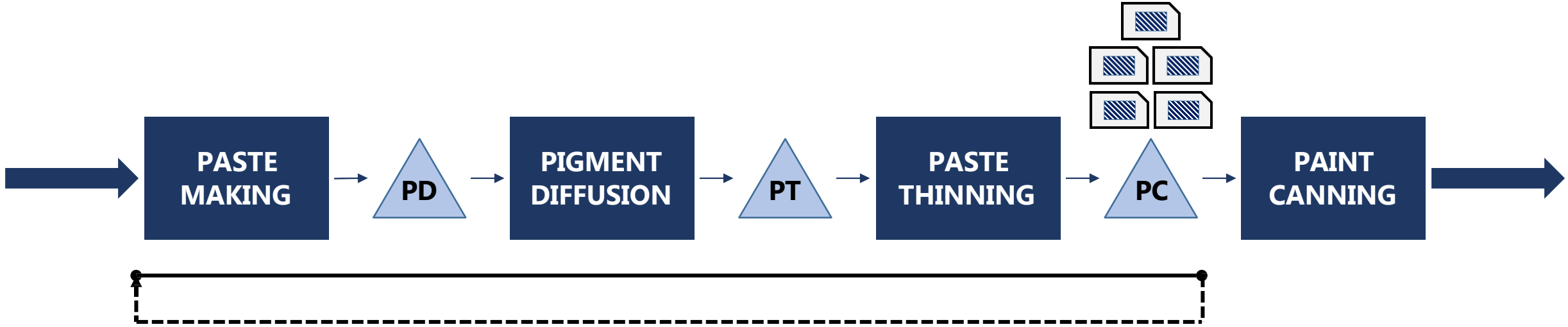
WHAT IS CONWIP?

AIGPE

5. WHEN THIS NEW JOB IS COMPLETE AT PASTE THINNING, IT JOINS THE INVENTORY AT PC.



6. AND THE SYSTEM RETURNS TO ITS RESTING STATE.



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WHAT IS CONWIP?

AIGPE

A CONWIP SYSTEM IS ONE OF THE SIMPLEST EXAMPLES OF A PULL SYSTEM.

IT IS A METHOD THAT HELPS SUSTAIN WORK-IN-PROGRESS AT A CONSTANT LEVEL BY DEMANDING A SIGN FROM THE PROCESS THAT INITIATES ACTION.

THE SIGN IS SENT TO THE BEGINNING OF THE OPERATING PROCESS TO BEGIN THE PRODUCTION OF A NEW PRODUCT.

THE INITIATION ACTION CAN EITHER BE AT THE BOTTLENECK PROCESS OR THE LAST OPERATION.

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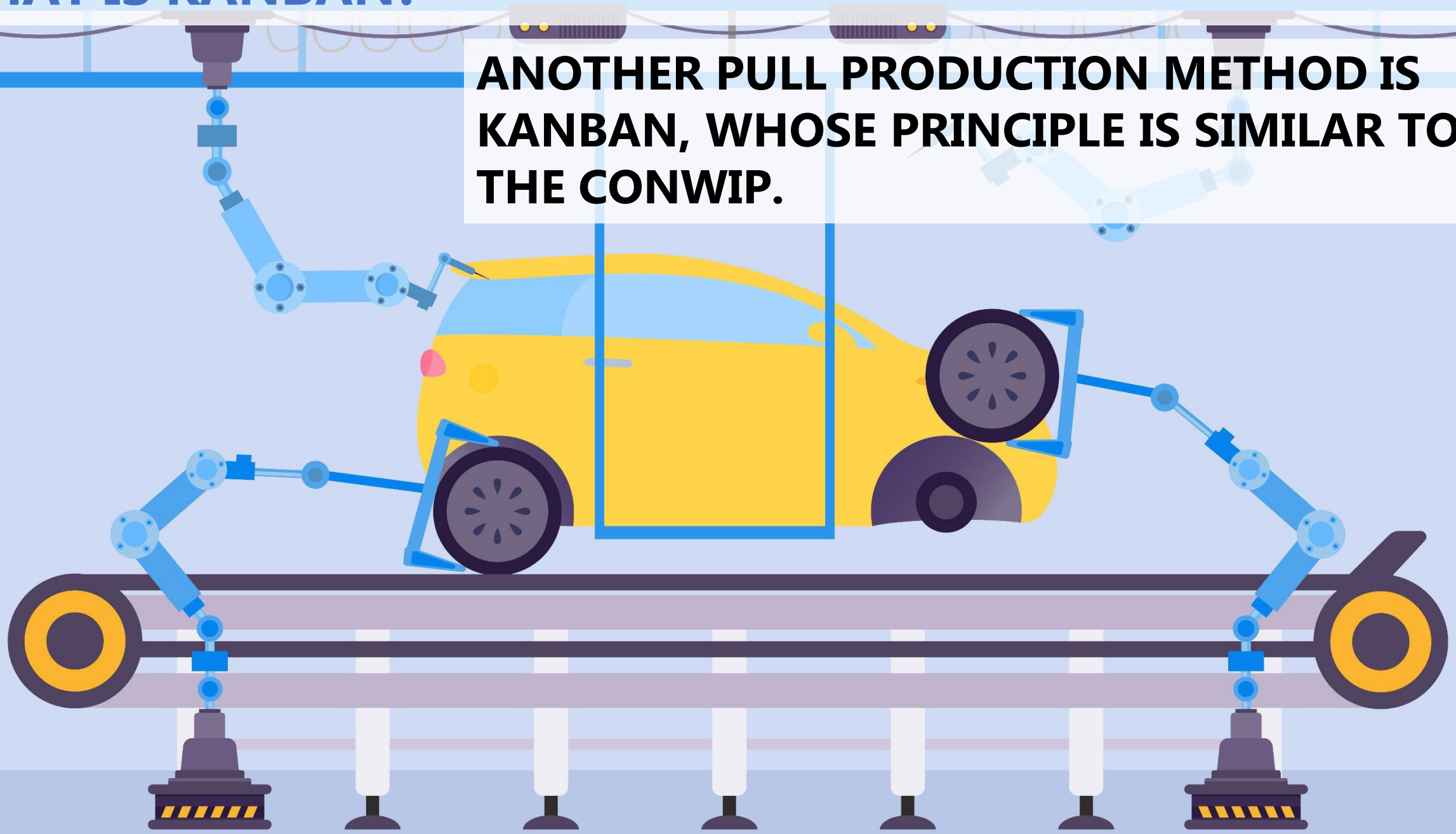
WHAT IS KANBAN?



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WHAT IS KANBAN?

ANOTHER PULL PRODUCTION METHOD IS KANBAN, WHOSE PRINCIPLE IS SIMILAR TO THE CONWIP.



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WHAT IS KANBAN? DYNAMIC LTD. CASE STUDY

AIGPE

FOR OUR DYNAMIC LTD. CASE STUDY:

LET'S UNDERSTAND KANBAN USING THE DYNAMIC LTD. CASE STUDY

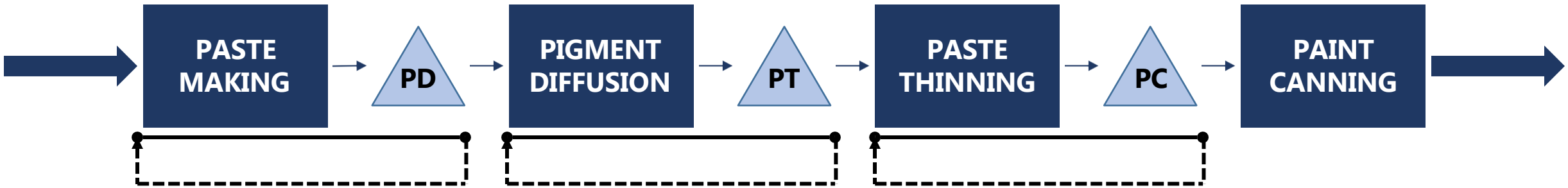


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WHAT IS KANBAN?

AIGPE

THIS IS THE KANBAN SYSTEM. IN THIS SYSTEM, THE FUNCTION OF THE CARDS IS SIMILAR TO WHAT WE HAD SEEN IN CONWIP. BUT NOW THERE ARE MULTIPLE CARD LOOPS. EACH CARD LOOP IS SEPARATELY DEDICATED TO WORK CENTERS PD, PT, AND PC, UNLIKE CONWIP, WHERE THERE WAS ONLY ONE SET OF CARDS AT THE PROCESS THAT WAS A BOTTLENECK IN THE SYSTEM.





- THIS SYMBOL REPRESENTS A FREE CARD THAT DOES NOT HAVE A JOB ATTACHED TO IT.



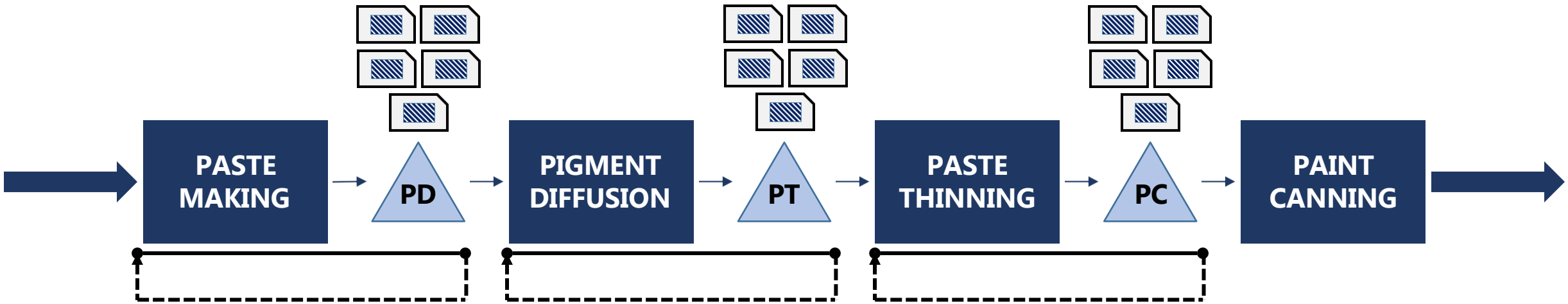
- THIS SYMBOL REPRESENTS THE CONWIP CARD ATTACHED TO A JOB.

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WHAT IS KANBAN?

AIGPE

BECAUSE OF THE SEPARATE CARD LOOPS, THE AT-REST INVENTORY, WHICH WAS UNEVEN AND ACCUMULATED AT THE END IN CONWIP, IS EVENLY DISTRIBUTED BETWEEN ALL WORK CENTERS.



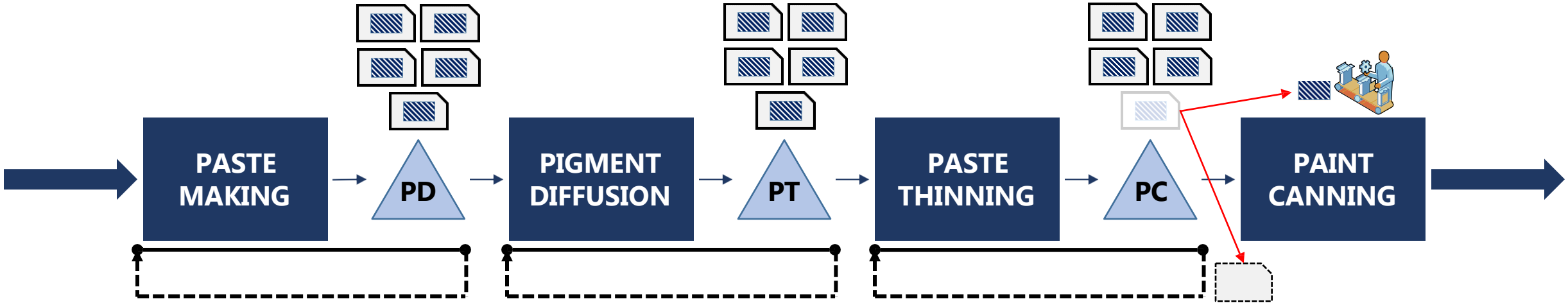
FOR COMPARISON TO THE CONWIP SYSTEM, WE NOW HAVE THREE SEPARATE CARD SETS OF FIVE CARDS EACH. AT REST, THERE ARE FIVE UNITS OF INVENTORY AT PD, PT, AND PC.

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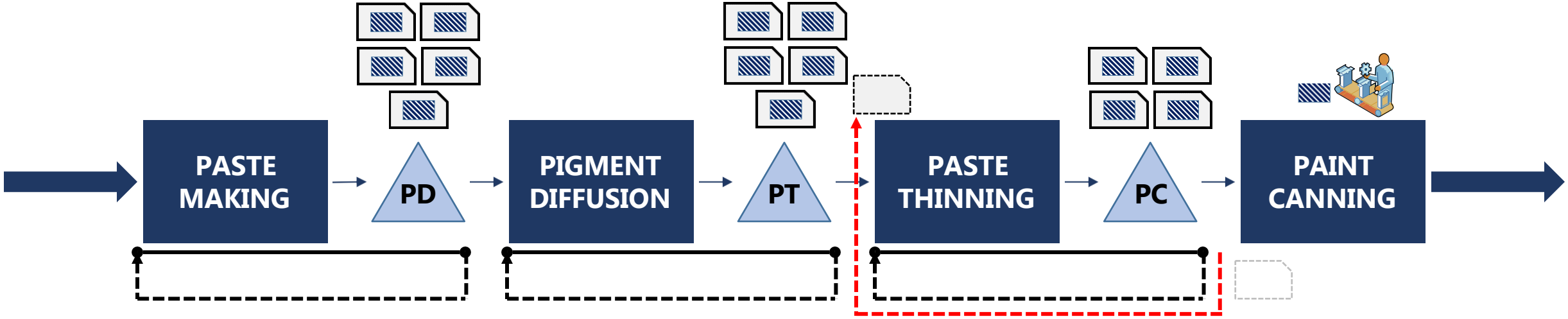
WHAT IS KANBAN?

AIGPE

1. NOW THE ACTION IS INITIATED AT PAINT CANNING, WHERE WE SEE CONSUMPTION OF INVENTORY AT PC.



2. BUT NOW THE DETACHED CARD TRAVELS BACK TO WORK CENTER PASTE THINNING.

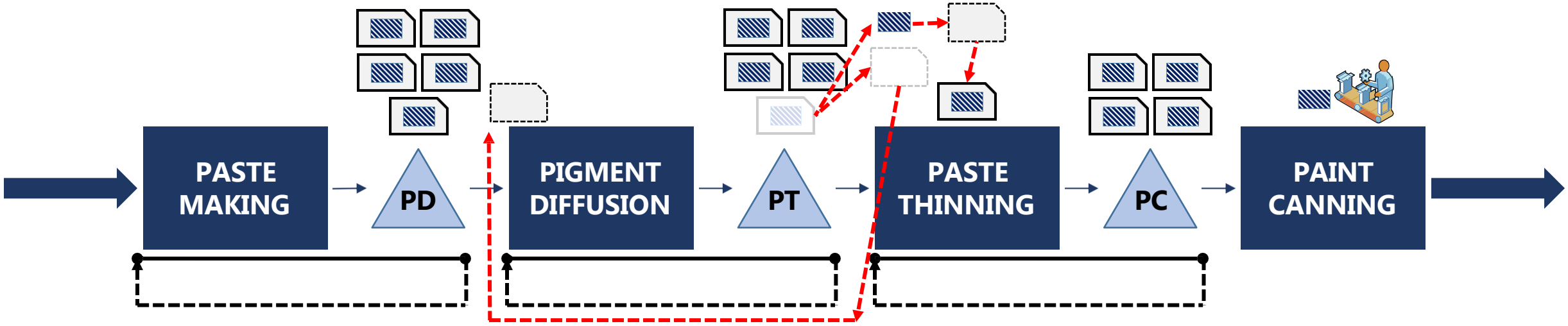


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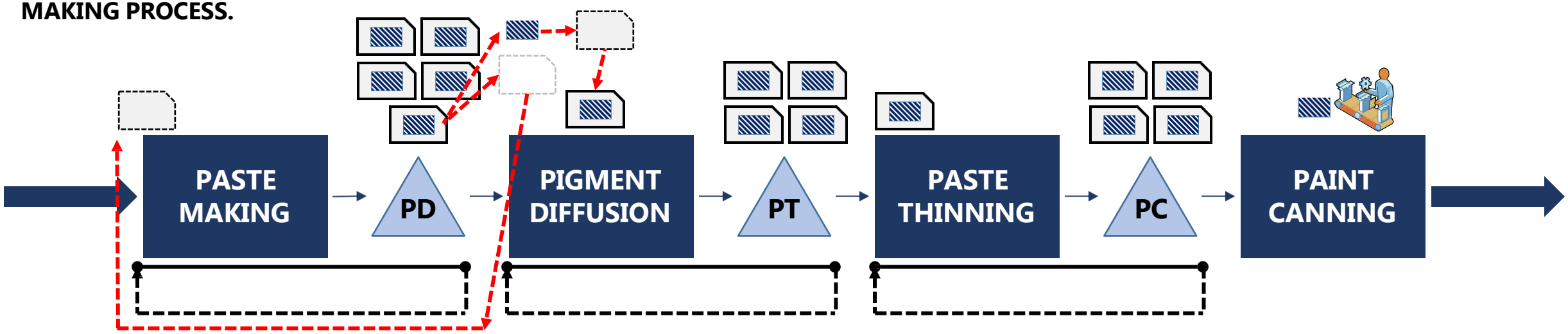
WHAT IS KANBAN?

AIGPE

3. **WHERE IT ATTACHES TO WORK-IN-PROGRESS AT PT. AT THE MOMENT OF ATTACHMENT, THIS JOB RELEASES ITS PT LOOP CARD. THE PT LOOP CARD IS NOW FREE TO TRAVEL BACK TO THE WORK CENTER PIGMENT DIFFUSION.**



4. **WHERE IT ATTACHES TO WORK-IN-PROGRESS AT PD. AT THE MOMENT OF ATTACHMENT, THIS JOB RELEASES ITS PD LOOP CARD. THIS CARD IS NOW FREE TO TRAVEL BACK TO THE BEGINNING OF THE SYSTEM AND ATTACH ITSELF TO NEW WORK ENTERING THE PASTE MAKING PROCESS.**

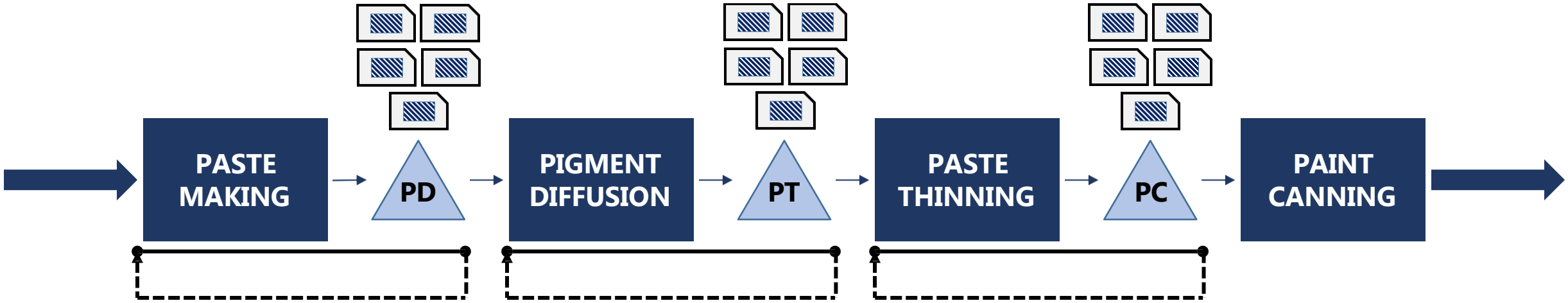


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WHAT IS KANBAN?

AIGPE

5. WHEN THE WORK ON EACH JOB IS COMPLETE, THE SYSTEM RETURNS TO ITS STATE OF REST WITH FIVE PRODUCTS IN EACH FINISHED WORK INVENTORY.



WHAT MAKES KANBAN DIFFERENT IS THAT RATHER THAN THE SIGN COMING FROM THE BOTTLENECK PROCESS, IT IS SENT FROM ONE OPERATION TO THE NEXT.

THIS IS MADE VISIBLE BY THE KANBAN ARROWS AND SYMBOLS AS THEY MOVE UPSTREAM BETWEEN EACH OPERATION.

KANBAN EXISTS IN VARIOUS VARIATIONS.

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HOW TO READJUST THE PROCESS?



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HOW TO READJUST THE PROCESS

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ADJUSTING ONE SECTION OF THE FUTURE STATE MAP CREATES AN IMBALANCE IN OTHER AREAS.



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HOW TO READJUST THE PROCESS

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THE IMBALANCE THEN CAUSES EITHER IDLE TIME OR OVERPRODUCTION.



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HOW TO READJUST THE PROCESS

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IF YOU FIND ANY EXTRA TIME FOR THE EMPLOYEES, IT CAN BE BETTER SPENT AS RECOVERY TIME, PERFORMING INVENTORY, MAINTENANCE, OR ASSISTING OTHERS.



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WHAT IS ALTERING THE BATCH SIZES?



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THE PROS AND CONS OF PRODUCING IN BATCH SIZES

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PROS

- BATCH SIZES INCREASE THE TOTAL CAPACITY OF OPERATIONS IN AREAS WITH CHANGEOVER.
- INCREASES PRODUCTION BECAUSE BATCH SIZES DO NOT NEED PHYSICAL CHANGEOVERS.

CONS

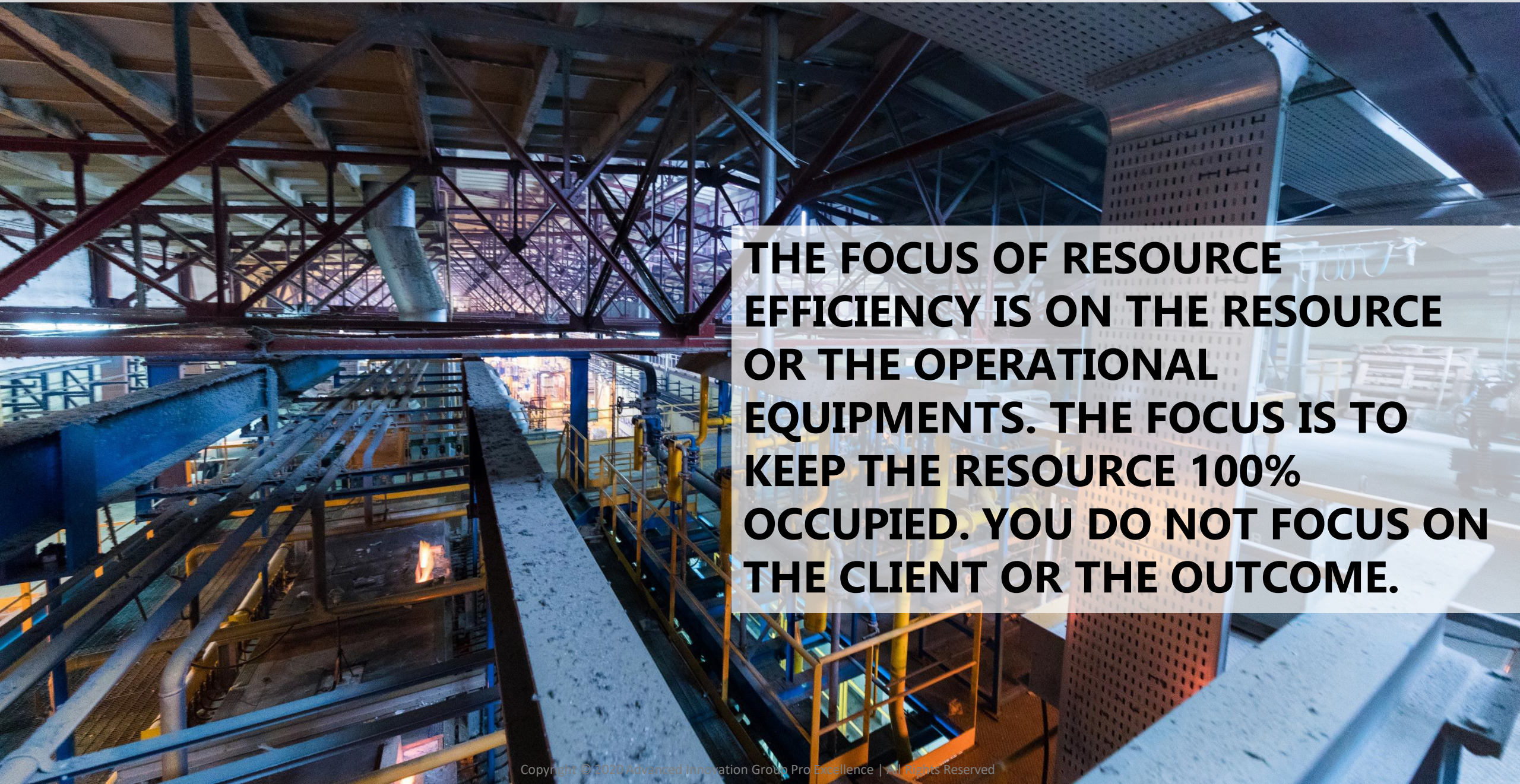
- BATCH SIZES CAN LEAD TO MENTAL CHANGEOVERS THAT AFFECT EFFICIENCY NEGATIVELY.
- LITTLE OR NO FLEXIBILITY RESULTS IN INCREASED LEAD TIME.
- INCREASE IN INVENTORY LEADS TO QUEUES, WHICH LEAD TO EVEN LONGER LEAD TIMES.

TO FIGURE OUT THE RIGHT BATCH PRODUCTION, WE NEED TO DISCUSS RESOURCE EFFICIENCY AND FLOW EFFICIENCY.

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RESOURCE EFFICIENCY

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THE FOCUS OF RESOURCE EFFICIENCY IS ON THE RESOURCE OR THE OPERATIONAL EQUIPMENTS. THE FOCUS IS TO KEEP THE RESOURCE 100% OCCUPIED. YOU DO NOT FOCUS ON THE CLIENT OR THE OUTCOME.

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FLOW EFFICIENCY

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WITH FLOW EFFICIENCY, YOU MAY HAVE A NUMBER OF PEOPLE OR PROCESSES WORKING ON ONE OUTCOME AND THE IDEA IS TO ACHIEVE THE OUTCOME IN THE COMMITED DELIVERY TIME WINDOW.

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HOW TO SMOOTHEN OUT PRODUCTION?

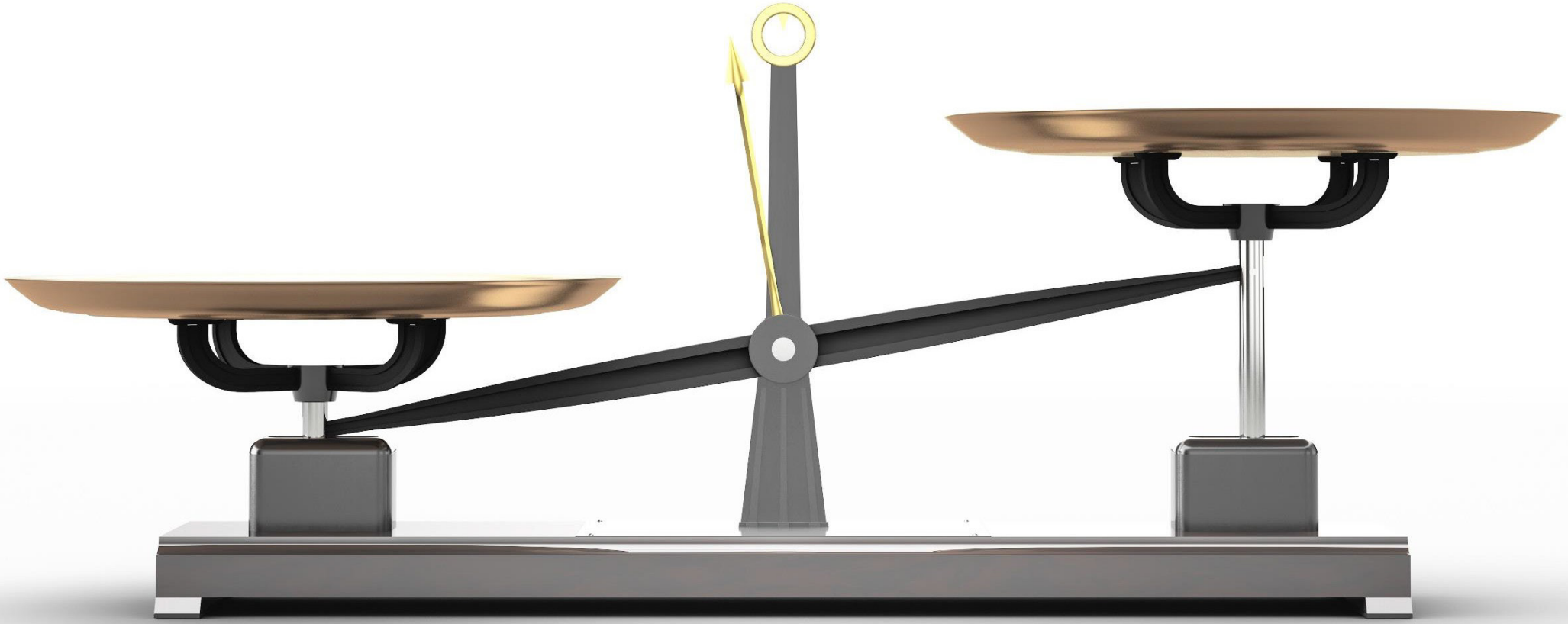


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HOW TO SMOOTHEN OUT PRODUCTION?

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AN UNEVEN AND JERKY PRODUCTION PROCESS OCCURS AS A RESULT OF A HUGE TIME DIFFERENCE IN THE CYCLE TIME BETWEEN OPERATION PROCESSES.



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HOW TO SMOOTHEN OUT PRODUCTION?

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THE APPLICATION OF THE PRODUCTION LEVELING PRINCIPLE OR THE HEIJUNKA FIXES THAT PROBLEM EASILY.

HEIJUNKA IS A JAPANESE TERM OF LOAD LEVELING.

ONE OF THE MOST COMMON WAYS TO USE HEIJUNKA IS: LEVELLING BY VOLUME



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HEIJUNKA: LEVELING BY VOLUME

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LEVELING BY VOLUME IS ALL ABOUT LEVELING THE PRODUCTION ACTIVITIES BY THE AVERAGE VOLUME OF THE ORDERS.



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HEIJUNKA: LEVELLING BY VOLUME? DYNAMIC LTD. CASE STUDY

AIGPE

FOR OUR DYNAMIC LTD. CASE STUDY:

LET'S UNDERSTAND HEIJUNKA: LEVELLING BY VOLUME USING THE DYNAMIC LTD. CASE STUDY



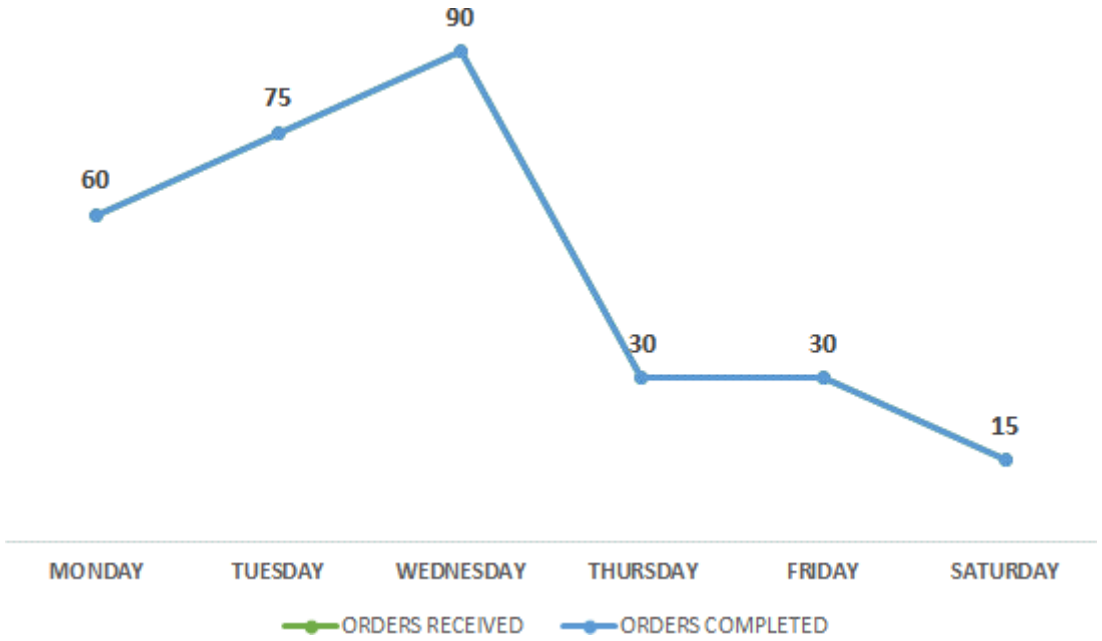
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HEIJUNKA: LEVELING BY VOLUME

AIGPE

THE CUSTOMER DEMAND IS 300 PAINT CANS PER WEEK. THE CURRENT ORDER COMPLETION SCHEDULE IS BASED ON INCOMING VOLUME EACH DAY.

DAY	ORDERS RECEIVED	ORDERS COMPLETED
MONDAY	60	60
TUESDAY	75	75
WEDNESDAY	90	90
THURSDAY	30	30
FRIDAY	30	30
SATURDAY	15	15



AS YOU CAN SEE IN THE CHART, THE CURRENT ORDER COMPLETION SCHEDULE IS UNEVEN AND JERKY. ON DAYS SUCH AS MONDAY, TUESDAY, AND WEDNESDAY, EMPLOYEES ARE BURDENED WITH TIGHT SCHEDULE TO COMPLETE ORDERS. AND ON THURSDAY, FRIDAY, AND SATURDAY, THERE IS A HIGH IDLE TIME DUE TO LOWER NUMBER OF ORDERS. THE GREEN LINE OF ORDERS RECEIVED IS NOT VISIBLE BECAUSE THE BLUE ORDERS COMPLETED LINE OVERLAPS THE GREEN LINE.

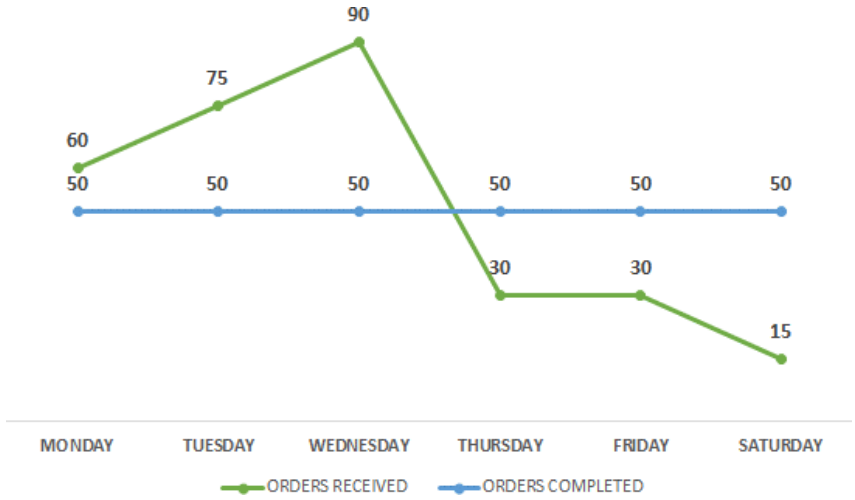
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HEIJUNKA: LEVELING BY VOLUME

AIGPE

<u>USING HEIJUNKA TO LEVEL THE WORKLOAD BY VOLUME:</u> YOU CAN PRODUCE AN AVERAGE OF 50 UNITS OF PAINT CANS PER DAY. AND YOU CAN STORE AN INVENTORY OF 50 PAINT CANS PER DAY.

DAY	ORDERS RECEIVED	ORDERS COMPLETED	BORROWED FROM INVENTORY	INVENTORY LEFT BY THE END OF THE DAY
MONDAY	60	50	10	40
TUESDAY	75	50	25	15
WEDNESDAY	90	50	15	-25
THURSDAY	30	50	-20	-5
FRIDAY	30	50	-20	15
SATURDAY	15	50	-35	50



BY END OF SATURDAY, YOUR INVENTORY OF FRIDAY AND SATURDAY WILL ADD UP TO SERVE THE INVENTORY REQUIREMENT FOR MONDAY. THUS, USING THE HEIJUNKA, LOAD LEVELING BY VOLUME, YOU WILL HAVE A STABLE PRODUCTION OF 50 PAINT CANS PER DAY. THIS WILL ALLOW YOU TO SMOOTHEN OUT THE PRODUCTION PROCESS AND EASILY MEET CUSTOMER DEMAND.

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WHAT IS HEIJUNKA BOX?

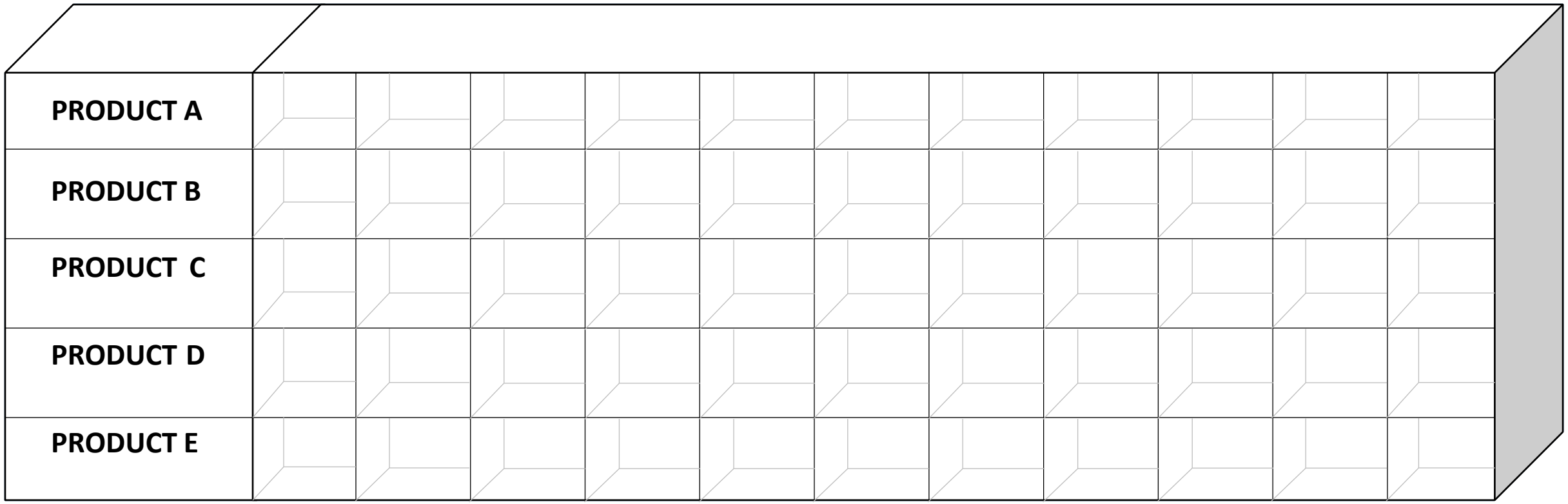


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WHAT IS HEIJUNKA BOX?

AIGPE

HEIJUNKA BOX IS ANOTHER TECHNIQUE FOR LOAD LEVELLING AND WORKS BEST WHEN YOU HAVE MULTIPLE PRODUCTS IN A PRODUCT FAMILY.



IN THIS, YOU WILL HAVE A PHYSICAL WOODEN OR PLASTIC BOX WITH PIGEON-HOLE KIND OF SHELVES IN IT. THE SHELF WOULD LOOK SOMETHING LIKE THE ABOVE DIAGRAM.

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WHAT IS HEIJUNKA BOX?

AIGPE

THE DIFFERENT ROWS INDICATE THE DIFFERENT TYPES OF PRODUCTS IN A PRODUCT FAMILY. IN AN EXAMPLE, LET'S CALL THEM PRODUCTS A TO E.

Paint Product A						
Paint Product B						
Paint Product C						
Paint Product D						
Paint Product E						

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WHAT IS HEIJUNKA BOX?

AIGPE

IN OUR EXAMPLE, WE ARE CONSIDERING A SIX-DAY WORKING IN THIS MANUFACTURING PROCESS. HENCE, WE HAVE SIX COLUMNS DESIGNATED AS MONDAY THROUGH SATURDAY.

	Mon	Tue	Wed	Thu	Fri	Sat
Paint Product A						
Paint Product B						
Paint Product C						
Paint Product D						
Paint Product E						

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WHAT IS HEIJUNKA BOX?

AIGPE

EACH PIGEON-HOLE BOX HAS A DESIGNATED CAPACITY. E.G., THE TOP LEFT CORNER OF THE STRUCTURE IS FOR PRODUCT A TO BE PRODUCED ON MONDAY. THE NEXT ONE BELOW THAT IS FOR PRODUCT B TO BE PRODUCED ON MONDAY.

	Mon	Tue	Wed	Thu	Fri	Sat
Paint Product A	Capacity: 10	Capacity: 12	Capacity: 15	Capacity: 7	Capacity: 9	Capacity: 6
Paint Product B	Capacity: 5	Capacity: 9	Capacity: 8	Capacity: 4	Capacity: 2	Capacity: 1
Paint Product C	Capacity: 12	Capacity: 10	Capacity: 12	Capacity: 15	Capacity: 7	Capacity: 9
Paint Product D	Capacity: 3	Capacity: 7	Capacity: 6	Capacity: 2	Capacity: 1	Capacity: 1
Paint Product E	Capacity: 6	Capacity: 10	Capacity: 12	Capacity: 5	Capacity: 4	Capacity: 3

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WHAT IS HEIJUNKA BOX?

AIGPE

THE CAPACITY IS DETERMINED BASED ON THE AVAILABILITY OF RESOURCES SUCH AS WORKFORCE, RAW MATERIAL INVENTORY, AS WELL AS UPTIME OF THE MACHINERY.

	Mon	Tue	Wed	Thu	Fri	Sat
Paint Product A	Capacity: 10 	Capacity: 12 	Capacity: 15 	Capacity: 7 	Capacity: 9 	Capacity: 6 
Paint Product B	Capacity: 5 	Capacity: 9 	Capacity: 8 	Capacity: 4 	Capacity: 2 	Capacity: 1 
Paint Product C	Capacity: 12 	Capacity: 10 	Capacity: 12 	Capacity: 15 	Capacity: 7 	Capacity: 9 
Paint Product D	Capacity: 3 	Capacity: 7 	Capacity: 6 	Capacity: 2 	Capacity: 1 	Capacity: 1 
Paint Product E	Capacity: 6 	Capacity: 10 	Capacity: 12 	Capacity: 5 	Capacity: 4 	Capacity: 3 

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WHAT IS HEIJUNKA BOX?

AIGPE

EACH BOX ALSO HOLDS A KANBAN CARD. THE CARD HAS TWO COLORS, GREEN AND RED.

Available Capacity

No Capacity

	Mon	Tue	Wed	Thu	Fri	Sat
Paint Product A	Capacity: 10	Capacity: 12	Capacity: 15	Capacity: 7	Capacity: 9	Capacity: 6
Paint Product B	Capacity: 5	Capacity: 9	Capacity: 8	Capacity: 4	Capacity: 2	Capacity: 1
Paint Product C	Capacity: 12	Capacity: 10	Capacity: 12	Capacity: 15	Capacity: 7	Capacity: 9
Paint Product D	Capacity: 3	Capacity: 7	Capacity: 6	Capacity: 2	Capacity: 1	Capacity: 1
Paint Product E	Capacity: 6	Capacity: 10	Capacity: 12	Capacity: 5	Capacity: 4	Capacity: 3

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ADDING THE GREEN KANBAN CARD

AIGPE

IF A GREEN KANBAN CARD IS PLACED INSIDE THE BOX, IT INDICATES THAT THE PRODUCTION HAS NOT REACHED ITS MAXIMUM CAPACITY LEVEL, AND MORE PRODUCTS OF THAT TYPE CAN BE REQUESTED TO BE PRODUCED ON THAT DAY.

	Mon	Tue	Wed	Thu	Fri	Sat
Paint Product A	Capacity: 10	Capacity: 12	Capacity: 15	Capacity: 7	Capacity: 9	Capacity: 6
Paint Product B	Capacity: 5	Capacity: 9	Capacity: 8	Capacity: 4	Capacity: 2	Capacity: 1
Paint Product C	Capacity: 12	Capacity: 10	Capacity: 12	Capacity: 15	Capacity: 7	Capacity: 9
Paint Product D	Capacity: 3	Capacity: 7	Capacity: 6	Capacity: 2	Capacity: 1	Capacity: 1
Paint Product E	Capacity: 6	Capacity: 10	Capacity: 12	Capacity: 5	Capacity: 4	Capacity: 3

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ADDING THE RED KANBAN CARD

AIGPE

IF A RED KANBAN CARD IS PLACED INSIDE THE BOX, IT INDICATES THAT THERE IS NO CAPACITY ANYMORE FOR THAT PRODUCT TYPE ON THAT DAY.

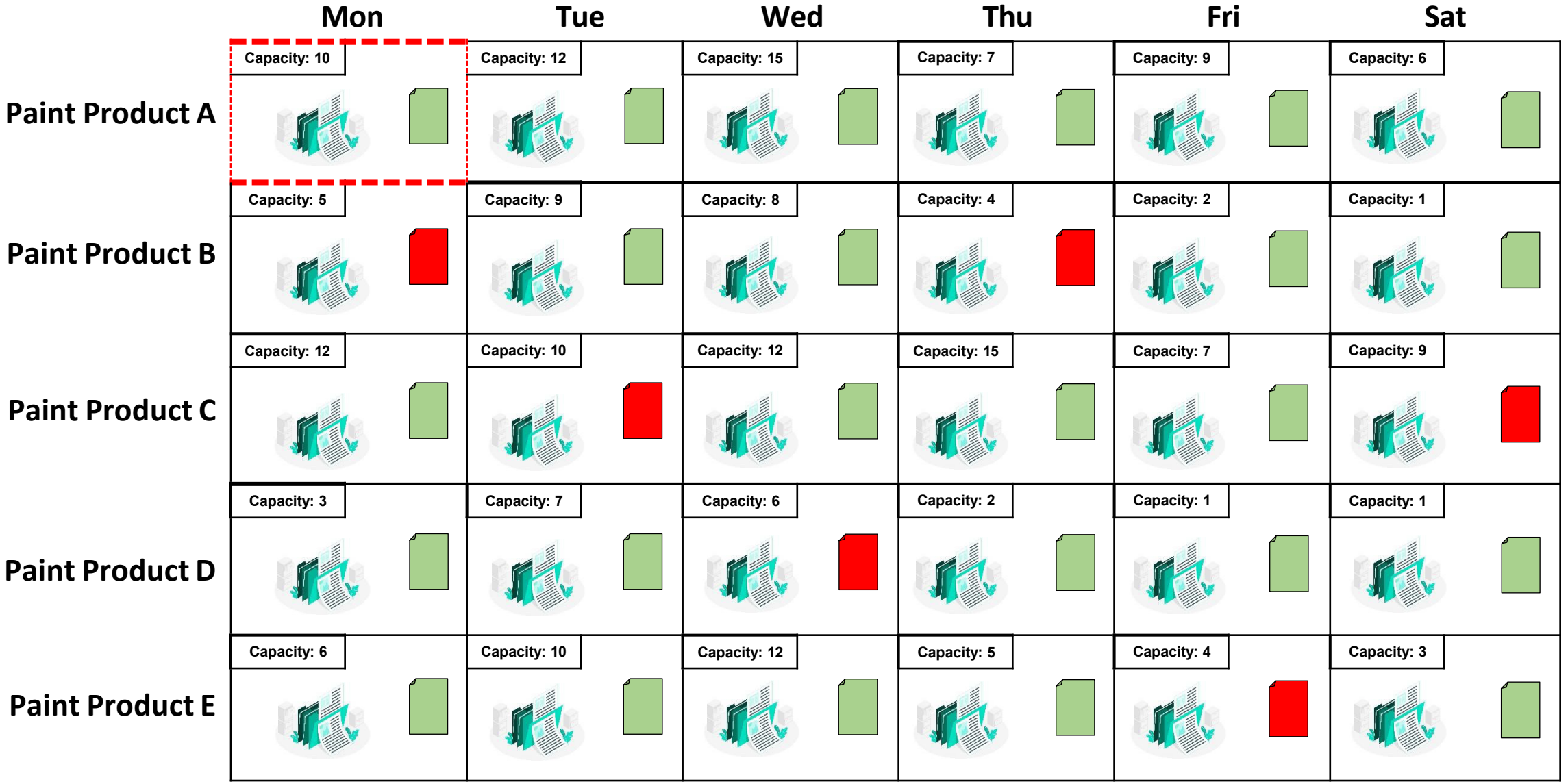
	Mon	Tue	Wed	Thu	Fri	Sat
Paint Product A	Capacity: 10	Capacity: 12	Capacity: 15	Capacity: 7	Capacity: 9	Capacity: 6
Paint Product B	Capacity: 5	Capacity: 9	Capacity: 8	Capacity: 4	Capacity: 2	Capacity: 1
Paint Product C	Capacity: 12	Capacity: 10	Capacity: 12	Capacity: 15	Capacity: 7	Capacity: 9
Paint Product D	Capacity: 3	Capacity: 7	Capacity: 6	Capacity: 2	Capacity: 1	Capacity: 1
Paint Product E	Capacity: 6	Capacity: 10	Capacity: 12	Capacity: 5	Capacity: 4	Capacity: 3

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EXAMPLE: ADDING THE GREEN KANBAN CARD

AIGPE

THUS, IN THE EXAMPLE, ON MONDAY, FOR PAINT PRODUCT A, THERE IS A GREEN KANBAN CARD. THIS INDICATES, MAXIMUM CAPACITY HAS NOT BEEN REACHED, AND MORE PRODUCTS OF PAINT A CAN BE PRODUCED ON MONDAY.

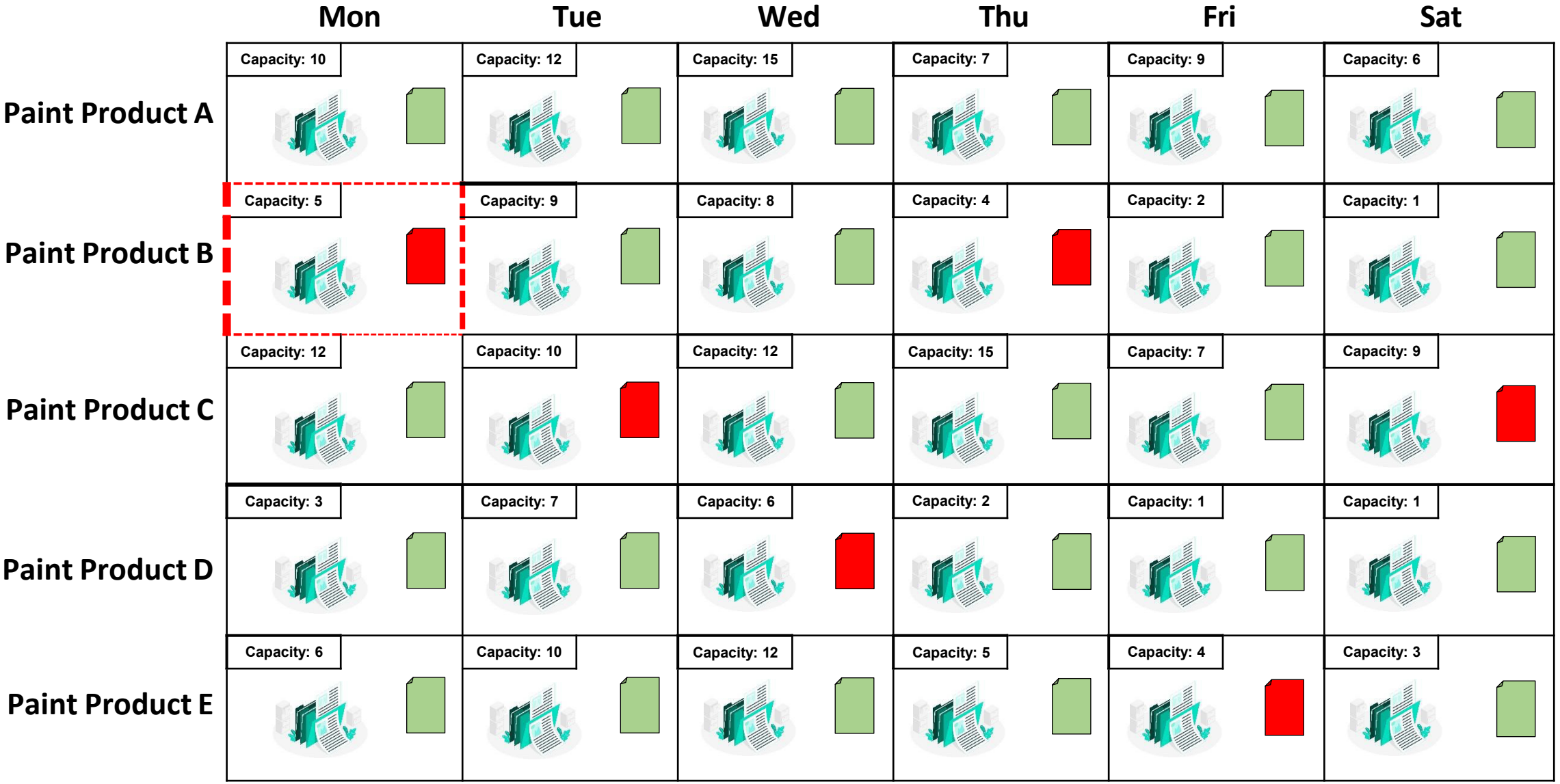


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EXAMPLE: ADDING THE RED KANBAN CARD

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SIMILARLY, ON MONDAY, FOR PAINT PRODUCT B, THERE IS A RED KANBAN CARD. THIS INDICATES MAXIMUM CAPACITY HAS REACHED FOR THAT PRODUCT TYPE ON THAT DAY.



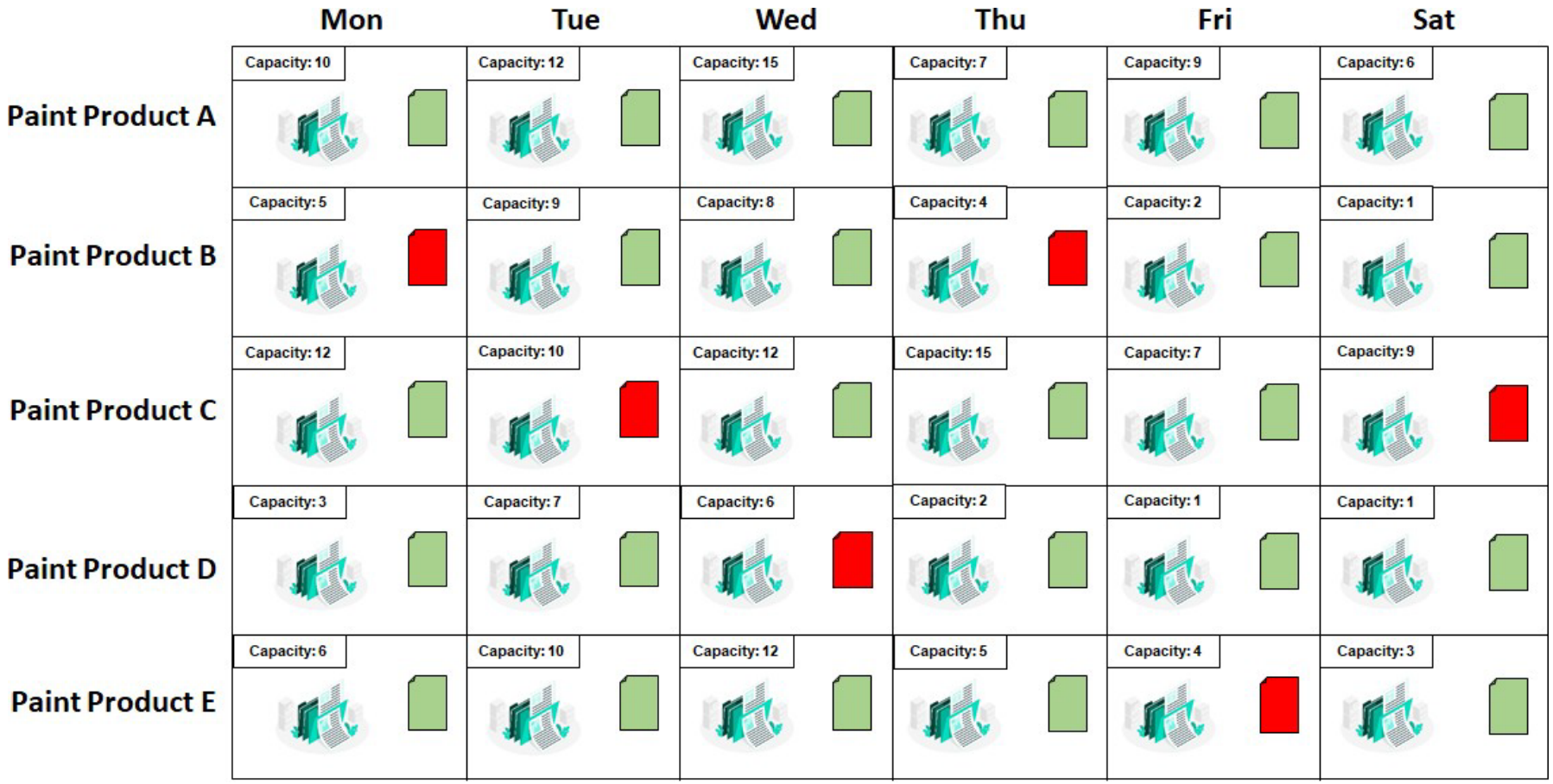
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HEIJUNKA BOX

AIGPE

HEIJUNKA BOX IS AN EXCELLENT INDICATOR TO THE SALES TEAM WHO CAN CHECK ON A REAL-TIME BASIS WHETHER THERE IS AVAILABLE CAPACITY TO FULFILL CLIENT ORDERS FOR DESIGNATED SPECIFIC PRODUCT TYPES.

IT IS ALSO A GOOD INDICATOR FOR A SHIFT SUPERVISOR TO IDENTIFY AVAILABLE CAPACITY AND FILL IT WITH CUSTOMER ORDERS.



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HOW DO YOU CUT BACK THE CHANGEOVER TIME?

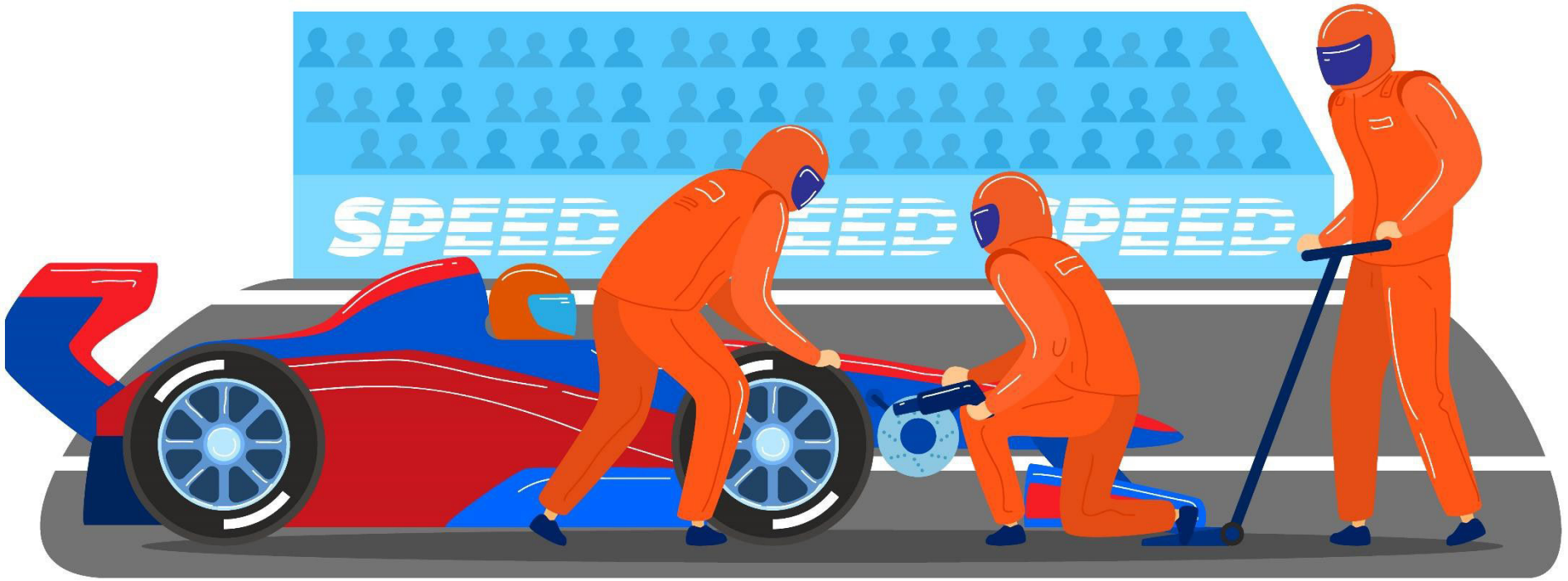


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WHAT IS CHANGEOVER TIME?

AIGPE

THE CHANGEOVER TIME IS THE TIME NEEDED TO PREPARE AN OPERATION FOR A NEW PRODUCT.



EXAMPLE: AN F1 RACE PITSTOP WHERE THE RACING CAR STOPS FOR A FEW MOMENTS, GETS REFUELLED, CHANGES TIRES, AND SMALLER PROBLEMS ARE HANDLED BEFORE IT ROLLS OUT FOR RACING ANOTHER FEW LAPS.

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WHAT IS SMED?

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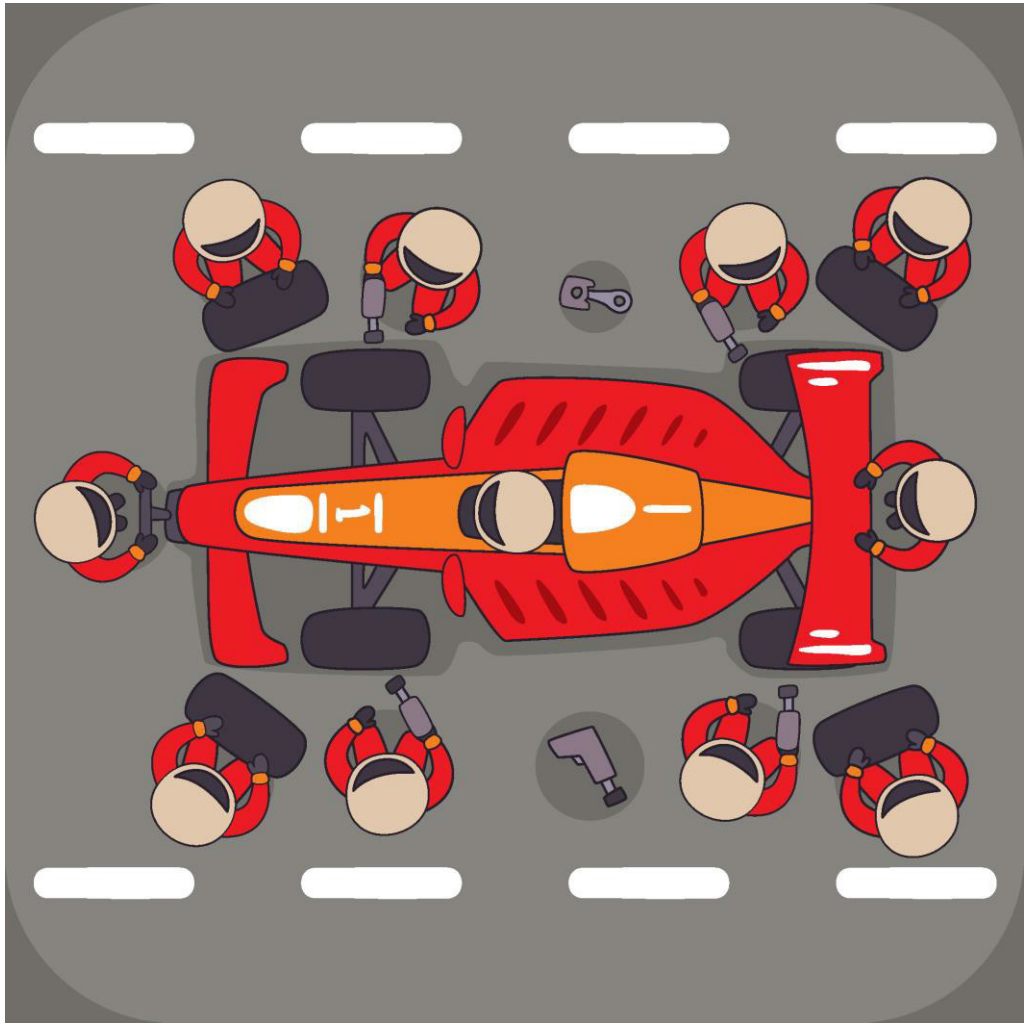
IN THE F1 RACE, THE CAR HAS TO GET OUT OF THE RACETRACK FOR THE CHANGEOVER. IT STOPS RACING FOR THOSE FEW MOMENTS.

WHEN THE EQUIPMENT CHANGEOVER OCCURS IN ANY OPERATION, PRODUCTION MAY HAVE TO STOP, LEADING TO A LOSS OF EFFICIENCY.

HENCE, CHANGEOVERS OFTEN NEED ADJUSTMENTS AND IMPROVEMENTS.

A GREAT WAY TO ACHIEVE THESE ADJUSTMENTS AND IMPROVEMENTS IS TO USE THE SINGLE MINUTE EXCHANGE OF DIE, ALSO KNOWN AS SMED.

SMED IS A COMPREHENSIVE SYSTEM THAT FOCUSES ON SIGNIFICANTLY IMPROVING THE TIME TAKEN FOR EQUIPMENT CHANGEOVERS.



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HOW SMED WORKS?

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IN MOST OPERATIONS (MANUFACTURING AND SERVICE), THE CHANGEOVER OF EQUIPMENT IS PART OF THE OPERATIONAL ACTIVITY.

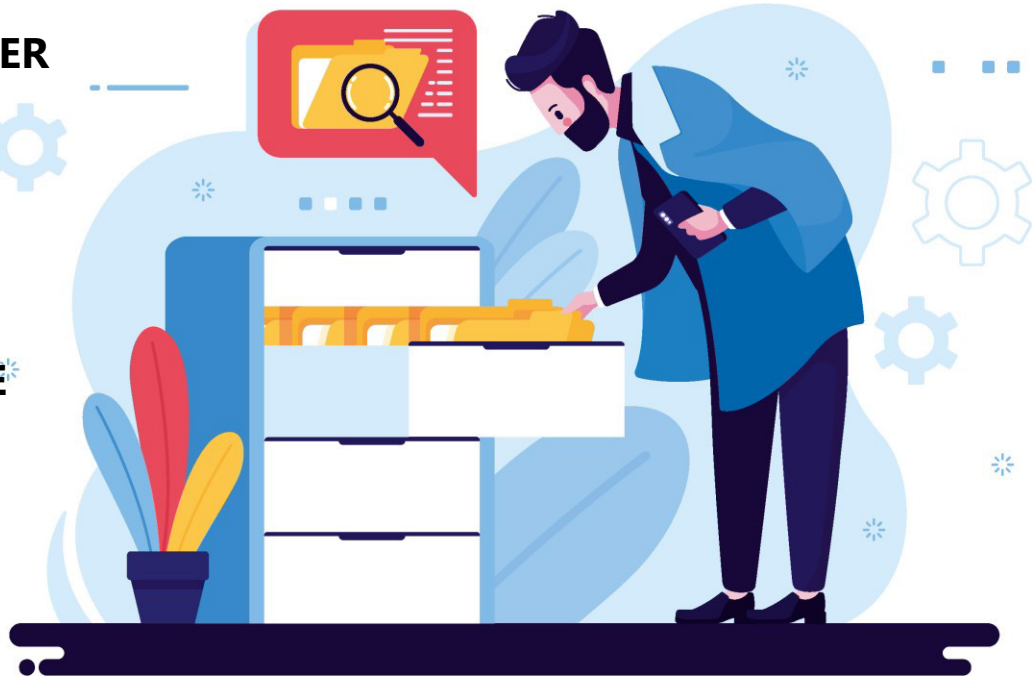
SMED RECOMMENDS THAT WE CHANGE THIS INTERNAL CHANGEOVER TO AN EXTERNAL ONE AS MUCH AS POSSIBLE.

IT SUGGESTS THAT:

1. **YOUR TEAM SHOULD PREPARE FOR AS MANY STEPS AS POSSIBLE BEFORE THE CHANGEOVER BEGINS (EXTERNAL CHANGEOVER)**

2. **DURING THE CHANGEOVER, THEY MUST COORDINATE AND PERFORM PARALLEL STEPS TO EXECUTE THE CHANGEOVER**

3. **AND THE FOCUS OF THIS EFFORT SHOULD BE TO CREATE A STANDARDIZED AND HIGHLY OPTIMIZED PROCESS**



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STEPS FOR SMED

AIGPE

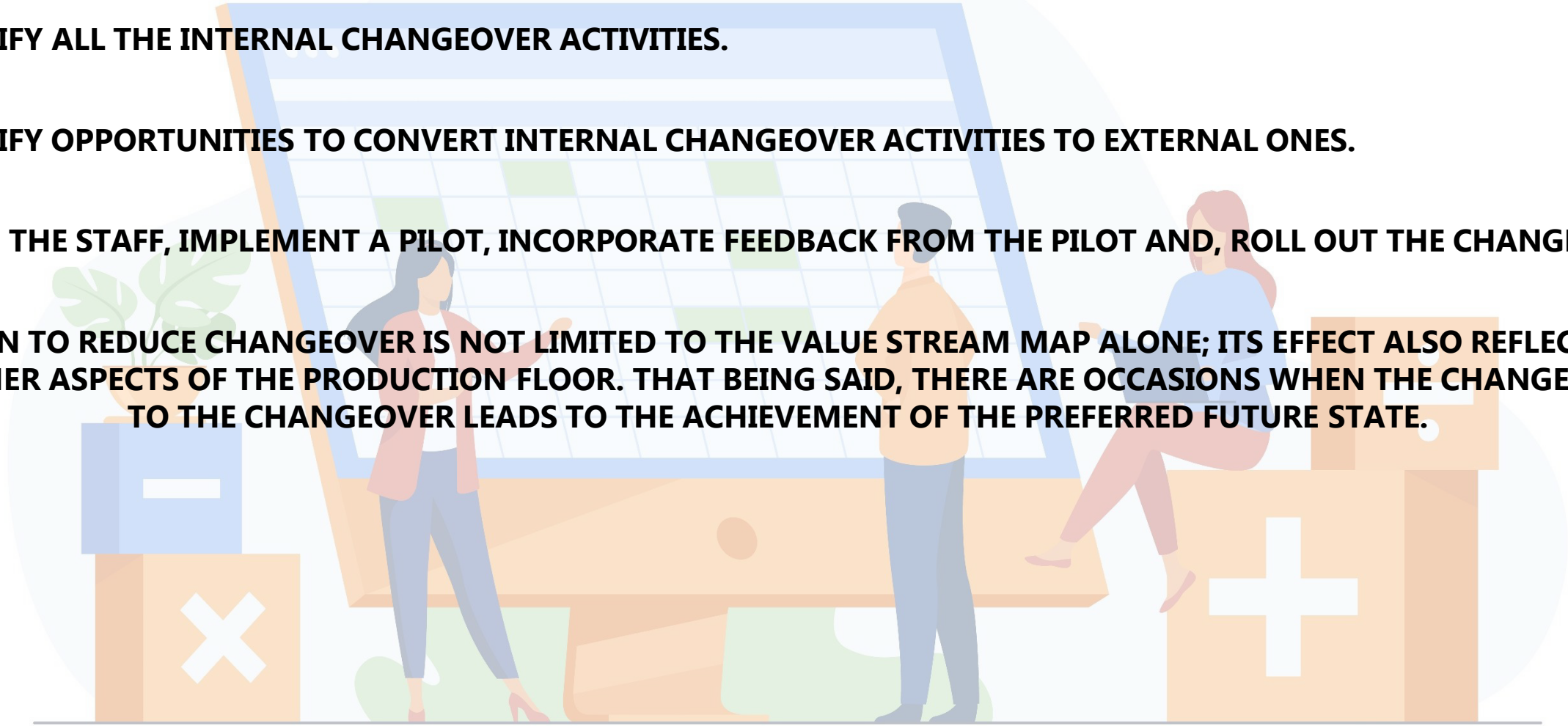
1. STUDY THE EXISTING ACTIVITIES IN YOUR PROCESS. PREFERABLY DO A VIDEO RECORDING OF THOSE STEPS. A VIDEO RECORDING CAN BE PLAYED MULTIPLE TIMES TO UNDERSTAND EACH STEP BETTER.

2. IDENTIFY ALL THE INTERNAL CHANGEOVER ACTIVITIES.

3. IDENTIFY OPPORTUNITIES TO CONVERT INTERNAL CHANGEOVER ACTIVITIES TO EXTERNAL ONES.

4. TRAIN THE STAFF, IMPLEMENT A PILOT, INCORPORATE FEEDBACK FROM THE PILOT AND, ROLL OUT THE CHANGES

TO PLAN TO REDUCE CHANGEOVER IS NOT LIMITED TO THE VALUE STREAM MAP ALONE; ITS EFFECT ALSO REFLECTS ON THE OTHER ASPECTS OF THE PRODUCTION FLOOR. THAT BEING SAID, THERE ARE OCCASIONS WHEN THE CHANGES MADE TO THE CHANGEOVER LEADS TO THE ACHIEVEMENT OF THE PREFERRED FUTURE STATE.



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HOW TO IMPROVE QUALITY?



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HOW TO IMPROVE QUALITY?

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TO IMPROVE QUALITY IS TO REDUCE THE NUMBER OF DEFECTIVE PRODUCTS.



DEFECTIVE PRODUCTS CAN BE REDUCED ONCE THE PROBLEMS THAT LEAD TO DEFECTIVE PRODUCTS IN THE FIRST PLACE ARE RECOGNIZED.

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HOW TO IMPROVE QUALITY?

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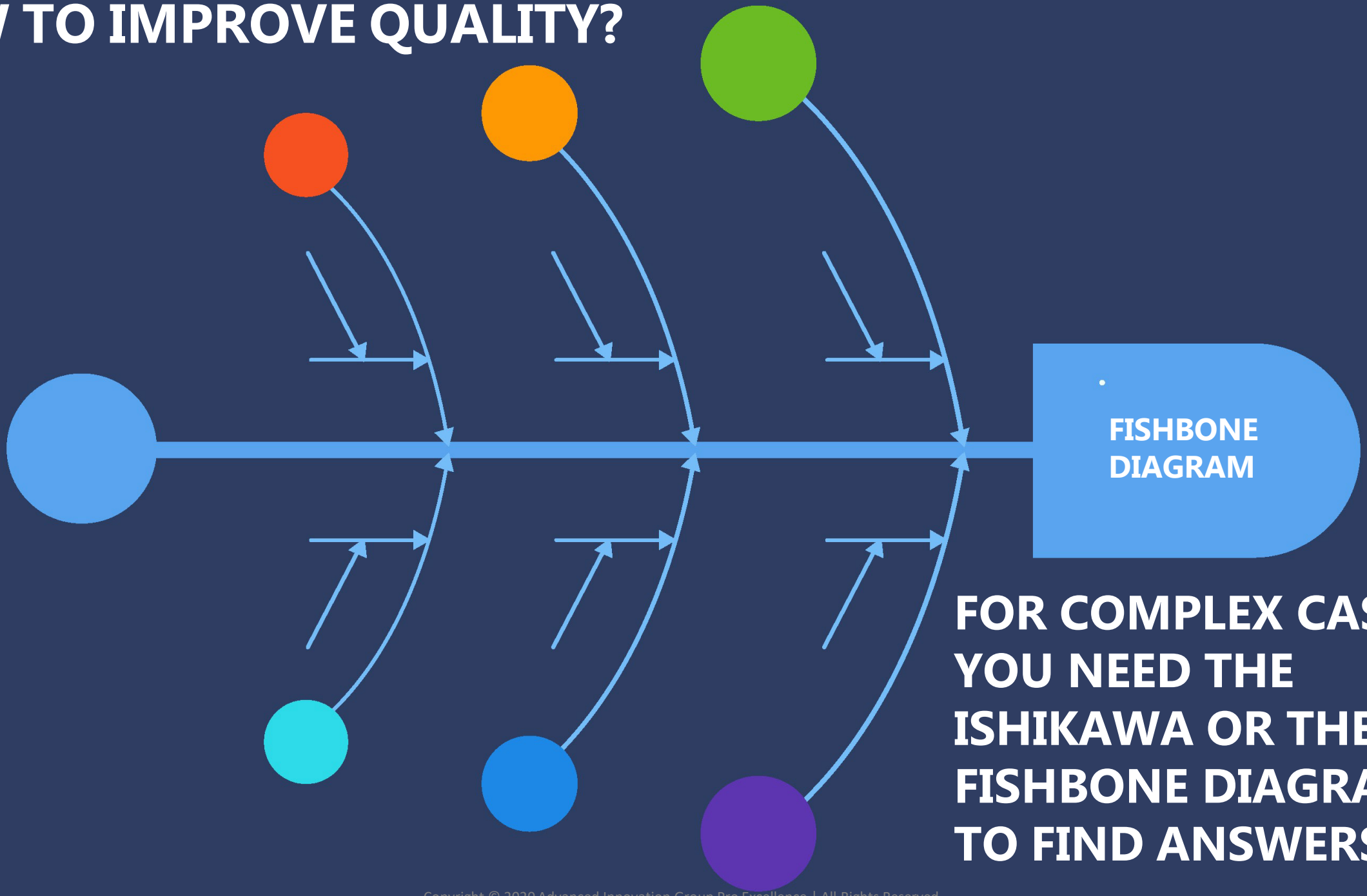
SIMPLE CASES CAN BE SOLVED USING THE "5 WHY" ANALYSIS



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HOW TO IMPROVE QUALITY?

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FOR COMPLEX CASES, YOU NEED THE ISHIKAWA OR THE FISHBONE DIAGRAM TO FIND ANSWERS.

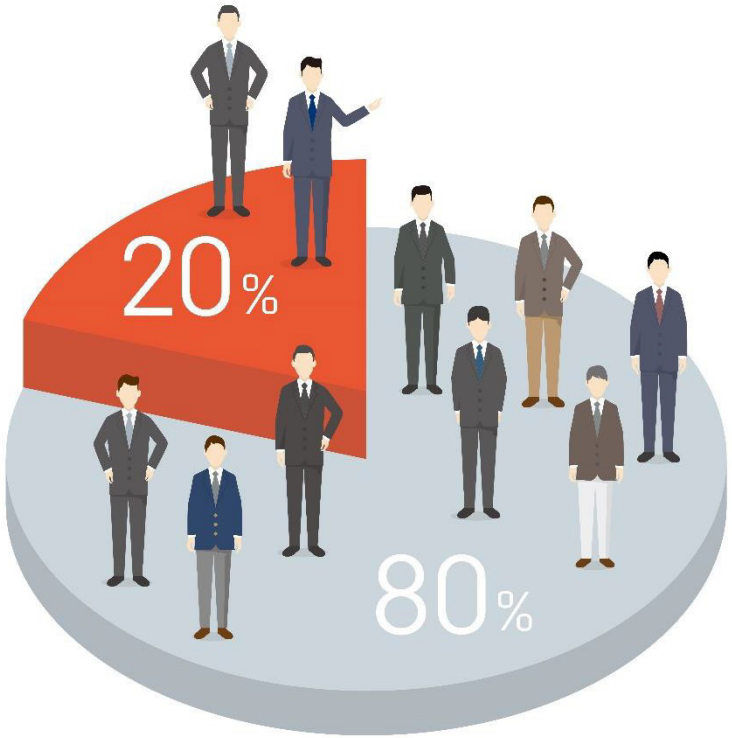
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HOW TO IMPROVE QUALITY?

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THE PARETO CHART HELPS YOU DRAW OUT THE REGULARITY OF THE SEVERAL PROBLEMS AS WELL AS THEIR ROOT CAUSES.

WITH THE PROBLEMS AND ROOT CAUSES IDENTIFIED, THE PARETO CHARTS CAN BE USED TO OBSERVE THE ESSENTIAL VARIABLES IN RESPONSE TO TIME.



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HOW TO IMPROVE QUALITY?

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IN ADDITION TO THESE BASIC TOOLS, YOUR CURRENT STATE VALUE STREAM MAP CAN SHOW YOU OPPORTUNITIES TO EXECUTE:

KAIZEN EVENTS

LEAN

SIX SIGMA

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SECTION CONCLUSION:

HOW TO LAY A STRONG FOUNDATION TO BUILD YOUR FUTURE STATE VSM?



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SECTION CONCLUSION

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EXAMINING THE CURRENT STATE MAP INCLUDED:

1. IDENTIFYING WHETHER YOUR PROCESS CAN MEET CUSTOMER DEMAND WITH EXISTING NUMBER OF EMPLOYEES
2. IDENTIFYING IF AN INCREASE IN THE NUMBER OF EMPLOYEES RESULT IN MEETING THE CUSTOMER DEMAND
3. ANALYZING THE PROCESS BALANCE
4. LOOKING FOR WASTE IN THE PROCESS

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SECTION CONCLUSION

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LAYING A STRONG FOUNDATION TO BUILD THE FUTURE STATE MAP INCLUDED:

1. **CHANGING THE PROCESS TO CONTINUOUS FLOW USING PRODUCTION TAKT VS. CONSUMER TAKT, CONWIP, AND KANBAN**
2. **READJUSTING THE PROCESS**
3. **ALTERING THE BATCH SIZES**
4. **SMOOTHENING OUT PRODUCTION USING HEIJUNKA/HEIJUNKA BOX**
5. **CUTTING BACK THE CHANGEOVER TIME USING SMED**
6. **IMPROVING QUALITY USING TOOLS SUCH AS 5 WHYS, FISHBONE DIAGRAM, PARETO CHARTS, KAIZEN, LEAN OR SIX SIGMA**

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SECTION INTRODUCTION:
BUILDING THE FUTURE STATE VSM



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VALUE STREAM MAPPING (VSM) MASTERCLASS

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Introduction to Value Stream Map (VSM)

- Course Introduction
- What is Value Stream Mapping (VSM)?
- Why is Value Stream Mapping so popular?
- Section Conclusion

Terminologies and Symbols used in VSM0

- Section Introduction
- Terminologies used in VSM
- Symbols used in VSM Process
- Material Flow Symbols
- Information Flow Symbols
- General Symbols
- Activity: Terminologies and Symbols used in VSM
- Activity Solution: Terminologies and Symbols used in VSM
- Section Conclusion

Steps to Create the Current State VSM

- Section and Case Study Introduction
- Step 1. Pick a Product Family
- Step 2. Choose Your VSM Team
- Step 3: Figure out Customer Demand
- Step 4: Map and Design the Process Flow
- Step 5: Map the Material Flow
- Step 6: Map the Information Flow
- Case Study Recap: Current State VSM
- Perform Calculations | Analyze Current State VSM
- Calculate Total Time | Lead Time | Takt Time
- Calculate Process Time | Cycle Time
- Activity: Steps to Create Current State VSM
- Activity Solution: Steps to Create Current State VSM
- Section Conclusion

Examine the Current State VSM

- Section Introduction
- Meeting Customer Demand with existing No. of Operators
- Meeting Customer Demand with increased No. of Operators
- Analyze the Process Balance
- Watch for Waste
- Section Conclusion

Strong Foundations for Future State VSM

- Section Introduction
- Change to Continuous Flow
- CONWIP
- Kanban
- Readjusting Your Process
- Alter the Batch Sizes
- Smoothening out Production
- Heijunka | Heijunka Box
- Cut back Changeover Time
- Improve Quality
- Putting it all Together

Key Questions to ask for Future State VSM

- Section Introduction
- What is the Takt Time?
- Are you building to a warehouse or directly to shipping?
- Where should Continuous Flow Control the Production?
- Controlling area of upstream processes
- Where will you implement the Pull System?
- How do you intend to level the production mix?
- What process improvements do you intend to achieve?
- What is the Implementation Plan?
- Closure of the Case Study
- Section Conclusion

VSM Guidelines

- Section Introduction
- How to get the most of your VSM?
- What are the Pitfalls to avoid?
- What are the advantages of VSM?
- What are the disadvantages of VSM
- Section Conclusion

Course Conclusion

- Course Summary
- Practice Questions
- Become a Certified VSM Practitioner!

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WHAT IS TAKT TIME?



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RECALL: TAKT TIME



WE HAD DONE THIS CALCULATION EARLIER, HOWEVER, SINCE WE ARE BUILDING THE FUTURE STATE VALUE STREAM MAP, LET'S LOOK AT THE CALCULATION AGAIN:

$$T = \frac{T_a}{T_d}$$

T = TAKT TIME

T_a = TOTAL AVAILABLE TIME

T_d = CUSTOMER DEMAND

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CALCULATE TAKT TIME

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FOR OUR DYNAMIC LTD. CASE STUDY:

LET'S DO THE CALCULATION OF THE TAKT TIME:



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TAKT TIME OF THE PROCESS

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CALCULATE TOTAL AVAILABLE TIME:

- **TOTAL AVAILABLE TIME = 8 HOURS WITH A BREAK OF 15 MINUTES (AT LEAST 3 TIMES)**
- **TOTAL AVAILABLE TIME = (8 X 60 MINUTES X 60 SECONDS) – [(15 MINUTES + 15 MINUTES + 15 MINUTES) X 60 SECONDS]**
- **TOTAL AVAILABLE TIME = (28,800 – 2,700)**
- **TOTAL AVAILABLE TIME = 26,100 SECONDS**
- **T_a = 26,100 SECONDS**

IDENTIFY CUSTOMER DEMAND:

- **CUSTOMER DEMAND = 300 CANS PER DAY**
- **T_d = 300**

TAKT TIME:

- $T = \frac{T_a}{T_d} = \frac{26{,}100}{300}$ **= 87 SECONDS**

THIS INDICATES THAT IT SHOULD TAKE NO MORE THAN 87 SECONDS TO PRODUCE ONE UNIT OF THE PAINT CAN TO MEET THE CUSTOMER REQUIREMENT OF 300 CANS PER DAY.

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TAKT TIME OF THE PROCESS

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YOU MAY POTENTIALLY ASK - WE HAD DONE THE TAKT TIME CALCULATIONS WHEN ANALYZING THE CURRENT STATE, SO WHY DO WE NEED TO DO THIS CALCULATION AGAIN?

THE ANSWER IS SIMPLE - TAKT TIME IS THE BASE OF MOST IMPROVEMENT ACTIONS YOU WILL TAKE IN THE FUTURE STATE VALUE STREAM MAP.

THERE IS A POSSIBILITY THAT YOU MAY NOT CALCULATE THE TAKT TIME IN PREVIOUS STEPS.

SO, ANSWERING THIS QUESTION FIRST BECOMES ESSENTIAL WHEN CREATING THE FUTURE STATE VALUE STREAM MAP.

HENCE, I HAVE ENSURED THAT WE REPEAT THE CALCULATION IN THIS LECTURE.

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ARE YOU BUILDING TO A WAREHOUSE OF DIRECTLY TO SHIPPING?



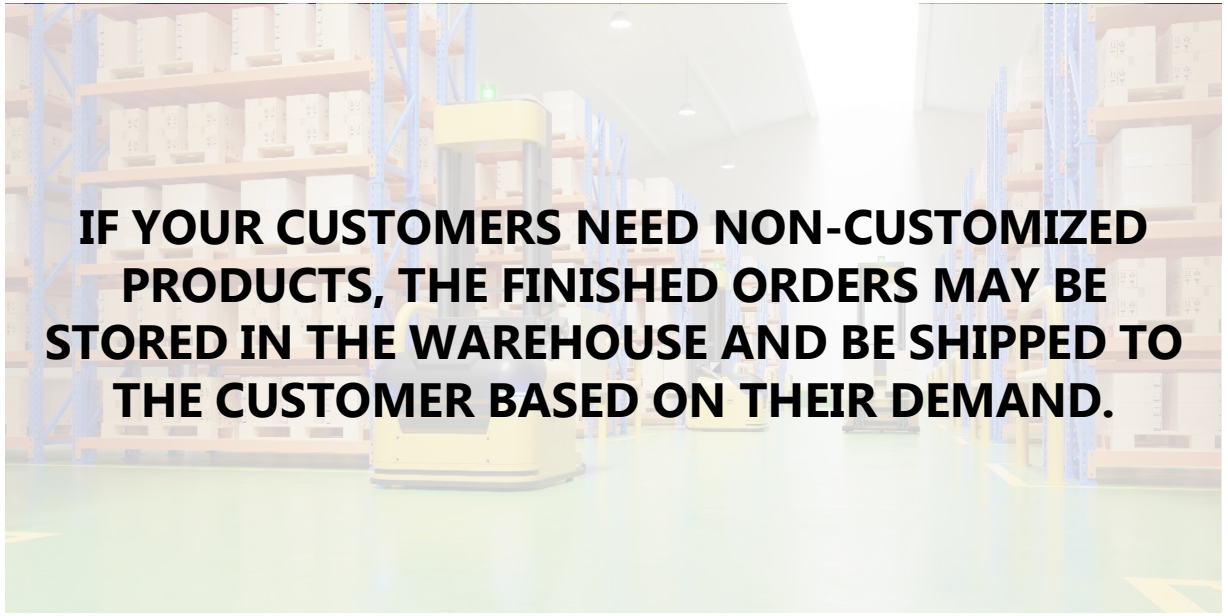
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ARE YOU BUILDING TO A WAREHOUSE OR SHIPPING?

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WHETHER YOU WILL STOCK PHYSICAL FINISHED PRODUCTS IN A WAREHOUSE OR DIRECTLY SHIP THEM TO THE CUSTOMER?

BUILDING TO A WAREHOUSE



IF YOUR CUSTOMERS NEED NON-CUSTOMIZED PRODUCTS, THE FINISHED ORDERS MAY BE STORED IN THE WAREHOUSE AND BE SHIPPED TO THE CUSTOMER BASED ON THEIR DEMAND.

SHIPPING DIRECTLY TO THE CUSTOMER



IF YOUR CUSTOMERS NEED CUSTOMIZED PRODUCTS, THE FINISHED ORDERS MAY NOT BE STORED IN THE WAREHOUSE AND DIRECTLY SHIPPED TO THE CUSTOMER.

IN THE CASE OF DYNAMIC LTD., THE CUSTOMER REQUIRES A NON-CUSTOMIZED PRODUCT. SO, THE FINISHED ORDERS CAN BE STORED IN THE WAREHOUSE BEFORE THEY GET SHIPPED TO THE CUSTOMER. HOWEVER, YOU CAN ASK THIS QUESTION:

IS THERE A NEED TO STORE THE FINISHED PAINT PRODUCTS FOR FIVE DAYS IN THE WAREHOUSE? CAN YOU SHIP THEM EARLIER?

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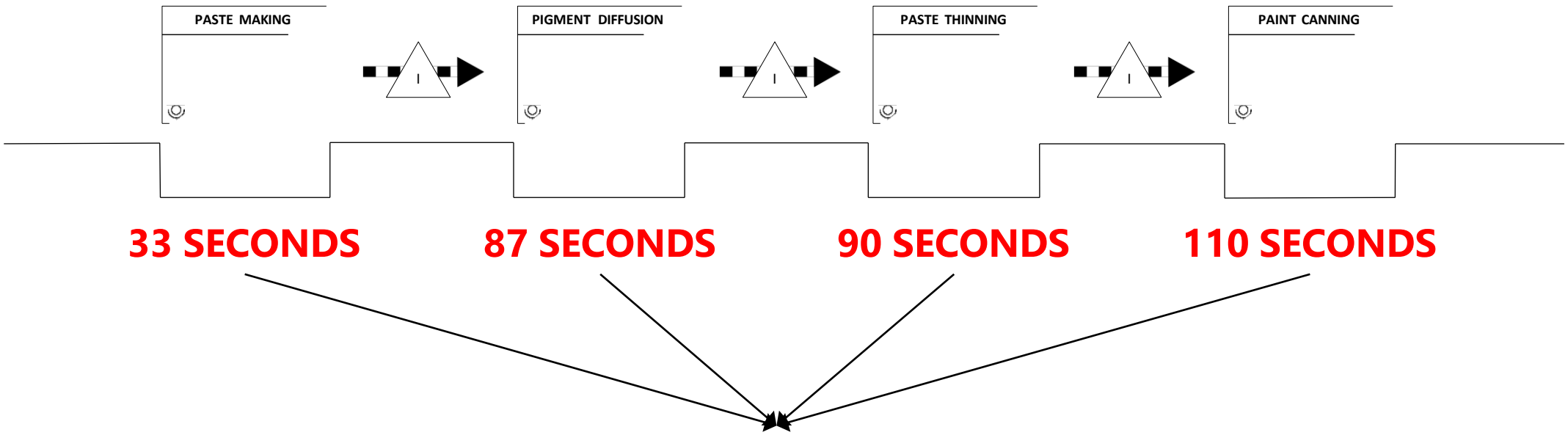
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WHERE SHOULD YOUR CONTINUOUS FLOW PROCESSING CONTROL THE PRODUCTION OF UPSTREAM PROCESSES?



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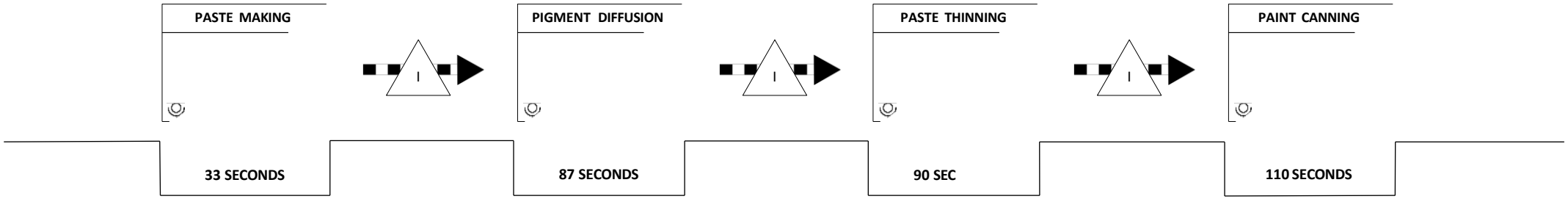
UNBALANCED CYCLE TIME

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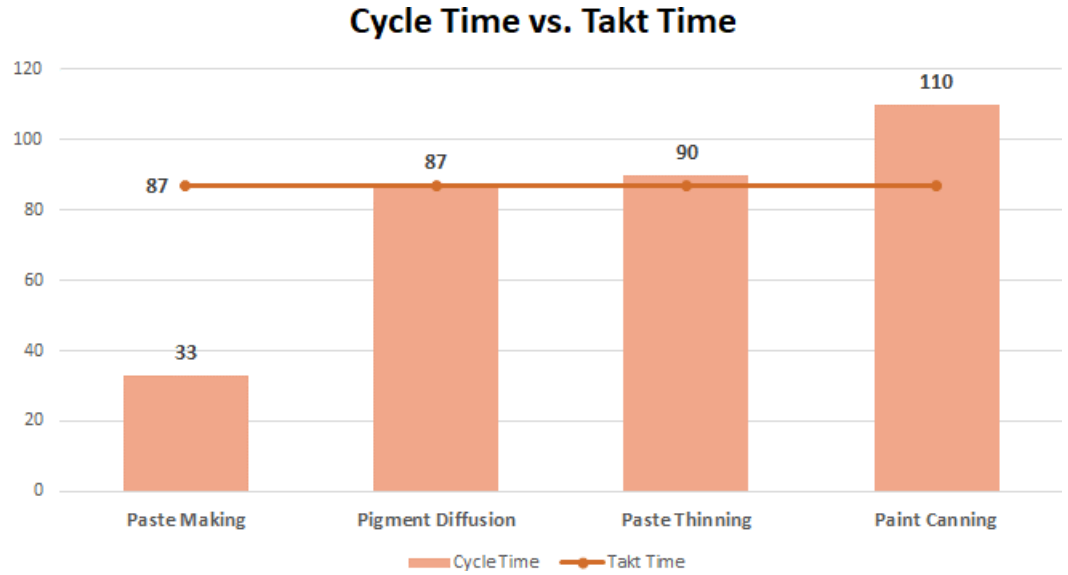
WHERE SHOULD YOUR CONTINUOUS FLOW PROCESSING CONTROL THE PRODUCTION?

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OBSERVED THAT THE PASTE MAKING, PIGMENT DIFFUSION, PASTE THINNING, AND PAINT CANNING PROCESSES HAVE CYCLE TIMES OF 33, 87, 90, AND 110 SECONDS, RESPECTIVELY. THESE CYCLE TIMES ARE UNBALANCED.



IF YOU PUT THIS ON THE CHART, YOUR GRAPH WOULD LOOK LIKE THIS. THE TAKT TIME IS THE LINE PASSING THROUGH ALL THE BARS AT 87 SECONDS.



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WHERE SHOULD YOUR CONTINUOUS FLOW PROCESSING CONTROL THE PRODUCTION?

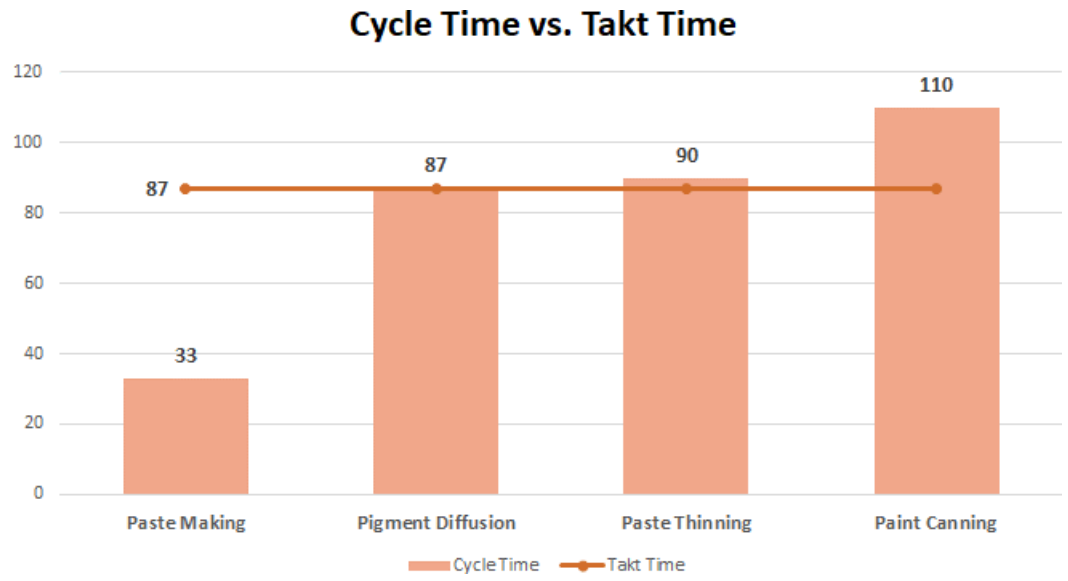
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WITH THREE OPERATORS, THE CYCLE TIME IS 106.7 SECONDS. IF YOU ADD A FOURTH OPERATOR, THE CYCLE TIME REDUCES TO 80 SECONDS, LOWER THAN THE TAKT TIME OF 87 SECONDS.

RECALL THE DISCUSSION OF PRODUCTION TAKT VS. CONSUMER TAKT. THE 87 SECONDS IS A CONSUMER TAKT. PRODUCTION TAKT IS LESSER THAN THE CONSUMER TAKT.

THIS INDICATES THAT YOU MUST REEXAMINE THE NATURE OF EACH OPERATION AND BALANCE THE TIMES PER OPERATION TO KEEP IT LESSER THAN THE DECIDED PRODUCTION TAKT TIME.

PASTE MAKING TAKES 33 SECONDS, PIGMENT DIFFUSION TAKES 87 SECONDS, PASTE THINNING TAKES 90 SECONDS, AND PAINT CANNING TAKES 110 SECONDS. YOU MAY WANT TO MOVE SOME ACTIVITIES EXECUTED IN PIGMENT DIFFUSION, PASTE THINNING, AND PAINT CANNING TO THE PASTE MAKING PROCESS.



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WHERE SHOULD YOUR CONTINUOUS FLOW PROCESSING CONTROL THE PRODUCTION?

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ACTION # 1: THE RAW MATERIAL USED FOR PIGMENT DIFFUSION AND PASTE THINNING PROCESSES CAN BE ADDED IN THE PASTE MAKING PROCESS. THIS CAN SPEED UP THE PROCESS OF PIGMENT DIFFUSION AND PASTE THINNING ACTIVITIES.

ACTION # 2: THE OPERATORS IN THE PASTE MAKING PROCESS CAN ALSO NOTE THE WEIGHT OF THE PASTE THINNING FINISHED MATERIAL, SO THIS ACTIVITY IS NOT REPEATED IN THE LATER PROCESSES.

ACTION # 3: THE ACTIVITY OF APPLYING STICKERS ON PASTE CANS CAN BE INITIATED IN THE PASTE MAKING PROCESS SINCE THE OPERATORS WILL KNOW THE ESSENTIAL DETAILS OF THE CUSTOMER ORDER WHEN THEY INITIATE THE PASTE MAKING PROCESS.

THESE CHANGES WOULD BETTER BALANCE THE CYCLE TIME AT EACH PROCESS AND KEEP IT LOWER THAN THE PRODUCTION TAKT TIME.

THIS WILL ALSO CONSEQUENTLY CREATE A CONTINUOUS FLOW WITH THE LEAST AMOUNT OF BOTTLENECKS.

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WHERE WILL YOU IMPLEMENT A PULL SYSTEM IN YOUR FUTURE STATE VSM?



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WHERE WILL YOU IMPLEMENT A PULL SYSTEM IN YOUR FUTURE STATE VSM?

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WHAT IS THE NEED TO IMPLEMENT A PULL SYSTEM IN ANY PROCESS?

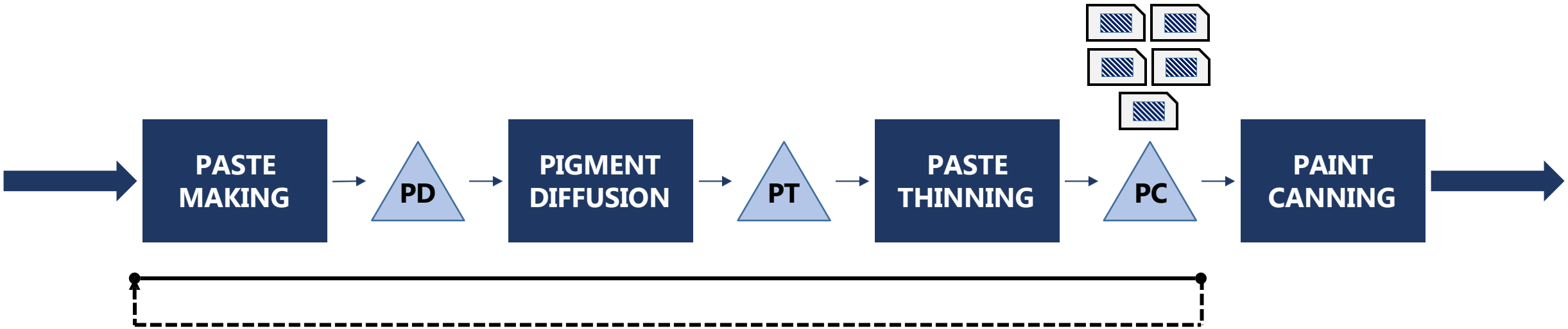
IT IS DIFFICULT TO SET UP THE CONTINUOUS FLOW IN YOUR BUSINESS OPERATIONS. IN THESE SITUATIONS, YOU CAN DEPLOY A PULL SYSTEM. THE BENEFIT OF A PULL SYSTEM IS THAT IT REGULATES THE UPSTREAM PRODUCTION ACCORDING TO THE DOWNSTREAM REQUIREMENTS.

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RECALL: CONWIP

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RECALL THE DISCUSSION OF CONWIP AND KANBAN. CONWIP IS ONE OF THE SIMPLEST PULL SYSTEMS. IN CONWIP, A SINGLE SET OF PHYSICAL CARDS ENFORCES ONE WORK-IN-PROGRESS LIMIT ACROSS THE WHOLE SYSTEM.

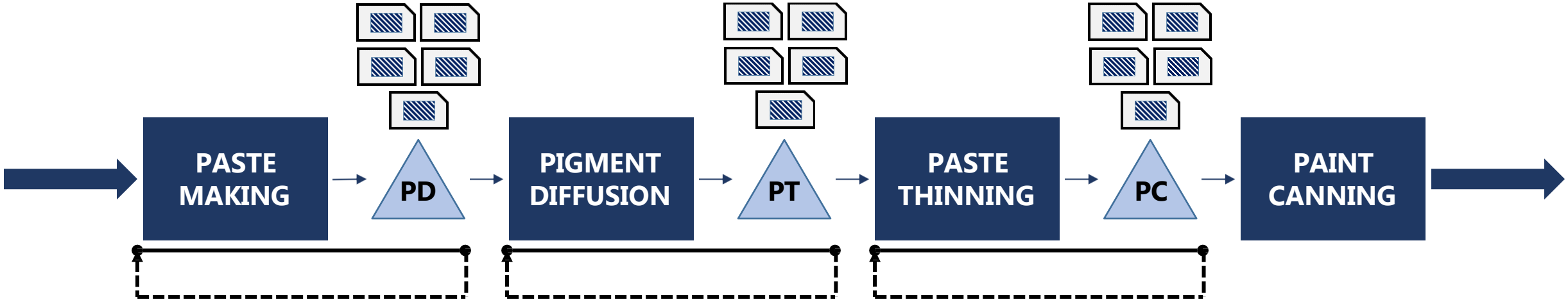


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RECALL: KANBAN

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KANBAN IS A MORE ADVANCED PULL SYSTEM. IN IT, THE FUNCTION OF THE CARDS IS SIMILAR TO CONWIP, BUT THERE ARE MULTIPLE CARD LOOPS. EACH CARD LOOP IS SEPARATELY DEDICATED TO DIFFERENT OPERATIONAL PROCESSES.



IN DYNAMIC LTD., CASE STUDY EXAMPLE, KANBAN PULL CAN BE IMPLEMENTED AT EACH OPERATIONAL PROCESS TO ENSURE THE RAW MATERIAL SUPPLY IS ALWAYS AVAILABLE AT EACH PROCESS. THIS WILL ALSO HELP US ENSURE EACH PROCESS IS OPTIMIZED AS WE HAD SEEN THAT THE CYCLE TIME, EVEN AFTER REBALANCING, WAS TOO CLOSE TO THE TAKT TIME.

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HOW DO YOU INTEND TO LEVEL THE PRODUCTION MIX?



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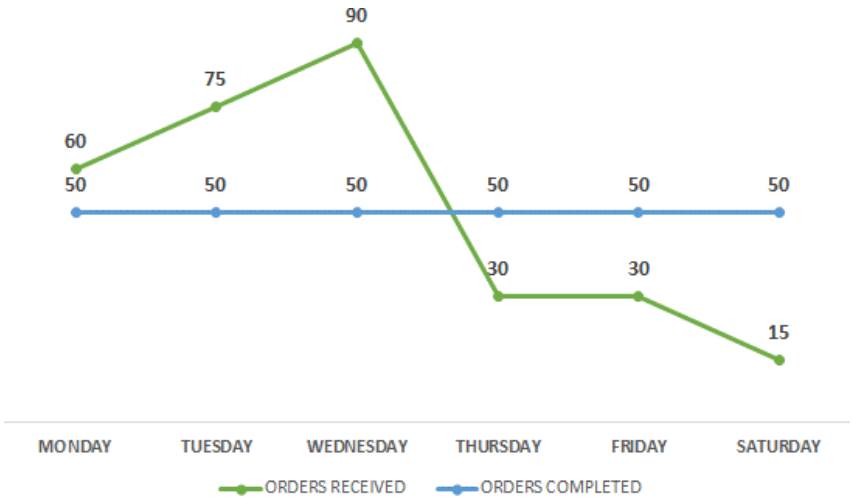
HOW DO YOU INTEND TO LEVEL THE PRODUCTION MIX?

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THE PRODUCTION MIX CAN BE LEVELLED BY IMPLEMENTING TOOLS SUCH AS HEIJUNKA OR HEIJUNKA BOX. RECALL THE DISCUSSION ON THESE TWO TOPICS.

YOU CAN LEVEL THE WORKLOAD BY VOLUME. SO, IF THE CUSTOMER DEMAND IS 300 PAINT CANS PER WEEK, YOU CAN MAINTAIN A STEADY PRODUCTION OF 50 PAINT CANS PER DAY WITH AN INVENTORY OF 50 PAINT CANS TO HELP ADDRESS ANY VARIATIONS.

DAY	ORDERS RECEIVED	ORDERS COMPLETED	BORROWED FROM INVENTORY	INVENTORY LEFT BY THE END OF THE DAY
MONDAY	60	50	10	40
TUESDAY	75	50	25	15
WEDNESDAY	90	50	15	-25
THURSDAY	30	50	-20	-5
FRIDAY	30	50	-20	15
SATURDAY	15	50	-35	50

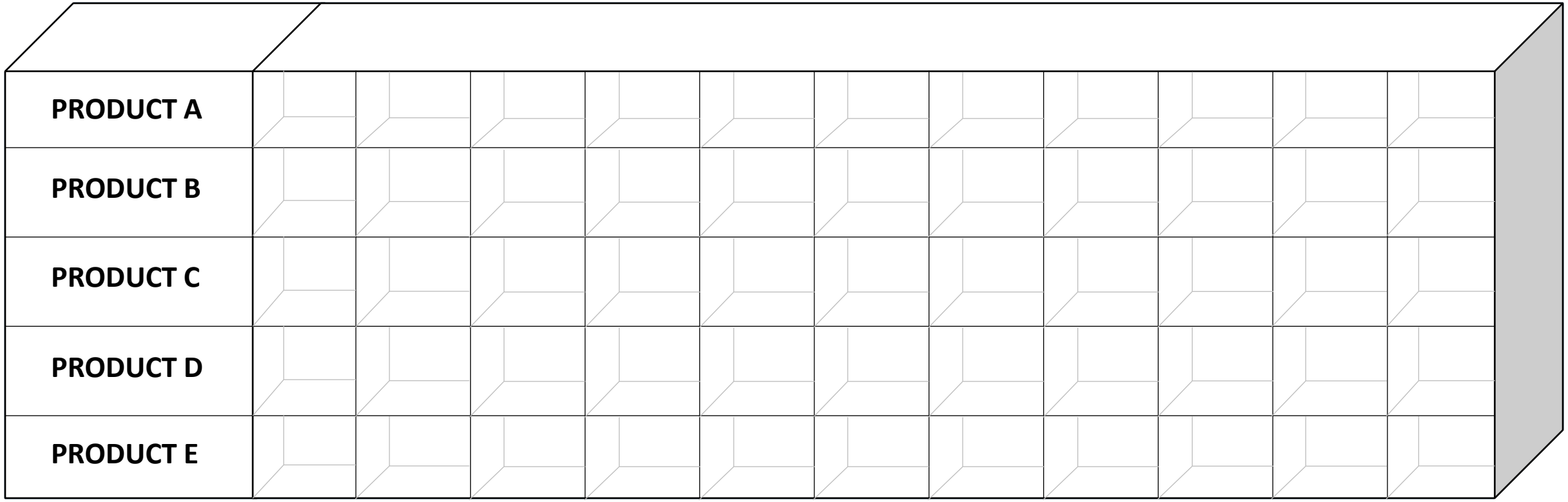


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HOW DO YOU INTEND TO LEVEL THE PRODUCTION MIX?

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THE HEIJUNKA BOX MAY BE USED IF YOU ARE DEALING WITH MULTIPLE PRODUCTS IN A PRODUCT FAMILY. IT WILL HELP SMOOTHEN YOUR PRODUCTION PROCESS.



FOR DYNAMIC LTD., WE ARE ONLY DEALING WITH ONE PRODUCT; HENCE, WE CAN USE THE HEIJUNKA TECHNIQUE OF LEVELING THE WORKLOAD BY VOLUME. HEIJUNKA BOX MAY NOT BE USED IN THIS SCENARIO.

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WHAT PROCESS IMPROVEMENTS DO YOU INTEND TO ACHIEVE WITH THE FUTURE STATE VSM?



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WHAT PROCESS IMPROVEMENTS DO YOU INTEND TO ACHIEVE?

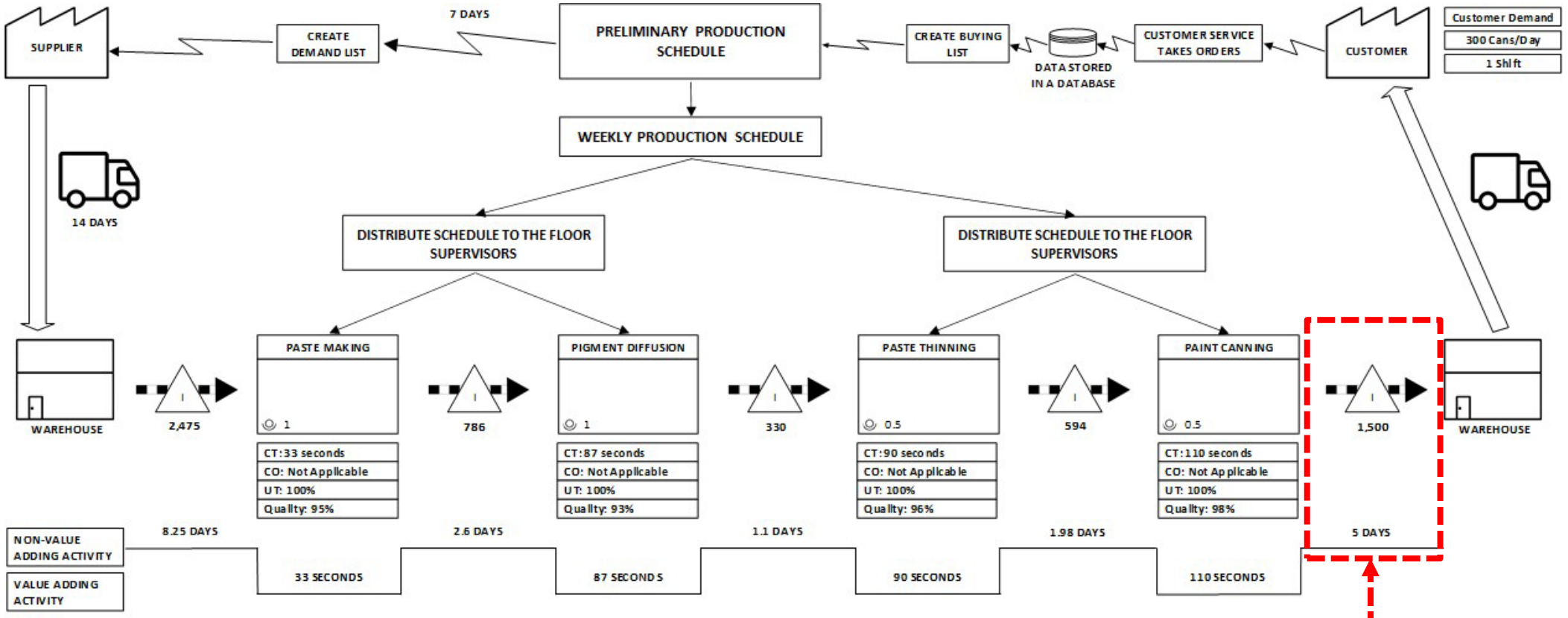
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THIS QUESTION, "WHAT PROCESS IMPROVEMENTS DO YOU INTEND TO ACHIEVE WITH THE FUTURE STATE MAP?" IS ASKED WITH A VIEW TO IDENTIFY IF THERE ARE GAPS IN THE CURRENT STATE VALUE STREAM MAP THAT ARE STILL NOT ADDRESSED.

LET'S REVIEW AND IDENTIFY WHAT ACTIONS ARE TO BE EXECUTED:

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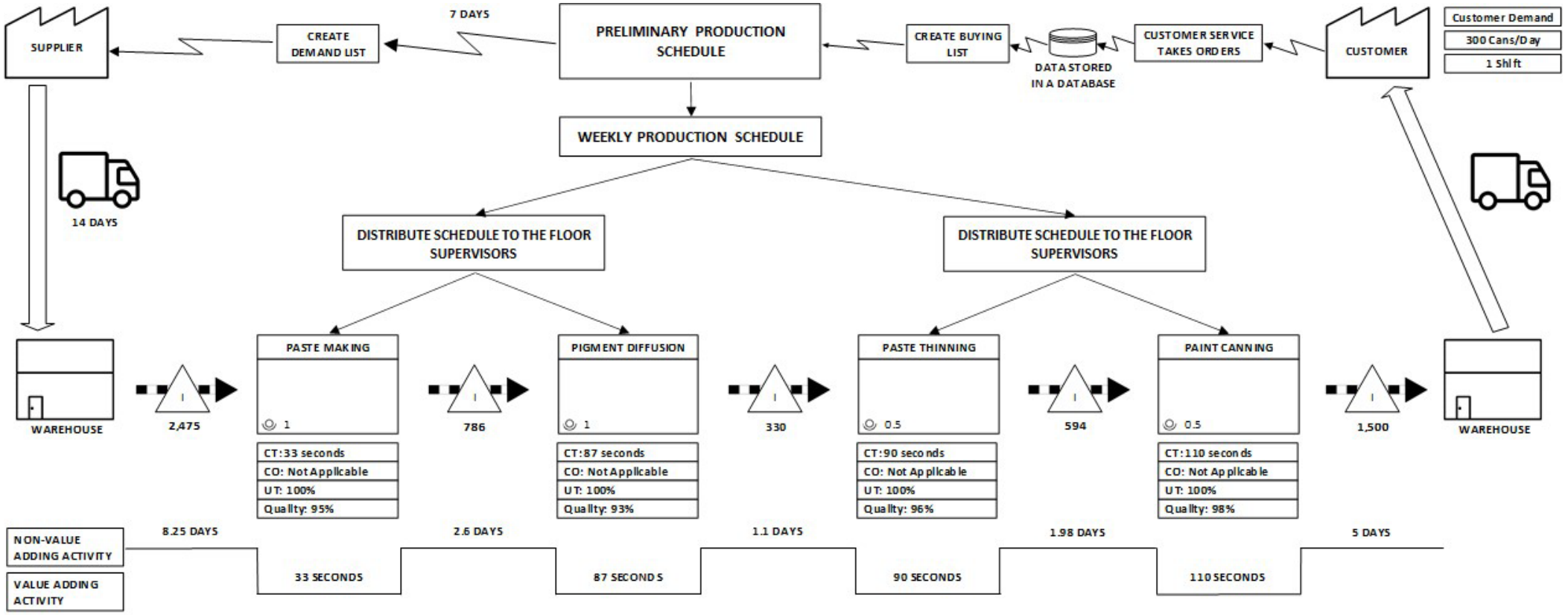
WHAT PROCESS IMPROVEMENTS DO YOU INTEND TO ACHIEVE?



CURRENTLY, THE FINISHED PAINT CANS ARE KEPT IN THE WAREHOUSE FOR FIVE DAYS. THAT MEANS THE PRODUCTS ARE DISPATCHED PER WEEK. CAN YOU REDUCE THE STORAGE OF PAINT CANS IN THE WAREHOUSE FROM FIVE DAYS TO THREE DAYS? IF YOU DO SO, YOU WILL INCREASE THE SHIPPING COST. BUT ON THE BRIGHT SIDE, CUSTOMERS WILL RECEIVE THEIR ORDERS EARLIER, AND YOU CAN REPURPOSE THE ADDITIONAL STORAGE SPACE OF THE WAREHOUSE. IF REQUIRED, YOU CAN REDUCE THAT ADDITIONAL STORAGE SPACE AND SAVE ON THOSE EXPENSES.

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WHAT PROCESS IMPROVEMENTS DO YOU INTEND TO ACHIEVE?

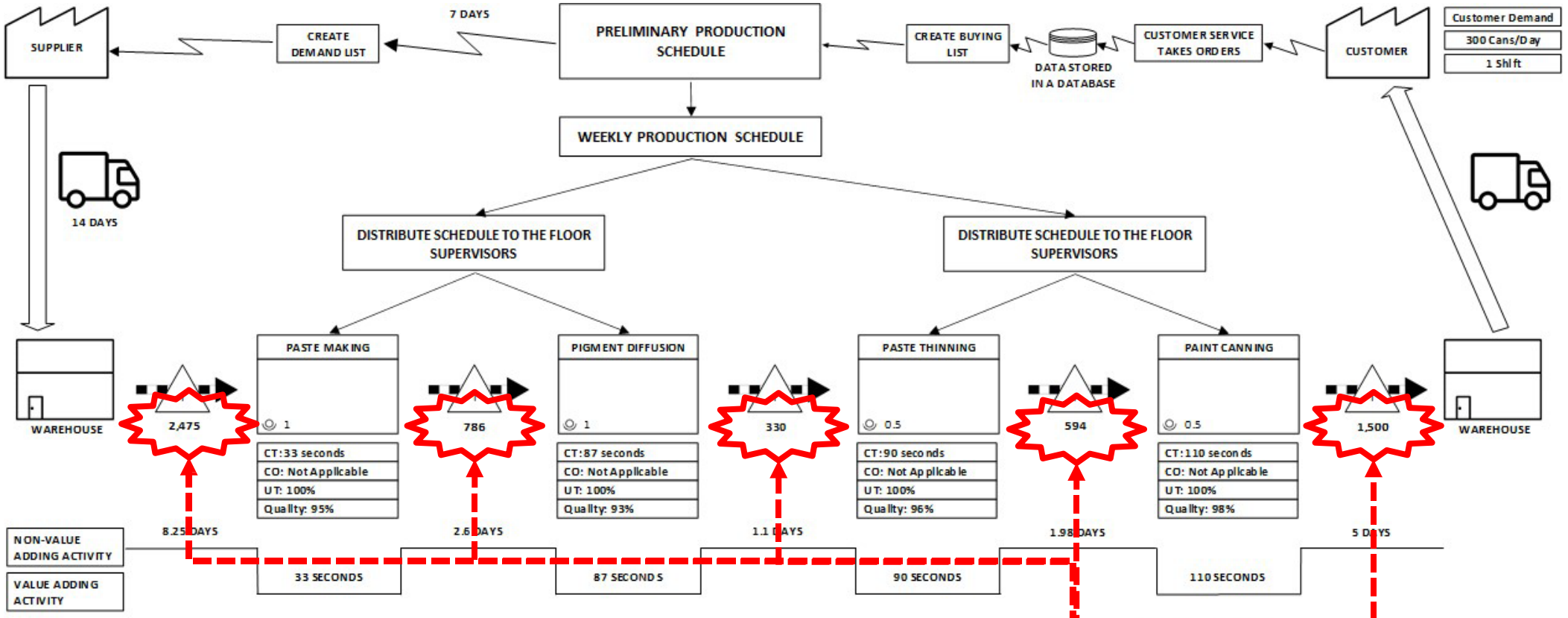


NEXT, YOU HAVE TO BALANCE THE OPERATIONAL PROCESSES (AS WE DISCUSSED IN THE THIRD QUESTION). YOUR CONSUMER TAKT TIME IS 87 SECONDS, AND YOU HAVE TO DECIDE THE PRODUCTION TAKT TIME THAT IS LESSER THAN 87. BY IMPROVING HOW TO EXECUTE EVENTS AND ACTIVITIES, YOU HAVE TO PUT THE PROCESS IN A CONTINUOUS FLOW TO ARRIVE AT A CYCLE TIME PER PROCESS THAT IS LESS THAN THE PRODUCTION TAKT TIME.

Total Time = 39.93 days
Lead Time = 18.99 days
Takt Time = 87 seconds
Process/Processing/Value Added Time = 320 seconds
Cycle Time = 106.7 seconds

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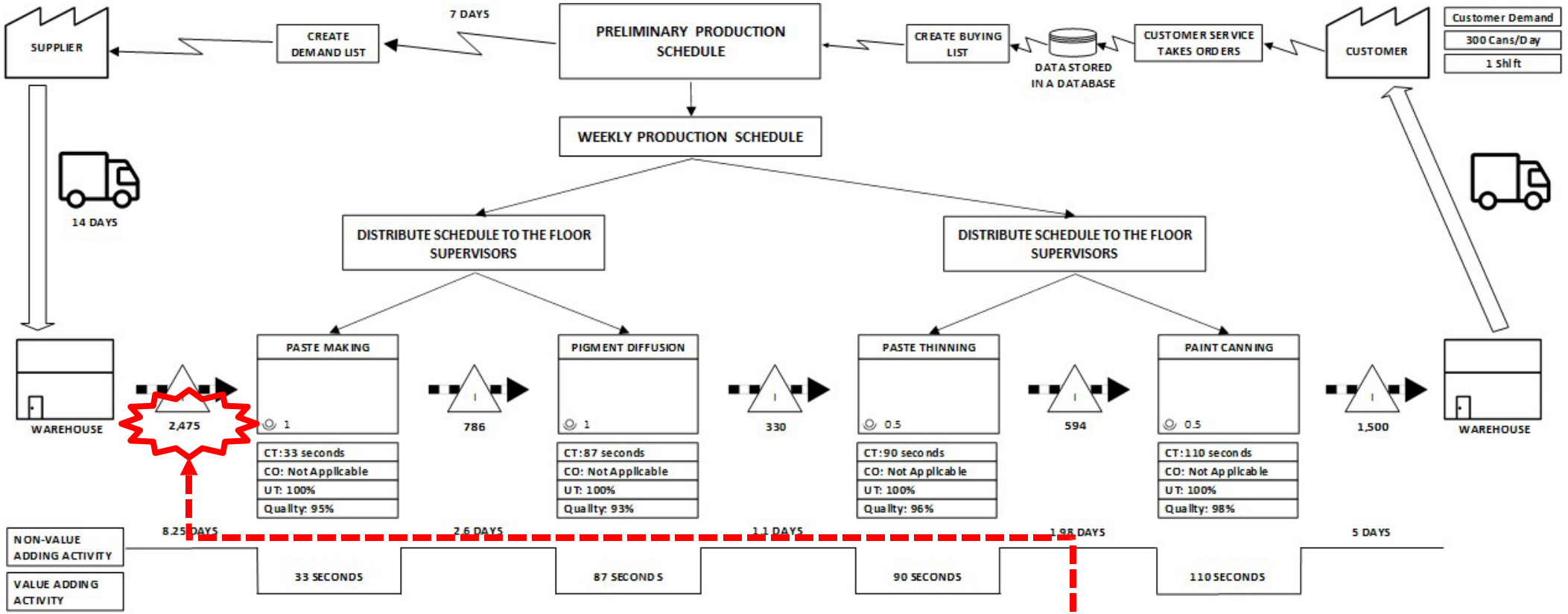
WHAT PROCESS IMPROVEMENTS DO YOU INTEND TO ACHIEVE?



NEXT, YOU ARE IMPLEMENTING THE KANBAN PULL SYSTEM AT EACH PROCESS, WHICH WILL REDUCE THE INTERMEDIATE STOCK REQUIREMENT TO A GREAT EXTENT, MAYBE ABOUT FOUR ITEMS IN-BETWEEN PROCESSES.

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WHAT PROCESS IMPROVEMENTS DO YOU INTEND TO ACHIEVE?

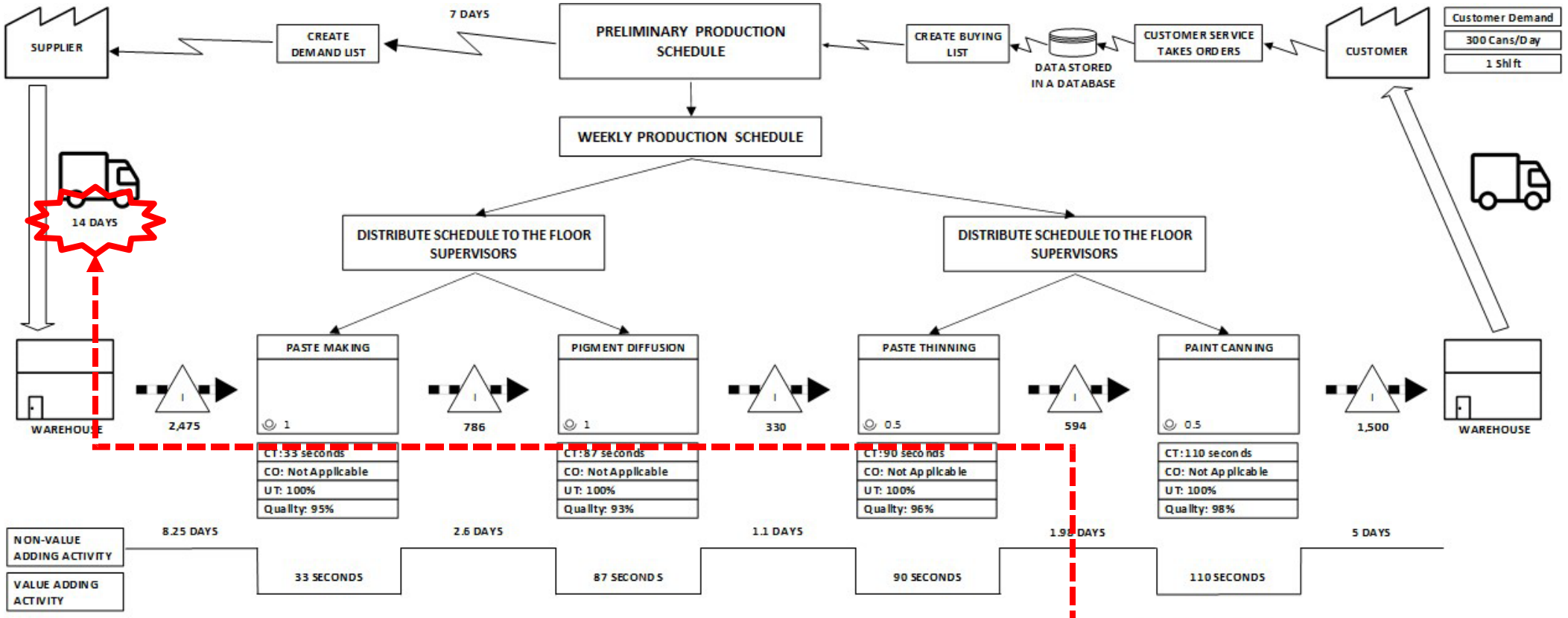


YOU SHOULD ALSO REVIEW THE STOCK AT THE BEGINNING OF THE FLOW. TODAY, THERE ARE 2,475 PIECES I.E., 8.25 DAYS OF ORDERS IN STOCK FOR 300 CANS PER DAY. WE PLAN TO DELIVER TWICE PER WEEK. SO, EVEN IF WE PLAN FOR A POTENTIAL SITUATION WHERE SHIPPING IS NOT SENT ONCE IN THE WEEK DUE TO UNFORESEEN SITUATIONS, YOU CAN THINK OF REDUCING THE STOCK TO 1,500 I.E., FROM 8.25 DAYS TO 5 DAYS WORTH OF ORDERS IN STOCK.

Total Time = 39.93 days
Lead Time = 18.93 days
Takt Time = 87 seconds
Process/Processing/Value Added Time = 320 seconds
Cycle Time = 106.7 seconds

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WHAT PROCESS IMPROVEMENTS DO YOU INTEND TO ACHIEVE?



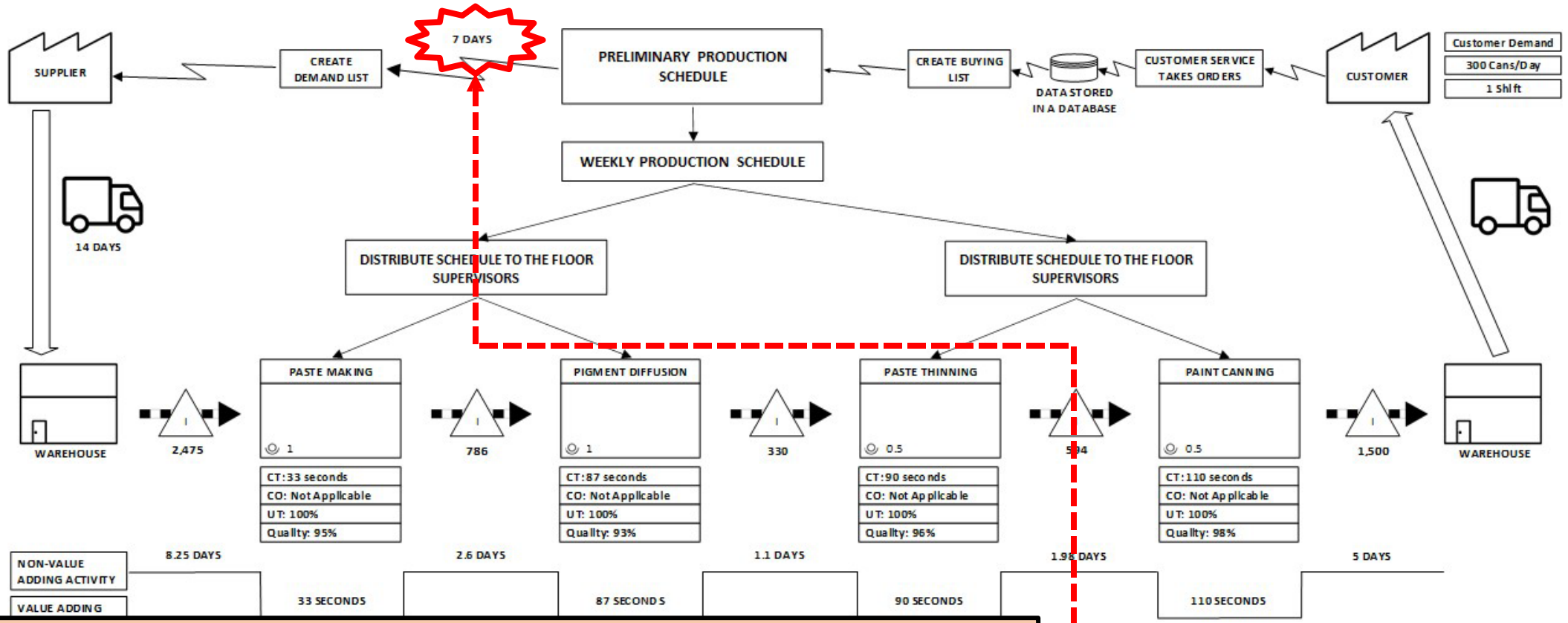
SAFETY STOCK: ACCORDING TO THE CURRENT STATE MAP, IT TAKES 14 DAYS FOR THE ORDERED RAW MATERIALS TO ARRIVE. THE PRESENCE OF SAFETY STOCK TO KICKSTART PRODUCTION WHILE WAITING FOR THE ORDERS TO ARRIVE FROM THE SELLER WILL SHAVE 14 DAYS FROM THE WAITING PERIOD.

Total Time = 39.93 days
Lead Time = 18.93 days
Takt Time = 87 seconds
Process/Processing/Value Added Time = 320 seconds
Cycle Time = 106.7 seconds

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WHAT PROCESS IMPROVEMENTS DO YOU INTEND TO ACHIEVE?

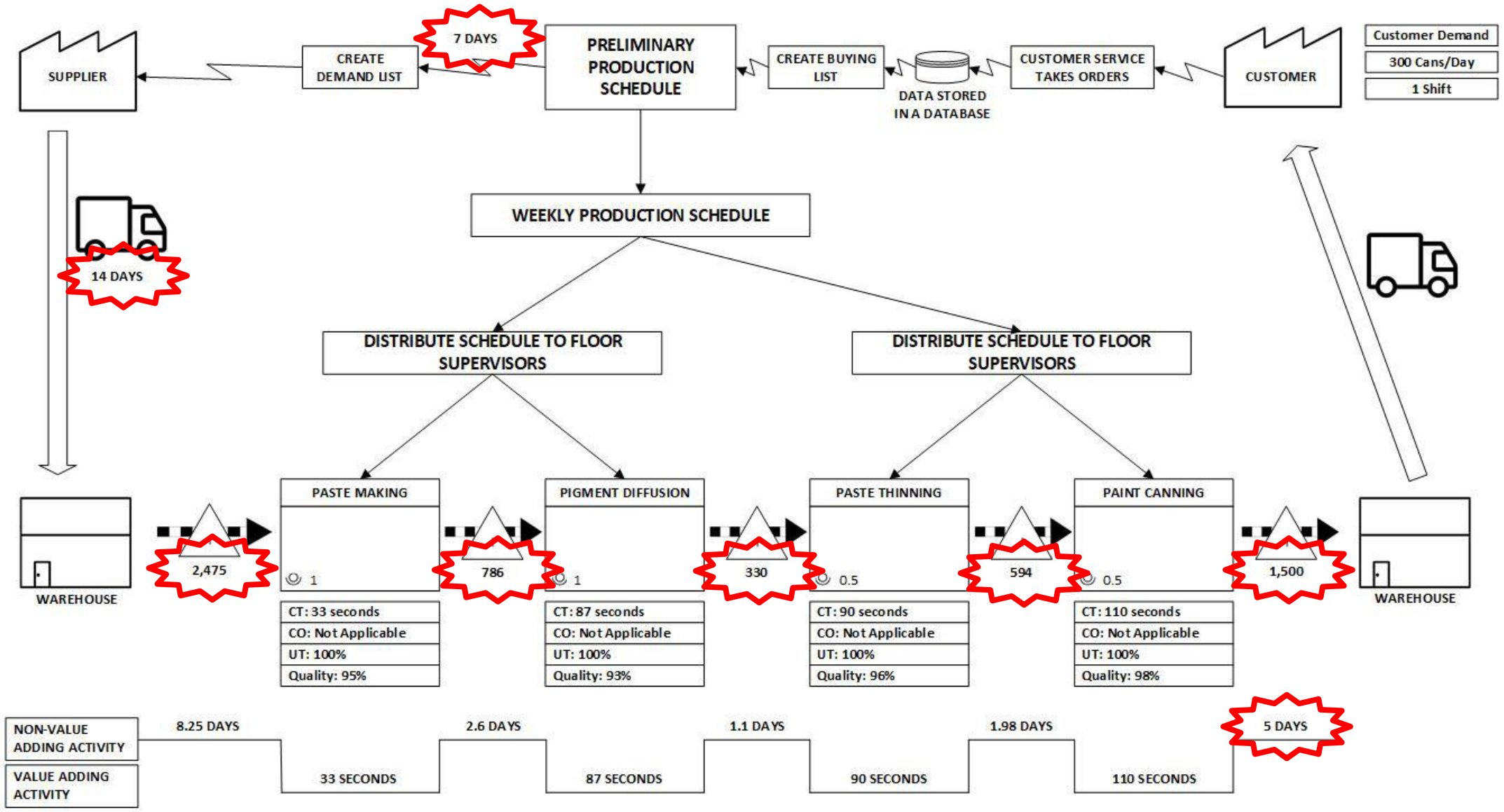
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THE SEVEN DAYS WAIT BETWEEN THE TIME THE CUSTOMER SERVICES RECEIVES THE ORDER AND THE TIME IT ARRIVES AT THE PURCHASING DEPARTMENT CAN ALSO BE REMOVED. A DATA STORAGE AND RETRIEVAL SYSTEM IS ALREADY IN PLACE. AN AUTOMATION CAN TRACK THE EXISTING INVENTORY AND THE CUSTOMER REQUIREMENTS TO IDENTIFY THE RAW MATERIALS THAT NEED TO BE PROCURED FROM THE SUPPLIER. THIS REDUCES THE INVOLVEMENT OF THE PURCHASING DEPARTMENT TO WAIT FOR 7 ADDITIONAL DAYS TO CREATE A DEMAND LIST.

Total Time = 39.93 days
Lead Time = 18.93 days
Takt Time = 87 seconds
Process/Processing/Value Added Time = 320 seconds
Cycle Time = 106.7 seconds

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Total Time = 39.95 days
Lead Time = [illegible] days
Takt Time = 87 seconds
Process/Processing/Value Added Time = 320 seconds
Cycle Time = 106.7 seconds

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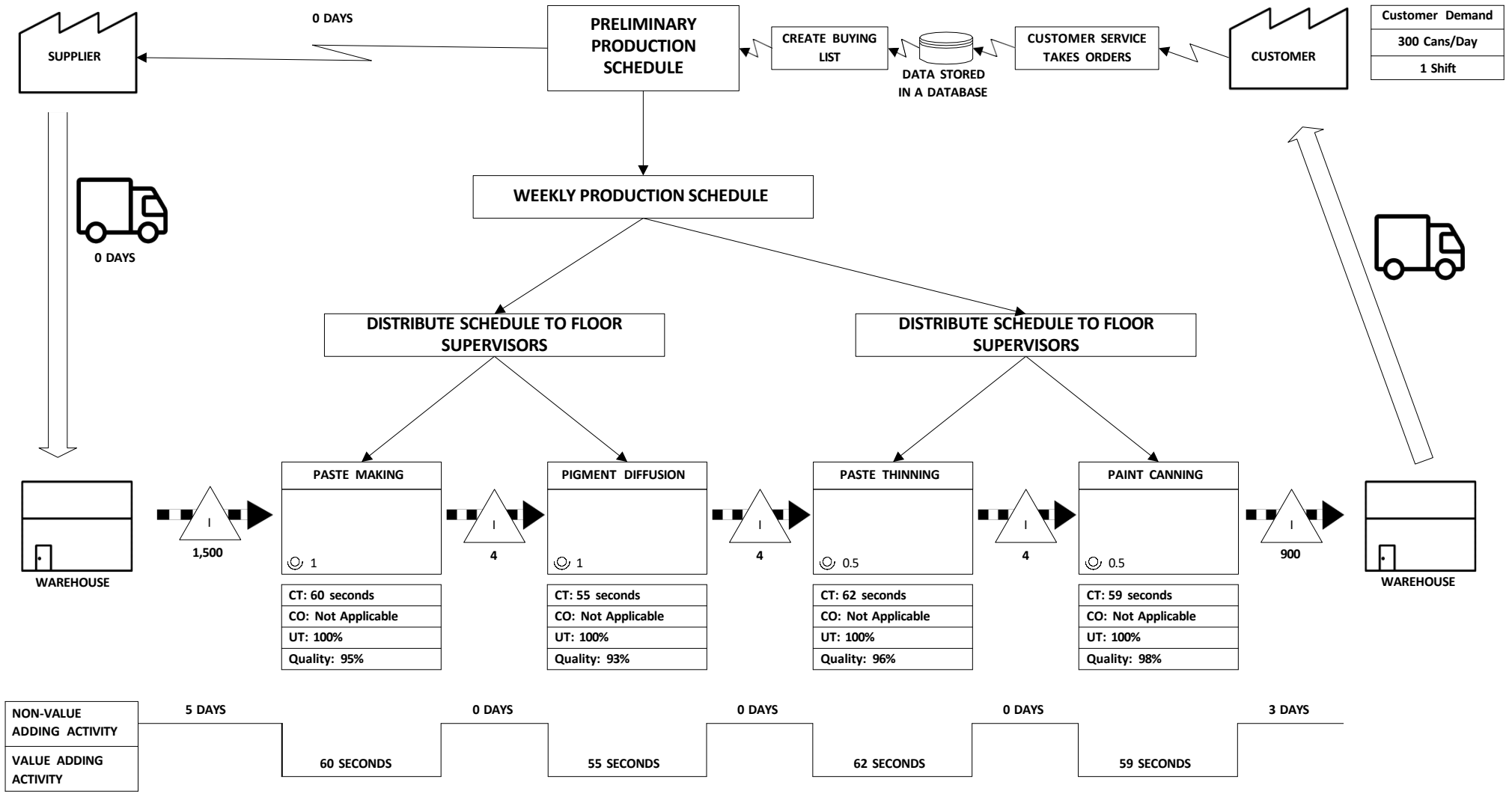
CLOSURE OF THE CASE STUDY



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FUTURE STATE MAP

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Total Time = 8 days
Lead Time = 8 days
Takt Time = 87 seconds
Process/Processing/Value Added Time = 236 seconds
Cycle Time = 78.67 seconds

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TOTAL TIME OF THE PROCESS

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		# OF DAYS	SECONDS
1.	THE AGEING OF RAW MATERIALS AT THE WAREHOUSE (BEFORE THE PASTE MAKING PROCESS) IS 5 DAYS	5	
2.	THE PASTE MAKING PROCESS TAKES 60 SECONDS		60
3.	THE PIGMENT DIFFUSION PROCESS TAKES 55 SECONDS		55
4.	THE PASTE THINNING PROCESS TAKES 62 SECONDS		62
5.	THE PAINT CANNING PROCESS TAKES 59 SECONDS		59
6.	AND THE FINISHED GOODS ARE KEPT IN THE WAREHOUSE FOR 3 DAYS BEFORE BEING SHIPPED TO THE CUSTOMER	3	
		8 DAYS	236 SECONDS

SINCE 236 SECONDS IS INFINITESIMALLY SMALLER THAN 8 DAYS, WE CAN IGNORE THIS NUMBER.

SO, THE TOTAL TIME IS 8 DAYS FROM ORDER TO DELIVERY.

AS COMPARED TO THE TOTAL TIME OF ~40 DAYS IN THE CURRENT STATE, THE FUTURE STATE SHOWS A SIGNIFICANT REDUCTION OF 32 DAYS. THAT IS 80% OF REDUCTION FROM ORDER TO DELIVERY TIME.

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LEAD TIME OF THE PROCESS

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		# OF DAYS	SECONDS
1.	THE AGEING OF RAW MATERIALS AT THE WAREHOUSE (BEFORE THE PASTE MAKING PROCESS) IS 5 DAYS	5	
2.	THE PASTE MAKING PROCESS TAKES 60 SECONDS		60
3.	THE PIGMENT DIFFUSION PROCESS TAKES 55 SECONDS		55
4.	THE PASTE THINNING PROCESS TAKES 62 SECONDS		62
5.	THE PAINT CANNING PROCESS TAKES 59 SECONDS		59
6.	AND THE FINISHED GOODS ARE KEPT IN THE WAREHOUSE FOR 3 DAYS BEFORE BEING SHIPPED TO THE CUSTOMER	3	

THE LEAD TIME WILL BE THE SAME AS TOTAL TIME OF 8 DAYS. THIS HAS ALSO SHOWN A SIGNIFICANT REDUCTION OF 11 DAYS I.E. THE LEAD TIME REDUCED FROM ~19 DAYS TO 8 DAYS. A REDUCTION OF 58%.

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TAKT TIME OF THE PROCESS

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CUSTOMER DEMAND HAS NOT CHANGED AND WE HAVE NOT MADE ANY CHANGES TO THE SHIFT TIMINGS, ETC. HENCE, TAKT TIME CONTINUES TO REMAIN THE SAME AT 87 SECONDS.

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PROCESS TIME

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THE EARLIER PROCESS TIME WAS 320 SECONDS. IT WAS CALCULATED BY ADDING THE PROCESS TIME OF PASTE MAKING, PIGMENT DIFFUSION, PASTE THINNING, AND PAINT CANNING PROCESS.

		# OF DAYS	SECONDS
1.	THE PASTE MAKING PROCESS TAKES 60 SECONDS	0	60
2.	THE PIGMENT DIFFUSION PROCESS TAKES 55 SECONDS	0	55
3.	THE PASTE THINNING PROCESS TAKES 62 SECONDS	0	62
4.	THE PAINT CANNING PROCESS TAKES 59 SECONDS	0	59
		0 DAYS	236 SECONDS

THUS, THE PROCESS TIME HAS REDUCED FROM 320 TO 236 SECONDS WHICH IS A SIGNIFICANT REDUCTION OF 26%.

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CYCLE TIME

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CALCULATE PROCESS TIME:

- **PROCESS TIME = 60 SECONDS + 55 SECONDS + 62 SECONDS + 59 SECONDS (CALCULATED PREVIOUSLY)**
- **PROCESS TIME = 236 SECONDS**

IDENTIFY THE NUMBER OF OPERATORS:

- **NUMBER OF OPERATORS INVOLVED = 3**

CYCLE TIME:

$$\text{CYCLE TIME} = \frac{PROCESS\ TIME}{NUMBER\ OF\ OPERATORS} = \frac{236}{3}$$

CYCLE TIME = 78.67 SECONDS

IT TAKES 78.67 SECONDS WHICH IS A REDUCTION OF 26%.

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CASE STUDY CLOSURE

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A CUSTOMER WHO RECEIVED THE ORDERED PAINT CANS AFTER 40 DAYS NOW RECEIVES IT JUST AFTER 8 DAYS.

IMAGINE HOW HAPPY THE CUSTOMER WOULD BE.

IN A NUTSHELL, YOU HAVE REDUCED THE:

- **TOTAL TIME BY 80%**
- **THE LEAD TIME BY 58%**
- **THE PROCESS TIME BY 26%**
- **THE CYCLE TIME BY ANOTHER 26%**

ALTHOUGH NOT MENTIONED HERE, VALUE STREAM MAPPING WILL ALSO REDUCE THE MATERIAL HANDLING PERIODS, AND THAT WILL ALSO INCREASE EFFICIENCY ALONG WITH THE 80% CUT IN THE WAITING PERIOD.

LEAN MANUFACTURING WAS IMPLEMENTED IN EVERY FACET OF THE PRODUCTION PROCESS, LEADING TO AN IMPRESSIVE OVERALL REDUCTION IN THE NUMBER OF DAYS FROM THE MOMENT THE CUSTOMER ORDER IS PLACED TO THE MOMENT THE FINISHED PRODUCT IS SHIPPED.

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THE IMPLEMENTATION PLAN



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THE IMPLEMENTATION PLAN

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THE IMPLEMENTATION PLAN CREATES A GUIDELINE ON EXECUTING THE ADJUSTMENTS MADE ON THE CURRENT STATE MAP TO CREATE THE FUTURE STATE MAP.

YOU SHOULD HOLD REGULAR MEETINGS, AT LEAST ONCE EVERY WEEK, TO UPDATE AND TRACK THE PROGRESS OF THE CHANGES MADE WITH THE TEAM MEMBERS.

THE REGULAR MEETINGS ALSO STRESS TO THE MEMBERS THE IMPORTANCE OF THE CHANGES AND WHY THEY MUST BE IMPLEMENTED.

BY MAKING THEM ACCOUNTABLE FOR THE IMPLEMENTATION OF THE CHANGES, YOU WILL ENSURE THAT THE PROCESS IS SPEND ALONG.

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SECTION CONCLUSION:
BUILDING THE FUTURE STATE VSM



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SECTION CONCLUSION

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WHILE BUILDING THE FUTURE STATE VSM, YOU ADDRESSED THE FOLLOWING QUESTION:

1. **WHAT IS THE TAKT TIME?**
2. **ARE YOU BUILDING TO A WAREHOUSE OR DIRECTLY TO SHIPPING?**
3. **WHERE SHOULD YOUR CONTINUOUS FLOW PROCESSING CONTROL THE PRODUCTION OF UPSTREAM PROCESSES?**
4. **WHERE WILL YOU IMPLEMENT A PULL SYSTEM?**
5. **HOW DO YOU INTEND TO LEVEL THE PRODUCTION MIX AT THE PACER PROCESS?**
6. **WHAT PROCESS IMPROVEMENTS DO YOU INTEND TO ACHIEVE WITH THE FUTURE STATE MAP?**

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SECTION INTRODUCTION:
WHAT ARE THE VSM GUIDELINES?



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VALUE STREAM MAPPING (VSM) MASTERCLASS

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Introduction to Value Stream Map (VSM)

- Course Introduction
- What is Value Stream Mapping (VSM)?
- Why is Value Stream Mapping so popular?
- Section Conclusion

Terminologies and Symbols used in VSM0

- Section Introduction
- Terminologies used in VSM
- Symbols used in VSM Process
- Material Flow Symbols
- Information Flow Symbols
- General Symbols
- Activity: Terminologies and Symbols used in VSM
- Activity Solution: Terminologies and Symbols used in VSM
- Section Conclusion

Steps to Create the Current State VSM

- Section and Case Study Introduction
- Step 1. Pick a Product Family
- Step 2. Choose Your VSM Team
- Step 3: Figure out Customer Demand
- Step 4: Map and Design the Process Flow
- Step 5: Map the Material Flow
- Step 6: Map the Information Flow
- Case Study Recap: Current State VSM
- Perform Calculations | Analyze Current State VSM
- Calculate Total Time | Lead Time | Takt Time
- Calculate Process Time | Cycle Time
- Activity: Steps to Create Current State VSM
- Activity Solution: Steps to Create Current State VSM
- Section Conclusion

Examine the Current State VSM

- Section Introduction
- Meeting Customer Demand with existing No. of Operators
- Meeting Customer Demand with increased No. of Operators
- Analyze the Process Balance
- Watch for Waste
- Section Conclusion

Strong Foundations for Future State VSM

- Section Introduction
- Change to Continuous Flow
- CONWIP
- Kanban
- Readjusting Your Process
- Alter the Batch Sizes
- Smoothening out Production
- Heijunka | Heijunka Box
- Cut back Changeover Time
- Improve Quality
- Putting it all Together

Key Questions to ask for Future State VSM

- Section Introduction
- What is the Takt Time?
- Are you building to a warehouse or directly to shipping?
- Where should Continuous Flow Control the Production?
- Controlling area of upstream processes
- Where will you implement the Pull System?
- How do you intend to level the production mix?
- What process improvements do you intend to achieve?
- What is the Implementation Plan?
- Closure of the Case Study
- Section Conclusion

VSM Guidelines

- Section Introduction
- How to get the most of your VSM?
- What are the Pitfalls to avoid?
- What are the advantages of VSM?
- What are the disadvantages of VSM
- Section Conclusion

Course Conclusion

- Course Summary
- Practice Questions
- Become a Certified VSM Practitioner!

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HOW TO GET THE MOST OUT OF YOUR VALUE STREAM MAP (VSM)?



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HOW TO GET THE MOST OUT OF YOUR VSM?

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BUILD YOUR VSM AROUND CUSTOMER DEMANDS

CAPTURE THE PROCESS AS IT CURRENTLY OPERATES, NOT AS IT OUGHT TO OPERATE

ASSIGN A VALUE STREAM MAP MANAGER TO LEAD THE MAPPING EFFORT

PERFORM A WALKTHROUGH TO ENSURE THE CORRECT INFORMATION IS PUT IN ABOUT THE INFORMATION AND MATERIAL FLOW

FOCUS ON ONE SMALL STEP AT A TIME

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WHAT ARE THE PITFALLS TO AVOID FOR YOUR VSM?



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PITFALLS TO AVOID FOR YOUR VSM

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VALUE STREAM MAPPING IN ITSELF IS NOT THE GOAL

MANY PEOPLE BELIEVE THAT THE CREATION OF THE VALUE STREAM MAP ALONE IS WHAT YOU NEED TO DO WITHOUT TAKING THE EXTRA STEP OF IMPLEMENTING THE CHANGES NOTED. DO NOT MAKE THAT MISTAKE.

STARTING A DETAILED DOCK-TO-DOCK FLOW

BY BEGINNING WITH A DETAILED ANALYSIS OF THE PRODUCTION PROCESS, YOU WILL FIND YOURSELF BOGGED DOWN WITH THE MINUTE DETAILS AND WILL LOSE SIGHT OF THE OVERALL FLOW PERSPECTIVE.

OVEREMPHASIS ON TALLYING INVENTORY

FINDING AND TALLYING THE ACCUMULATION OF INVENTORY CAN BE FUN, BUT YOU SHOULD STRIVE TO MAKE SURE IT DOES NOT OVERSHADOW WHY THE INVENTORY IS THERE.

DO NOT GO TOO FAR INTO THE FUTURE

WHEN YOU SEE MORE THAN SIX KAIZEN LIGHTENING BURSTS ON YOUR MAP, YOU NEED TO TAKE A BREAK. INSTEAD OF THINKING TOO FAR AHEAD, YOU CAN SKETCH WHERE YOU WOULD LIKE TO BE IN THE NEXT COUPLE OF YEARS TO GIVE YOURSELF AND THE ORGANIZATION A SENSE OF DIRECTION.

CONTINUOUS IMPROVEMENT NEEDS THE CHALLENGE OF TARGET CONDITIONS

YOU MUST ANSWER THESE TWO QUESTIONS; 'WHAT IS STOPPING YOU FROM ACCOMPLISHING THE TARGET CONDITION IN THIS PROCESS' AND 'WHAT STEP ARE YOU TAKING TO ACHIEVE THE TARGET CONDITION'?

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WHAT ARE THE ADVANTAGES OF VALUE STREAM MAPPING?



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ADVANTAGES OF VSM

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- VALUE STREAM MAPPING MAKES USE OF THE BIG PICTURE AND COGENT COMMUNICATION TO HELP YOU BETTER UNDERSTAND HOW TO IMPLEMENT THE NEW DATA AND ACTIONS PROPERLY.
- THE DISTRIBUTION PATHWAYS AND THE SUPPLY CHAINS ARE LINKED BY VALUE STREAM MAPPING. BY CONNECTING ALL THE BODIES OF THE PRODUCTION PROCESS, THE INFORMATION FLOW AND MATERIAL FLOW ARE ALSO INCLUDED.
- ALL OF THE PRODUCTION CONTROL AND SCHEDULING FUNCTIONS (PCS) LIKE PRODUCTION PLANNING AND DEMAND FORECASTING, PRODUCTION SCHEDULING, SHOP FLOOR CONTROL, AMONG OTHERS, ARE LINKED WITHIN THE VALUE STREAM MAP BY THE USE OF THE OPERATING STRICTURES IN THE MANUFACTURING SYSTEM.
- WITH VALUE STREAM MAPPING, YOU CAN INCORPORATE A LOT OF THE INDUSTRIAL ENGINEERING (IE) TECHNIQUES TO ANALYZE FLOW.
- WITH VALUE STREAM MAPPING, THE INFORMATION FOR THE OPERATION AND STORAGE PROCESSES ARE WELL DESCRIBED.
- VALUE STREAM MAPPING OFFERS YOU THE FOUNDATION TO SUCCESSFULLY IMPLEMENT LEAN MANUFACTURING AS YOU NOW HAVE A PRODUCTION SYSTEM THAT HIGHLIGHTS A TOTAL DOCK FLOW OF A PRODUCT.
- VALUE STREAM MAPPING OFFERS YOU A PROTOTYPE OF YOUR PRODUCTION PROCESS, WHICH MAKES IT EASIER TO STRATEGIZE AND MAKE USE OF THE LEAN PRINCIPLES.

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WHAT ARE THE DISADVANTAGES OF VALUE STREAM MAPPING?



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DISADVANTAGES OF VSM

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- UNFORTUNATELY, VALUE STREAM MAPPING DOES NOT WORK FOR SEVERAL PRODUCTS THAT HAVE DIFFERENT MAPS.
- VALUE STREAM MAPPING FAILS TO ACCOUNT FOR THE DELAYS AND CHANGES THAT OCCUR AS A RESULT OF POOR STRUCTURAL LAYOUT AND THE MISMANAGEMENT OF MATERIALS THAT ARISE AS A RESULT OF OPERATION STRUCTURES.
- JUST LIKE THE FLOW PROCESS CHARTING TECHNIQUE THAT IS USED BY INDUSTRIAL ENGINEERING (IE), VSM LACKS ECONOMIC MEASURE FOR VALUE SUCH AS PROFITS, INVENTORY EXPENSES, OPERATING COSTS, AND SO ON.
- THE LACK OF A THREE-DIMENSIONAL STRUCTURE OF THE PLANT LEADS TO A SENSE OF IMPRACTICALITY WHEN IMPLEMENTING CHANGES TO THE FUTURE MAP.
- IT IS PROBABLE THAT ONLY THE PRODUCTS THAT OFFER A HIGH VOLUME AND LOW VARIETY MANUFACTURING SYSTEMS ARE CONSIDERED WITH RESPECT TO CONTINUOUS FLOW
- FLOOR SPACE IS AN UNDERCOOKED AND OFTEN IGNORED ASPECT OF VSM. THIS LEADS TO THE UNDERUTILIZATION OF RESOURCES THAT AFFECTS VARIOUS PRODUCTION PROCESSES, SUCH AS WIP STORAGE, SUPPORT, AND PRODUCTION.
- VSM DOES NOT ADEQUATELY EXPLAIN ITS ROLE IN ORDER THROUGHOUT, OPERATING EXPENSES, AND WIP.
- NOT ALL FACTORS THAT ARE RELATED TO THE PRODUCTION PROCESS ARE INVOLVED IN THE VALUE STREAM MAPPING. SOME SUCH OVERLOOKED FACTORS INCLUDE THE QUEUING DELAYS, THE CAPACITY CONSTRAINTS, THE ORDER THROUGHOUT, AMONG OTHERS.

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DISADVANTAGES OF VSM

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- **THE MANUAL CREATION OF VALUE STREAM MAPS MEAN THAT THE MAPS CANNOT DEVELOP RAPIDLY, NOR CAN THERE BE SEVERAL CONTINGENCY PLANS FOR EITHER TIME OR BUDGET LIMITATIONS.**

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SECTION CONCLUSION:
WHAT ARE THE VSM GUIDELINES?



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SECTION CONCLUSION

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IN THIS SECTION, YOU LEARNED:

1. HOW TO GET THE MOST OUT OF YOUR VALUE STREAM MAP?
2. WHAT ARE THE PITFALLS TO AVOID WHEN BUILDING YOUR VSM?
3. WHAT ARE THE ADVANTAGES AND DISADVANTAGES OF A VALUE STREAM MAP?

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COURSE CONCLUSION



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